# Class 10
# Chemistry

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# 1. Chemical Reactions and Equations

## LEVEL-I

**1.** **Answer: (D)**
Zinc carbonate = **$ZnCO_3$**

**2.** Zinc oxide = **ZnO**

**3.** Carbon dioxide = **$CO_2$**
The correct equation is:
$ZnCO_3 \rightarrow ZnO + CO_2$
**Answer: (C)**
Methane has the molecular formula $CH_4$. Therefore, the correct symbol equation is:
$CH_4 + O_2 \rightarrow CO_2 + H_2O$
Answer: (D)
Balancing gives:
Carbon: $3 \rightarrow 3CO_2$
Hydrogen: $8 \rightarrow 4H_2O$
Oxygen atoms needed = 6 + 4 = 10 atoms = $5O_2$ molecules

**4.** **Answer: (A)**
A chemical equation represents reactants and products using chemical symbols and formulae.

**5.** **Answer: (B)**
According to the **Law of Conservation of Mass**, atoms are neither created nor destroyed during a **chemical reaction**. They are only rearranged to form new substances.

**6.** **Answer: (A)**
The substances present before the reaction begins are called **reactants**.

**7.** **Answer: (B)**
The new substances produced after the reaction are called **products**.

**8.** **Answer: (D)**
The arrow (→) means "**yields**" or "**forms**", indicating that reactants produce products.

**9.** **Answer: (B) Balanced equation**
A **balanced chemical equation** has an equal number of atoms of each element on both the reactant side and the product side. This follows the **Law of Conservation of Mass**, which states that matter is neither created nor destroyed during a chemical reaction.
**Example:**
Unbalanced equation: $H_2 + O_2 \rightarrow H_2O$
Balanced equation: $2H_2 + O_2 \rightarrow 2H_2O$
Here, the number of hydrogen and oxygen atoms is the same on both sides.

**10.** **Answer: (C) Skeletal equation**
A **skeletal chemical equation** (or unbalanced equation) shows only the correct chemical formulas of the reactants and products but does **not** have equal numbers of atoms on both sides.
**Example:**
Skeletal (unbalanced) equation: $Mg + O_2 \rightarrow MgO$

Balanced equation:$2Mg+O_2 \rightarrow 2MgO$
The skeletal equation is balanced by adding suitable coefficients so that the number of atoms of each element becomes equal on both sides.

**11. Answer: (B) N O Pb**

While balancing chemical equations:
Balance the element that appears in the **largest number of compounds first**.
Balance **oxygen last**, as it is present in several compounds.
For this equation:

1. Balance **Nitrogen (N)**
2. Balance **Oxygen (O)**
3. Check **Lead (Pb)**

Balanced equation:
$2Pb(NO_3)_2 \rightarrow 2PbO+4NO_2+O_2$
Hence the correct order is **N → O → Pb**

**12. Answer: (A) Al**

Count the atoms., Although sulphur appears equal, the equation as a whole is incomplete and the intended balanced element from the options is **Aluminium**, which is correctly balanced from the start.

**13. Answer: (A)**

Only option **A** has equal numbers of atoms of every element on both sides.

**14. Answer: (D)**

General rule:

1. Chromium
2. Potassium
3. Hydrogen
4. Sulphur
5. Oxygen (last)

Therefore,
**i → v → ii → iv → iii**

**15. Answer: (D)**

1 gram nitrogen + 3 grams hydrogen → 2 grams ammonia( Incorrect)
Actual masses:
1 mole $N_2$ = **28 g**;3 moles $H_2$ = **6 g**
Products:
2 moles $NH_3$ = **34 g**;Masses do **not** follow a **1 : 3 : 2 gram** ratio.

**16. Answer: (B)**

A **combination reaction** is one in which **two or more reactants combine to form a single product**.
Example:$C+O_2 \rightarrow CO_2$

**17. Answer: (D)**

A **combination reaction** is one in which **two or more substances combine to form a single product**.
Here,$2Mg+O_2 \rightarrow 2MgO$
Magnesium and oxygen combine to form only one product, magnesium oxide.

**18. Answer: (C)**

When magnesium burns in oxygen, it forms **magnesium oxide (MgO)**, a white powder.

Balanced equation:
$2Mg+O2 \rightarrow 2MgO$
This is also a **combination reaction**.

**19. Answer: (C)**
A **word equation** uses:
"**+**" to separate reactants,
"→" to show the formation of products.
Correct word equation:
Potassium hydroxide + Sulphuric acid → Potassium sulphate + Water
(Symbol equation: $2KOH+H_2SO_4 \rightarrow K_2SO_4+2H_2O$)

**20. Answer: (A)**
A **single displacement reaction** occurs when a more reactive element displaces a less reactive element from its compound.
Example:
$Fe+CuSO_4 \rightarrow FeSO_4+Cu$
Iron is more reactive than copper, so it displaces copper from copper sulphate.

**21. Answer: (D)**
Start with the skeletal equation:
$Fe_2O_3+H_2SO_4 \rightarrow Fe_2(SO_4)_3+H_2O$
Balance sulphate groups by placing 3 before $H_2SO_4$:
$Fe_2O_3+3H_2SO_4 \rightarrow Fe_2(SO_4)_3+H_2O$
Balance hydrogen and oxygen by placing 3 before $H_2O$:
$Fe_2O_3+3H_2SO_4 \rightarrow Fe_2(SO_4)_3+3H_2O$

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**Level-II**

**1. Answer: (A)**
The equation contains **five different elements**:
Aluminium (Al)
Carbon (C)
Oxygen (O)
Hydrogen (H)
Sulphur (S)
Thus, the numbers given under the headings correspond to the count of each distinct element symbol in the molecular formulas, matching:
**Al = 2, C = 2, O = 5, H = 2, S = 2**

**2. Answer: (B)**
$Mg+CO_2 \rightarrow MgO+C$
**Step 1: Balance oxygen**
$CO_2$ contains **2 oxygen atoms**, so place **2** before MgO.
$Mg+CO_2 \rightarrow 2MgO+C$
**Step 2: Balance magnesium**
Now there are **2 Mg** atoms on the right.
Place **2** before Mg.
$2Mg+CO_2 \rightarrow 2MgO+C$

**3. Answer: (B)**
$KClO_3 \rightarrow KCl+O_2$
**Step 1: Balance potassium and chlorine**
Place 2 before $KClO_3$ and KCl.
$2KClO_3 \rightarrow 2KCl+O_2$

**Step 2: Balance oxygen**

Left side:

2 × 3 = 6 oxygen atoms

Right side:Place **3** before $O_2$.

$2KClO_3 \rightarrow 2KCl + 3O_2$

**4. Answer: (A)**

Combustion of benzene is an exothermic reaction, meaning heat is released. Therefore, the enthalpy change (ΔH**) is** negative**.**

Balanced equation:

$2C_6H_6 + 15O_2 \rightarrow 12CO_2 + 6H_2O \Delta H = -ve$

**5. Answer: (C)**

The balanced equation is:

$Fe_3O_4 + 4H_2 \rightarrow 3Fe + 4H_2O$

Thus, the correct coefficients are **1 : 4 : 3 : 4**.

**6. Answer: (D)**

This reaction represents the roasting of iron pyrite. The balanced equation is:

$4FeS_2 + 11O_2 \rightarrow 2Fe_2O_3 + 8SO_2$

**7. Answer: (D)**

Copper reacts with **hot concentrated sulphuric acid** to produce copper sulphate, sulphur dioxide gas, and water. The balanced equation is:

$Cu + 2H_2SO_4 \rightarrow CuSO_4 + SO_2 + 2H_2O$

**8. Answer: (A)**

**Step 1: Write the equation**

$Pb(NO_3)_2 \rightarrow PbO + NO_2 + O_2$

**Step 2: Balance Pb**

There is **1 Pb** on both sides.

**Step 3: Balance N**

There are **2 nitrogen atoms** in $Pb(NO_3)_2$, so place **2 before $NO_2$**.

$Pb(NO_3)_2 \rightarrow PbO + 2NO_2 + O_2$

**Step 4: Count oxygen**

Left side:

$Pb(NO_3)_2$ = **6 O**

Right side:

PbO = 1

$2NO_2$ = 4

Total = 5

Need one more oxygen atom, which comes from **½ $O_2$**.

To remove the fraction, multiply the whole equation by 2.

$\boxed{2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2}$

**9. Answer: (B)**

**Step 1: Balance K and Cr**

Choose coefficient **2** for $K_2Cr_2O_7$.

$2K_2Cr_2O_7 + H_2SO_4 \rightarrow 2K_2SO_4 + 2Cr_2(SO_4)_3 + H_2O + O_2$

Now,

K = 4 on both sides

Cr = 4 on both sides

**Step 2: Balance sulphate groups**

Products contain:

$2K_2SO_4$ → 2 sulphates
$2Cr_2(SO_4)_3$ → 6 sulphates
Total sulphates = **8**
Hence use **$8H_2SO_4$**.
**Step 3: Balance H**
$8H_2SO_4$ contains **16 H**
So put **$8H_2O$**.
**Step 4: Balance oxygen**
Left side:
$2K_2Cr_2O_7$ = 14 O
$8H_2SO_4$ = 32 O
Total = **46 O**
Right side:
$2K_2SO_4$ = 8
$2Cr_2(SO_4)_3$ = 24
$8H_2O$ = 8
Total = 40
Need 6 more oxygen atoms = **$3O_2$**
Balanced equation:

$$\boxed{2K_2Cr_2O_7 + 8H_2SO_4 \rightarrow 2K_2SO_4 + 2Cr_2(SO_4)_3 + 8H_2O + 3O_2}$$

**10. Answer: (C)**
**Step 1: Balance aluminium**
There are **2 Al** on the left.
So put **2** before $Al(NO_3)_3$.
$Al_2(CO_3)_3 + HNO_3 \rightarrow 2Al(NO_3)_3 + CO_2 + H_2O$
**Step 2: Balance nitrate groups**
$2Al(NO_3)_3$ contains **6 nitrate ions**.
Therefore place **$6HNO_3$**.
**Step 3: Balance carbon**
There are **3 carbon atoms** in $Al_2(CO_3)_3$.
So put **$3CO_2$**.
**Step 4: Balance hydrogen**
$6HNO_3$ contains **6 H**
Therefore put **$3H_2O$**.
**Check oxygen**
Left side:
$Al_2(CO_3)_3$ = 9 O
$6HNO_3$ = 18 O
Total = **27 O**
Right side:
$2Al(NO_3)_3$ = 18
$3CO_2$ = 6
$3H_2O$ = 3
Total = **27 O**
Balanced equation:

$$\boxed{Al_2(CO_3)_3 + 6HNO_3 \rightarrow 2Al(NO_3)_3 + 3CO_2 + 3H_2O}$$

**11. Answer: (A,C,D)**
A balanced chemical equation provides:

Correct symbols and formulae
Physical states (g, l, s, aq)
Relative number of molecules
Reaction conditions (temperature, catalyst, pressure)
However, **it does not show the actual number of atoms and molecules** participating because the equation represents **relative ratios**, not actual numbers.

**12. Answer: (A,B,C)**

Option (A)
"8 atoms of reactants react to give 8 atoms of products."
Reactants:$N_2$ = 2 atoms
$3H_2$ = 6 atoms
Total = **8 atoms**
Products:$2NH_3$
Each $NH_3$ has 4 atoms.
2 × 4 = **8 atoms** Correct
**Option (B)**
"4 molecules of reactants react to give two molecules of product."
Reactants:
1 $N_2$ molecule
3 $H_2$ molecules
Total = **4 molecules**
Products:2 $NH_3$ molecules Correct
**Option (C)**
"34 g reactants give 34 g products."
Mass of reactants:
$N_2$ = 28 g
$3H_2$ = 6 g
Total = **34 g**
Products:$2NH_3$
2 ×17 = **34 g** Correct

**13. Answer: (B,C,D)**

**(B)**$2Mg + O_2 \rightarrow 2MgO$
Left: Mg=2;O=2
Right: Mg=2;O=2
Balanced
**(C)**$2CO + O_2 \rightarrow 2CO_2$
Left: C=2;O=4
Right: C=2;O=4 (Balanced)
**(D)**$Fe + S \rightarrow FeS$
Left: Fe=1;S=1
Right: Fe=1;S=1( Balanced)

14. **Answer: (A,B,C,D)**

**(A)**$\boldsymbol{Na_2CO_3 + 2HCl \rightarrow 2NaCl + CO_2 + H_2O}$
Na=2;Cl=2H=2;C=1;O=3(Balanced)
(B)$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$
Ca=1;C=1;H=2;O=4 (Balanced )
(C)$2H_2S + SO_2 \rightarrow 2H_2O + 3S$
Left: H=4;S=3;O=2

Right: H=4;S=3;O=2(Balanced )
(D)$4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$
Left: N=4;H=12;O=10
Right: N=4;H=12;O=10 (Balanced )

**15. Answer: (A), (B) and (C)**
$2K_2Cr_2O_7+8H_2SO_4 \rightarrow 2K_2SO_4+2Cr_2(SO_4)_3+8H_2O+3O_2$
All atoms match.( Balanced)
**(B)$3CaOCl_2 + 2NH_3 \rightarrow 3CaCl_2 + 3H_2O + N_2$**
Left:Ca=3;Cl=6 ;N=2;H=6;O=3
Right:Ca=3;Cl=6;N=2;H=6;O=3( Balanced)
**(C)$6FeSO_4 + 3H_2SO_4 + 2HNO_3 \rightarrow 3Fe_2(SO_4)_3 + 2NO + 4H_2O$**
All atoms balance.( Balanced)

**16. Answer: A, B and C**
Carbon:
Left =5
Right =1
Not balanced.
(B)$2NH_3 + N_2 \rightarrow 2N_2H_2 + 2H_2$
Nitrogen:
Left =4
Right =4
Hydrogen:
Left =6
Right =8
Not balanced.
(C)$2Cr(OH)_3 + 6Na_2O_2 \rightarrow 2Na_2CrO_4 + 2NaOH + 2H_2O$
Sodium:
Left =12
Right =6
Not balanced.

**Assertion and reason**

**17. Answer: (A)**
Many chemical reactions exhibit more than one observable change at the same time. For example:
Reaction of zinc with dilute hydrochloric acid:$Zn + 2HCl \rightarrow ZnCl_2 + H_2 \uparrow$
Observations:
Hydrogen gas is evolved.
The reaction mixture becomes warm because heat is released (exothermic reaction).
Thus, the reason correctly explains why some chemical reactions show multiple characteristics.

**18. a⟶R; b⟶S; c⟶Q; d⟶P**

**19.** Answer: B)
A balanced chemical equation uses the correct symbols and chemical formulae of the reactants and products.
Example: $2H_2 + 0_2 \rightarrow 2H_20$

**20. Answer: B)**
Besides showing reactants and products, a chemical equation may indicate:
Physical states: (s), (l), (g), (aq)
Heat changes (Δ or "Heat")

Catalyst
Temperature and pressure
Example:$CaCO_3(s) \xrightarrow{\Delta} CaO(s) + CO_2(g)$
This shows:
Physical states of substances.
Heat is supplied**.**

**21. Answer: C)**
A chemical equation clearly identifies:
Reactants (substances present before the reaction)
Products (substances formed after the reaction)
Example:$2H_2 + O_2 \rightarrow 2H_2O$

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## CHEMICAL REACTIONS

### Level-1

**1. Answer: (D)**
A chemical change forms new substances.
Glowing of a bulb – Physical change.
Crushing ice – Only size changes.
Magnetizing iron – Temporary physical change.
Burning a matchstick produces ash, carbon dioxide, etc.

**2. Answer: (A)**
Burning produces new substances like carbon dioxide and water.

**3. Answer: (C)**
A new substance is formed

**4. Answer: (A)**
Iron loses electrons.$Fe \rightarrow Fe^{3+} + 3e^-$
Loss of electrons = Oxidation.

**5. Answer: (D)**
Nitrogen gains hydrogen (reduction).
Hydrogen loses electrons (oxidation).
Nitrogen acts as the oxidising agent.
Therefore all statements are correct.

**6. Answer (C).**
$2KClO_3 \rightarrow 2KCl + 3O_2$
Type: Decomposition reaction.
Reason: A single compound, Potassium Chlorate ($KClO_3$), breaks down upon heating into two simpler substances: Potassium Chloride (KCl) and Oxygen gas ($O_2$).

**7. Answer: (D)**
Usually requires heat, electricity or light.
Hence mostly endothermic.
One compound breaks into simpler substances.
Both statements (B) and (C) are correct**.**

**8. Answer: (C)**
This is the definition of decomposition reaction.

**9. Answer: (D)**
Aluminium displaces iron from iron oxide.

**10. Answer: (A)**

One element replaces another.

**11. Answer: (C)**

Zinc displaces copper. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

**12. Answer: (D)**

First equation: $2K + 2H_2O \rightarrow 2KOH + H_2$

So I = $2H_2O$

Second equation: $Fe + 2HCl \rightarrow FeCl_2 + H_2$

II = $FeCl_2$; III = $H_2$

**13. Answer: (C)**

Double displacement is another name for double decomposition.

**14. Answer: (A)**

An insoluble solid formed is called a precipitate.

**15. Answer: (C)**

$HCl+NaOH \rightarrow NaCl+H_2O$

This is neutralisation.

**16. Answer: (B)**

Oxidation = Loss of electrons

**17. Answer: (B)**

Gain of electrons = Reduction

**18. Answer: (D)**

Formation of $CO_2$ releases a large amount of heat.

$C + O_2 \rightarrow CO_2 + Heat$

**19. Answer: (C)**

Heat is released.

**20. Answer: (A)**

Energy released > Energy absorbed

Hence heat is evolved.

**21. Answer: (B)**

Such reactions occur in both forward and backward directions.

**22. Answer: (A)**

Reaction proceeds only in one direction.

**23. Answer: (D)**

All are reversible in a closed system.

**24. Answer: (B)**

$N_2+3H_2 \rightleftharpoons 2NH_3$

The reaction proceeds in both directions.

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**Level-II**

**1. Answer: (D)**

Reaction: $\boxed{3BaCl_2 + 2(NH_4)_3PO_4 \rightarrow Ba_3(PO_4)_2 + 6NH_4Cl}$

Step 1: Check the ratio

3 moles $BaCl_2$ react with 2 moles $(NH_4)_3PO_4$.

The given quantities exactly match the balanced equation.

Step 2: Find $NH_4Cl$ formed

According to the equation,

3 moles $BaCl_2$ produce 6 moles $NH_4Cl$.

**2. Answer: (B)**

A reducing agent loses electrons and reduces another substance.

$HNO_3$ is a strong oxidising agent.

$H_2SO_3$ is a moderate reducing agent.
$SnCl_2$ is a reducing agent.
$H_2S$ is one of the strongest reducing agents because sulfur is in its lowest oxidation state (−2) and is easily oxidised.

**3. Answer: (C)**

$Cl_2+H_2O \rightarrow 2HCl+O_2$
Oxidation states:
Oxygen in $H_2O$ = −2
Oxygen in $O_2$ = 0
Oxygen is oxidised, so $H_2O$ acts as the reducing agent.

**4. Answer: (B)**

A reducing agent causes reduction by donating electrons.
Oxidation = Loss of electrons
Reduction = Gain of electrons

**5. Answer: (D)**

odium:$Na \rightarrow Na^+ + e^-$
Sodium loses electrons.
Hence sodium is oxidised.
Chlorine gains electrons.
Therefore chlorine acts as the oxidising agent.

**6. Answer: (C)**

H atoms form $3H_2$.
$\boxed{2Al + 6HCl \rightarrow 2AlCl_3 + 3H_2}$
Thus,$a = 6, b = 3$
Their values satisfy
$a = 6$
$b = 3$
$a/2 = 3$

**7. Answer: (C)**

I)$2H_2O \rightarrow 2H_2 + O_2$
Electrolysis
→ Electrolytic decomposition
(II)$3O_2 \xrightarrow{UV} 2O_3$
(Although this is actually a photochemical combination reaction, the question intends to associate UV light with photolytic processes.)
→ Photolytic
(III)$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$
Heat causes decomposition. → Thermal decomposition
Matching:(a) → III ;(b) → I ;(c) → II

**8. Answer: (D)**

Incorrect. Air is a homogeneous mixture of gases under normal conditions.

**9. Answer: (C)**

Step 1: Balance Carbon
6 carbon atoms → $6CO_2$
Step 2: Balance Hydrogen
14 hydrogen atoms → $7H_2O$
Step 3: Balance Oxygen
Right side:$6CO_2$ = 12 O ;$7H_2O$ = 7 O

Total = 19 oxygen atoms

Therefore, $\frac{19}{2}O_2$

Multiply the whole equation by 2: $\boxed{2C_6H_{14} + 19O_2 \rightarrow 12CO_2 + 14H_2O}$

Number of moles of $O_2$ = 19

**10. Answer: (B)**

(I)$Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3$

Aluminium loses electrons.

Reducing agent = Al

(II)$CuO + H_2 \rightarrow Cu + H_2O$

Hydrogen reduces CuO.

Reducing agent = $H_2$

(III)$H_2S + SO_2 \rightarrow 3S + 2H_2O$

$SO_2$ is reduced.

Oxidising agent = $SO_2$

(IV)$Mg + O_2 \rightarrow MgO$

Oxygen is reduced.

Oxidising agent = $O_2$

Thus, I → Al ;II → $H_2$ ;III → $SO_2$ ;IV → $O_2$

**11. Answer: (A)**

Copper reacts with:

Oxygen ($O_2$)

Water vapour ($H_2O$)

Carbon dioxide ($CO_2$)

forming green basic copper carbonate: $2Cu + H_2O + CO_2 + O_2 \rightarrow CuCO_3 \cdot Cu(OH)_2$

**12. Answer: (B)**

Rusting reaction:$4Fe + 3O_2 + xH_2O \rightarrow 2Fe_2O_3 \cdot xH_2O$

Iron combines with oxygen.

Hence it is a **combination reaction**.

**13. Answer: (B)**

Rust is hydrated iron(III) oxide. $\boxed{Fe_2O_3 \cdot xH_2O}$

14. Answer: (C)

Carbon removes oxygen from lead oxide.

Carbon itself is oxidised to CO.

Therefore Carbon is the **reducing agent**.

PbO is reduced to Pb.

**15. Answer: (D)**

$Fe+CuSO_4 \rightarrow FeSO_4+Cu$

Iron is more reactive than copper.

Iron displaces copper.

Blue $CuSO_4$ solution turns green due to $FeSO_4$.

Brown copper gets deposited on the nail.

**16. Answer: (B and C)**

(B) $2H_2O \rightarrow 2H_2 + O_2$— Decomposition reaction

(C) $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$— Decomposition reaction

**17. Answer: (C)**

Heat evolved or absorbed during a chemical reaction is commonly measured in kilojoules (kJ).

**18. Answer: (A,B)**

Reaction: $\boxed{CaCO_3 \xrightarrow{\Delta} CaO + CO_2}$

Products are:

Calcium oxide (CaO)

Carbon dioxide ($CO_2$)

**19. Answer: (D)**

An **inhibitor (negative catalyst)** slows down a chemical reaction.

**20. Answer: (A)**

Combination reactions include:

Element + Element

Element + Compound

Compound + Compound

A common definition is: $A + B \rightarrow AB$

**21. Answer: A, B and C**

A) Hydrogen burns in oxygen.( True)

**(B)**Two elements combine to form one compound.(True)

**(C)**Formation of water from hydrogen and oxygen is a combination reaction.

**22. Answer: (D)**

It is a decomposition reaction.

**23. A nswer: (A)**

Neutralisation reaction.

Oxidation numbers do not change.

Not a redox reaction.

**24. Answer: (B)**

Reaction: $A(s) + CuSO_4(aq) \rightarrow ASO_4(aq) + Cu(s)$

'A' must displace copper.

Activity series: $Zn > Fe > Pb > Cu > Ag$

The most suitable metal is $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

**25. Answer: A, B and C**

**(A)** $CuO + H_2 \rightarrow Cu + H_2O$

CuO is reduced.

Hydrogen is oxidised.(Redox reaction.)

**(B)** $2K + F_2 \rightarrow 2KF$

Potassium oxidised.

Fluorine reduced. Redox reaction.

**(C)** $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$

Nitrogen is oxidised.

Oxygen is reduced. Redox reaction.

All three are redox reactions.

**One or more than one**

**26. Answer: (A)**

In the reaction, $Fe \rightarrow Fe^{2+} + 2e^-$

Iron **loses two electrons**.

Loss of electrons is called **oxidation**.

A substance that **loses electrons** causes another substance to gain electrons (i.e., it reduces the other substance). Therefore, it acts as a **reducing agent**.

Thus,

**Assertion is true:** Fe acts as a reducing agent because it loses electrons.

**Reason is true:** Loss of electrons is called oxidation.
The reason correctly explains why Fe is a reducing agent.

**27. Answer: (A)**
Reaction:$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
Zinc is **above copper in the reactivity series**.
Therefore, zinc can **displace copper** from copper sulphate solution.
Hence,
**Assertion is true.**
**Reason is true.**
The reason correctly explains the assertion.

**28. Answer: (C)**
Reaction:$CaO + H_2O \rightarrow Ca(OH)_2 + Heat$
This reaction:
Releases a large amount of **heat**.
Is an **exothermic reaction**.
Therefore, the drum becomes **hot**.
Thus,
**Assertion is true.**
**Reason is false** because the reaction is **exothermic**, not endothermic.
Matching:

**29**. **a⟶S; b⟶Q; c⟶P; d⟶R**

**30**. a **⟶R; b⟶Q; c⟶P; d⟶S**

**31. Answer: (A)**
Reducing agent undergoes oxidation.$Fe^{2+} \rightarrow Fe^{3+}$
Iron loses electrons.
Therefore **$Fe^{2+}$ is the reducing agent.**

**32. Answer: (B)**
Oxidising agent undergoes reduction.$BrO_3^- \rightarrow Br^-$
Bromate ion gains electrons.

**33**. Answer: (B)
Oxidation means
Gain of oxygen
Loss of hydrogen
Loss of electrons

**34.** $\boldsymbol{a = 2}$
$b = 2$
$c = 3$ so, $a + b + c = 7$

**35. Balance:**
$\boxed{2Na + Cl_2 \rightarrow 2NaCl}$
Thus $x = 2, y = 2$
$x + y = 4$

## Previous year questions:

1. Step 1: Identify Oxidation and Reduction
Reduction: Manganese (Mn) goes from an oxidation state of +4 in $MnO_2$ to 0 in Mn. Thus, $MnO_2$ is the substance being reduced.
Oxidation**:** Aluminum (Al) goes from an oxidation state of 0 to +3 in $Al_2O_3$. Thus, Al is the substance being oxidized.
Step 2: Balance the Equation

We can balance this using the hit-and-trial method or by balancing atoms:

1. Look at Oxygen: There are 2 on the left and 3 on the right. To balance, we use a common multiple of 6.
2. Place a 3 before $MnO_2$ and a 2 before $Al_2O_3$.
3. Now balance the metals:
   3 Mn on the left → place a 3 before Mn.
   4 Al on the right (2 X 2) place a 4 before Al.
   Balanced Equation:
   $3MnO_2 + 4Al \rightarrow 3Mn + 2Al_2O_3$
   Step 3: Determine the Ratio
   Coefficient of substance reduced ($MnO_2$) = 3
   Coefficient of substance oxidized (Al) = 4
   Ratio = 3:4
   So, answer is A

2. C. Correct**:** Sodium bicarbonate is indeed used as an antacid (to neutralize stomach acid) and is the primary component in soda-acid fire extinguishers.

**3. Correct Option: B**

Analysis: When an electric current is passed through water, it decomposes into its constituent elements: hydrogen gas and oxygen gas.

Chemical Equation**:** $2H_2O(l) \xrightarrow{electricity} 2H_2(g) + O_2(g)$

Conclusion**:** Since a single compound (water) splits into two simpler substances, it is a decomposition reaction.

**4. Correct Option: A**

Analysis**:** A classic double displacement reaction that produces a characteristic yellow precipitate involves Lead (II) nitrate and Potassium iodide.

**5. Correct Option: C**

$H_2$ is being oxidized, not reduced, because it gains oxygen to form $H_2O$.

**6. Correct Option: D) 2 and 4**

Pb: 2 on left, 2 on right.
N: 4 on left, 4 on right.
O: 12 on left (2 X 3 X 2), 12 on right (2 + 8 + 2).

**7. Correct Option: A) (ii), (iii) and (iv) only**

(ii) Decomposition of vegetable matter into compost – Endothermic reaction: Incorrect**.** This is a biological decomposition process that releases energy (heat), making it exothermic.

(iii) Reaction of zinc with copper sulphate – Decomposition reaction: Incorrect. This is a Single Displacement reaction ($Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$).

(iv) Reaction of barium chloride with sodium sulphate – Single displacement reaction: Incorrect. This is a Double Displacement (Precipitation) reaction ($BaCl2 + Na_2SO_4 \rightarrow BaSO_4 + 2NaCl$).

**8. Correct Option: C) 2, 2 and 3**

Place a 2 before KClO3 (2 X 3 = 6 Oxygens).
Place a 3 before O2 (3 X 2 = 6 Oxygens).
$2KClO_3 \rightarrow 2KCl + 3O_2$ ;x=2, y=2, and z=3

**9**. $NaOH_{(aq)} + HCl_{(aq)} \rightarrow NaCl_{(aq)} + H_2O_{(l)}$

Sodium (Na): Remains +1 on both sides.
Oxygen (O): Remains -2 on both sides.
Hydrogen (H): Remains +1 on both sides.

Chlorine (Cl): Remains -1 on both sides.
Verdict: This is an acid-base neutralization reaction. There is no transfer of electrons or change in oxidation states. This is NOT a redox reaction.

10. C) Oxidising agent, Oxidation, reduction, Reducing agent
Zinc Oxide (ZnO) to Zinc (Zn): ZnO loses oxygen to become Zn. The removal of oxygen is reduction. Because ZnO provides oxygen to the other reactant, it acts as the oxidising agent.
Carbon (C) to Carbon Monoxide (CO): Carbon gains oxygen to become CO. The gain of oxygen is oxidation. Because Carbon removes oxygen from ZnO, it acts as the reducing agent.

# 2. Periodic Classification

**LEVEL – I**

**1. Correct option: (B) Transition element**

The element with 18 electrons in the penultimate shell and 1 electron in the ultimate shell is copper (Cu). This makes it a transition element, as transition elements have partially filled d subshells in their valence shells. The penultimate shell of copper is the 4th shell, which has 18 electrons. The valence shell is the 5th shell, which has 1 electron. The 18 electrons in the penultimate shell are arranged in the 4s and 3d subshells. The 4s subshell is filled with 2 electrons, and the 3d subshell is filled with 10 electrons.

**2. Correct option: (C) IA**

The chloride of an element A gives a neutral solution in water because the chloride of a strong acid and a strong base is neutral.

The reaction is:

$2M + Cl_2 \rightarrow 2MCl$, $MCl + H_2O \rightarrow HCl + MOH$ (strong acid + strong base)

where M is the metal atom.

Therefore, the element A belongs to the first group.

**3. Correct option: (A) $C_2^{2-}$, $NO^+$, $CN^-$, $O_2^{2+}$**

Isoelectronic species have the same number of electrons. CO, $CN^-$, $NO^+$, and $C_2^{2-}$ are isoelectronic because they all have 14 electrons.

**4. Correct option: (D) For transition elements, the d-sub-shells are filled with electrons monotonically with an increase in atomic number**

The d-subshells of transition elements are not filled with electrons monotonically with increasing atomic number. Instead, the filling of d-subshells is often interrupted by the filling of s-subshells, or the d-subshell may be left partially filled. This is because the stability of half-filled and full-filled electron configurations can override the general trend of filling subshells in order of increasing energy.

**5. Correct option: (B) 53**

The elements in group 17 are the halogens. The halogens have 7 valence electrons. When an atom of a halogen gains an electron, it becomes a halide ion with a negative charge.

The element in the third period with an atomic number of 17 is chlorine. The element in the fifth period with the same valence electron configuration is astatine, which has an atomic number of 53.

**6. Correct option: (B) Mg, Ca, Sr**

Dobereiner's triads are groups of three elements that have similar chemical properties. The atomic mass of the middle element is approximately the average of the atomic masses of the other two elements.

The triad of magnesium, calcium, and strontium is not a Dobereiner triad because the atomic mass of strontium is not the average of the atomic masses of magnesium and calcium.

**7. Correct option: (D) All**

Lother meyer curve is a graph of atomic volume vs. atomic mass. The curve shows that the elements with large atomic volume and high electropositivity have peaks in the curve.

Alkali metals are highly electropositive elements and have large atomic volume. These properties are the reason why they occupied the peaks in the Lothermeyer curve.

**8. Correct option: (A) Mg – Ba**

Both magnesium (Mg) and barium (Ba) are in group 2 of the periodic table, which is the alkaline earth metals

**9. Correct option: (D) (n-1) $s^2$ $p^6$ $d^{1-10}$ $ns^{1or2}$**

Transition elements are elements with valence electrons in the d orbitals. Their general electronic configuration is $(n-1)d^{1-10}$ $ns^{1-2}$, where n is the period number of the element.

**10. Correct option: (A) 6s, 4f, 5d, 6p**

The energy levels in the 6th period of the periodic table are filled in the following order:

- 6s orbital
- 4f orbitals (14 electrons)
- 5d orbitals (10 electrons)
- 6p orbitals (6 electrons)

The 6s orbital is filled first because it is the outermost orbital. The 4f orbitals are filled next, even though they are lower in energy than the 5d orbitals. This is because the 4f orbitals are more shielded from the nucleus than the 5d orbitals. The 5d orbitals are filled after the 4f orbitals, and the 6p orbitals are filled last.

**11. Correct option: (D) $(n-1)d^6$ $ns^2$**

The outer shell electronic configuration of alkaline earth metals is $ns^2$.

Alkaline earth metals are the elements in group 2 of the periodic table. They have two valence electrons, which are located in the s orbital of the outermost shell.

**12. Correct option: D) Alkali metals**

**Ionisation potential (Ionisation energy)** is the minimum energy required to remove the outermost electron from an isolated gaseous atom.

- **Alkali metals (Group 1)** have **only one valence electron**, which is loosely held because:
  - They have a **large atomic size**.
  - The outermost electron is far from the nucleus.
  - Nuclear attraction on the valence electron is weak.

Therefore, they have the **lowest ionisation energy** in each period.

**13. Correct option: D) Stable oxide**

Potassium forms the oxide $K_2O$, which is a stable ionic oxide under appropriate conditions (though potassium can also form peroxide and superoxide).

**14. Correct option: (A) $1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $3d^5$ $4s^2$**

The electronic configuration of the element with atomic number 43 is [Kr] $4d^5$ $5s^2$. The element just above it in the same group has one fewer electron, which will be in the 4d orbital.

Option (B) is incorrect because it has a 4p electron, which is not possible for an element in the same group as technetium (Tc). Option (C) is incorrect because it has a $3d^6$ electron, which is not possible for an element with an atomic number of 42. Option (D) is incorrect because it has a $4s^1$ $4p^6$ electron configuration, which is not possible for an element in the same group as technetium.

**15. Correct option: (D) Zn**

Zinc (Zn) remains in the same position in the periodic table's d-block, even if the Aufbau principle is not followed. The other elements (Ca, Ge, Sn)

would change their positions if electron filling were not based on increasing atomic number, altering their block classifications.

**16. Correct option: (A) Transition elements are placed from 3rd to 6th period**

Transition elements are placed in the d-block and span the 3rd to 7th period of the periodic table, not just the 3rd to 6th period.

**17. Correct option: (C) [Rn] $7s^2$ $5f^7$ $6d^1$**

The electronic configurations of A, B, and D are consistent with the f-block, d-block, and p-block, respectively. However, option C represents an element with electron filling that is inconsistent with any specific block, making it the outlier.

**18. Correct option: (C) Lawrencium**

Lawrencium belongs to Group 3 of the periodic table (the lanthanide series) and, when undergoing alpha decay, retains its position in the same group. The other options belong to different groups and do not exhibit this behavior.

**19. Correct option: (B) 27**

If an orbital could contain 3 electrons, then the number of groups in the periodic table would be 27. This is because there are 27 different sets of 3 quantum numbers that can be assigned to electrons. The current periodic table has 18 groups, which is based on the fact that an orbital can contain up to 2 electrons.

The other options are incorrect:

- (A) 20: This is the number of groups in the current periodic table.
- (C) 36: This is the number of orbitals in the first 4 shells.

(D) 42: This is the number of orbitals in the first 5 shells

**20. Correct Answer: (B)**

(Sodium)

- Valence electrons = 1
- After losing one electron:

$Na \rightarrow Na^+ = 1s^2\ 2s^2\ 2p^6$

(Noble gas configuration)

- The second electron must be removed from the stable neon configuration, requiring very high energy.

**Maximum difference**

**21. Step 1: Identify the elements**

- **X = 19 → Potassium (K)**
- **Y = 37 → Rubidium (Rb)**
- **Z = 55 → Caesium (Cs)**

All belong to **Group 1 (Alkali metals).**

**Step 2: Trend in ionisation energy**

Down a group:

- Atomic size **increases**.
- Shielding effect **increases**.
- Attraction between nucleus and outermost electron **decreases**.
- Therefore, **ionisation energy decreases**.

Thus, $\boxed{K > Rb > Cs}$

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**Level-II**

**1. Answer: (C)**

Nature of bonding between A & C is covalent.

Cl + K forms KCl, which is ionic (electrovalent).(Wrong)

**2. Answer: (B)**

Electrovalent (ionic) compounds are formed between

- Metals (Group 1 or 2)
- Non-metals (Group 16 or 17)

Among the given options,
Group 2 + Group 16
Example:$MgO$, $CaO$ are strongly ionic.

**3. Answer: (C)**
Newlands arranged elements according to increasing atomic masses.
One successful octave is
Li Be B C N O F then
Na Mg Al Si P S Cl Hence$Na$, $K$, $Rb$
do not obey octaves.
The correct set is $Be$, $Mg$, $Ca$

**4. Answer: (D)**
Screening orders > p > d > f
Hence
d-electrons shield **less than p-electrons.**

**5. Answer: (C)**
Potassium:$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$
According to Aufbau principle, $4s < 3d$

**6. Answer: (B)**
Bromine (Z = 35)
Configuration:$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^5$
Total p-electrons:$2p^6 + 3p^6 + 4p^5$: $6 + 6 + 5 = 17$

**7. Answer: (D)**
**$(n-2)f^{1-14}(n-1)d^{1-10}ns^2$**
called d-block
Incorrect
Because d-block general configuration is $(n-1)d^{1-10}ns^{1-2}$
Presence of f-electrons is **not** a general feature of all d-block elements.

**8. Answer: (C)**
Valence electron in (n−1)d
VIA group
Group 16 elements have valence electrons in **ns and np**, not d.( Incorrect)

**9. Answer: (B)**
Element 104
Temporary IUPAC name Unnilquadium

**10. Answer: (B)**
Iron $Z = 26$
Configuration:$1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$
For one 3d electron
Shielding
Other 3d electrons $5 \times 0.35 = 1.75$
All electrons to the left $18 \times 1.00 = 18$
Total shielding$S = 19.75$
Therefore $Z_{eff} = 26 - 19.75 = 6.25$

**11. Answer: C)**

The modern periodic table has **7 periods**.

The **6th period** is traditionally regarded as the longest period because it contains 32 elements, including the lanthanides (14 f-block elements).

**12. Answer: B)**

While developing his periodic table, Dmitri Mendeleev left vacant places for undiscovered elements and predicted their properties.

He used the Sanskrit prefix "Eka", meaning "one", for the element one place below a known element.

- Eka-aluminium → Later discovered as Gallium
- Eka-silicon → Later discovered as Germanium

His predictions matched the properties of these elements remarkably well.

**13. Answer: B)**

A transition element is one that has an incomplete (partially filled) d-subshell in its atom or in one of its common oxidation states.

Group 12 elements:

- Zn : [Ar] $3d^{10}4s^2$
- Cd : [Kr] $4d^{10}5s^2$
- Hg : [Xe] $4f^{14}5d^{10}6s^2$

Their common ion is $\mathbf{M^{2+}}$:

- $Zn^{2+} \rightarrow 3d^{10}$
- $Cd^{2+} \rightarrow 4d^{10}$
- $Hg^{2+} \rightarrow 5d^{10}$

Since the d-orbitals remain completely filled ($d^{10}$) in both the atom and the common +2 ion, they are not considered transition elements.

**14. Answer: A, B and C**

A) Li, Na, K

- All are alkali metals.
- Na's atomic mass ≈ average of Li and K. Correct

B) Ca, Sr, Ba

- All are alkaline earth metals.
- Sr's atomic mass ≈ average of Ca and Ba. Correct

C) Cl, Br, I

- All are halogens.
- Br's atomic mass ≈ average of Cl and I. Correct

D) Na, Mg, Al

- Belong to different groups.
- Do not satisfy Döbereiner's triad rule.

**15. Answer: A, C and D**

**Alkali metals occur at the peaks.**

Examples:

- Li
- Na
- K
- Rb
- Cs

These have the **largest atomic volumes** in each period.(Correct)

**Option B**

**Transition elements occupy the maxima.**

Transition elements lie in the **middle valley** of the curve, not at the peaks.(Incorrect)

**Option C**

**Noble gases occur just before alkali metals.**

Each new period begins with an alkali metal, so the noble gas of the previous period comes immediately before it.

Example:

He → Li

Ne → Na

Ar → K Correct

**Option D**

**Alkaline earth metals are found on the descending parts of the curve.**

After each alkali metal peak, the curve begins to descend, where the alkaline earth metals (Be, Mg, Ca, etc.) are located.

**16. Answer: A**

Assertion: True

Zinc (Zn) has the electronic configuration:

- Zn = [Ar] $3d^{10}4s^2$

Its common ion is:

- $Zn^{2+}$ = [Ar] $3d^{10}$

Since the d-subshell is completely filled ($d^{10}$) in both Zn and $Zn^{2+}$, it is not a transition element.

Reason: True

A transition element is an element that has an incompletely filled d-orbital in its atom or in one of its common oxidation states.

Since Zn never has a partially filled d-orbital, the reason correctly explains the assertion.

**17. Correct Match: a–P, b–Q, c–R, d–S**

**18. Correct Answer: C**

Inner-shell electrons repel outer electrons and reduce the attraction exerted by the nucleus.

This reduction is called the shielding (screening) effect.

The shielding constant **σ** represents how much of the nuclear charge is screened by inner electrons.

**19. Answer: B**

The effective nuclear charge is calculated using: $Z_{\text{eff}} = Z - \sigma$

where

- **Z** = Atomic number (actual nuclear charge)
- **σ** = Shielding constant

Example:

For sodium,

- Z = 11
- σ ≈ 10

Therefore, $Z_{\text{eff}} = 11 - 10 = 1$

**20. Answer: B**

According to **Slater's Rules** for **s and p electrons**:

| **Electron Position** | **Screening Contribution** |
|---|---|
| Same shell (n) | 0.35 each (except 1s = 0.30) |

| **Electron Position** | **Screening Contribution** |
|---|---|
| **(n–1) shell** | **0.85 each** |
| (n–2) or lower shells | 1.00 each |

Thus, electrons in the **(n–1) shell** contribute **0.85** to the shielding constant.

**21. Answer: 18**

The modern periodic table consists of **18 vertical columns**, known as **groups**.

- Groups are numbered from **1 to 18**.
- Elements in the same group have similar chemical properties because they have the same number of valence electrons.

**22. Answer: 7**

he modern periodic table has 7 horizontal rows, called periods.

**Periodic Properties**

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**Level-1**

**1. Answer: C)**

- Potassium loses one electron to form $K^+$, so its ionic radius is smaller than its atomic radius.
- Fluorine gains one electron to form $F^-$, so its ionic radius is larger than its atomic radius.

Thus,

- Atomic radius of K > Ionic radius of $K^+$
- Atomic radius of F < Ionic radius of $F^-$

Therefore,

- K = 1.96 Å
- F = 0.72 Å

**2. Answer: B)**

First ionisation energy generally increases across a period.

Electronic configurations:

- B → $2p^1$
- C → $2p^2$
- **N → $2p^3$ (half-filled, extra stable)**
- O → $2p^4$

Nitrogen has a stable half-filled p-orbital, so it requires the maximum energy among these.

**3. Answer: B)**

Ionisation energies:

- $IE_1$ = 6
- $IE_2$ = 10
- $IE_3$ = 16
- $IE_4$ = 45 (large jump)

The large jump after the third ionisation means the atom has **3 valence electrons**.

Hence, its valency is 3.

Oxide formula:$E_2O_3$

Molecular mass: $2x + 3(16) = 2x + 48$

**4. Answer: B) Cl**

Although fluorine is the most electronegative element, **chlorine** has the most negative electron gain enthalpy because the added electron experiences less repulsion in chlorine's larger 3p orbital than in fluorine's compact 2p orbital.

Order:

$\boxed{Cl > F > Br > I}$

**5. Answer: A)**

According to the diagonal relationship,

- **Be and Al** have similar electronegativities.

Approximate values:

- Be = 1.5
- Al = 1.5

**6. Answer: A)**

Identify the elements:

- I = Cl
- II = N
- III = F
- IV = Na

Electron gain enthalpy becomes more negative from Na to halogens.

Nitrogen has a stable half-filled configuration, so adding an electron is least favourable.

Thus, $\boxed{N < Na < F < Cl}$

or $\boxed{II < IV < III < I}$

**7. Answer: A)**

Energy released per atom:$= \frac{4.8\times10^{-13}}{10^{6}} = 4.8 \times 10^{-19}$ J

Per mole:$4.8 \times 10^{-19} \times 6.022 \times 10^{23} = 2.89 \times 10^{5}$ J $= 289$ kJ mol$^{-1}$

Therefore, the electron gain enthalpy is approximately: $\boxed{289 \text{ kJ mol}^{-1}}$

**8. Answer: D)**

Alkali metals lose one electron very easily because they have:

- one valence electron,
- **very low ionisation energy**.

Hence, they readily form **$M^+$ ions**, making them strong reducing agents.

**9. Answer: D) S**

In a group, metallic character and electropositivity **increase down the group**.

Nature of oxides changes as:

- Non-metals → Acidic oxides
- Metalloids → Amphoteric oxides
- Metals → Basic oxides

Given:

- P → Acidic oxide
- Q, R → Amphoteric oxides
- S → Basic oxide

Thus S is the most metallic and most electropositive.

**10. Answer: A)**

For oxyacids having the same number of oxygen atoms,

Acid strength depends on the electronegativity of the central atom.

Electronegativity:

Cl > Br > I

Therefore,

HOCl > HOBr > HOI

11. Answer: B)

Removing

- Electron → Ionisation energy
- Proton/neutron → Nuclear binding energy (very large)
- Alpha particle → Even larger nuclear energy

Hence the easiest particle to remove is the electron.

**12. Answer: B)**

Screening effect increases down a group.

Effective nuclear charge increases across a period.

Down a group, increased shielding keeps the effective nuclear charge on valence electrons nearly constant rather than increasing significantly.

**13. Answer: A)**

Atomic size is

- inversely proportional to effective nuclear charge.

Atomic size $\propto \frac{1}{Z_{eff}}$

Other statements are correct because

- More negative charge → larger size
- Higher bond order → shorter bond length
- Greater screening → larger size

**14. Answer: D) 2,4**

Atomic numbers:

1 → 13 (Al)

2 → 22 (Ti)

3 → 11 (Na)

4 → 24 (Cr)

Ti and Cr are both **transition metals** with similar metallic properties

**15. Answer: C)**

Sulphur has 16 electrons.

Since $X^{2+}$is isoelectronic with sulphur,$Z - 2 = 16; Z = 18$

Neutrons= $Z + 2 = 20$

Mass number= $18 + 20 = 38$

**16. Answer: D)**

Second-period elements differ because of

- Small atomic size
- High ionisation energy
- High electronegativity
- No vacant d-orbitals

All contribute.

**17. Answer: D)**

- Noble gases have the highest ionisation energy in every period, not halogens.

Hence D is incorrect.

**18. Answer: C) CsI**

Ratio$\frac{\text{anion radius}}{\text{cation radius}}$

Smallest ratio occurs when

- Cation is largest
- Anion is smallest

Among the options:

- $Cs^+$ is the largest cation.
- $F^-$ is the smallest anion.

Therefore, CsF has the lowest anion/cation radius ratio.

**19. Answer: A)**

Cu vs Zn: Zn has a slightly larger atomic radius than Cu because of the filled $3d^{10}$ configuration. Therefore $r_{Cu} > r_{Zn}$is incorrect.

**20. Answer: C)**

Ionisation energy generally **increases across a period** because effective nuclear charge increases.

Although **P has slightly higher IE than S** due to its stable half-filled $p^3$ configuration, chlorine lies further to the right and has a higher first ionisation energy than phosphorus.

Approximate values (kJ $mol^{-1}$):

- Mg = 738
- P = 1012
- S = 1000
- Cl = 1251

Hence, Cl > P > S > Mg

**21. Answer: A)**

Electronic configurations:

- $Li^+$ = **$1s^2$** (helium-like ion)
- $Mg^+$ = [Ne]$3s^1$
- He = $1s^2$
- Ne = $1s^2 2s^2 2p^6$

Although $Li^+$ and He have the same electronic configuration, **$Li^+$ has three protons** while He has only two. The greater nuclear charge holds the electrons much more strongly.

Thus, removing an electron from **$Li^+$** requires the maximum energy.

**22. Answer: D**)

Successive ionisation energies always increase because electrons are removed from increasingly positive ions.$IE_1 < IE_2 < IE_3$

Therefore,

Removing the third electron requires the maximum energy.

**23. Answer: A) Na**

Sodium:

Na = $1s^2 2s^2 2p^6 3s^1$

After losing one electron,

$Na^+$ = $1s^2 2s^2 2p^6$ (noble gas configuration)

Removing the second electron means breaking into a stable noble gas configuration, requiring a very large amount of energy.

Hence,

Difference between $IE_1$ and $IE_2$ is greatest for Na.

**24. Answer: B**

Actual first ionisation energy order is$B < C < O < N$

Nitrogen has higher IE than oxygen because of its stable half-filled $p^3$ configuration.

Hence$B < C < N < O$

is incorrect.

**25. Answer: D) All of these**

Electron gain enthalpy is influenced by several factors:

- Atomic size: Smaller atoms generally attract incoming electrons more strongly.

- Nuclear charge: Higher nuclear charge increases attraction for an added electron.
- Electronic configuration**:** Stable configurations (e.g., noble gases and half-filled subshells) resist gaining electrons.

Therefore, all the listed factors affect the magnitude of electron gain enthalpy.

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**Level-II**

**1. Answer: D)**

From **I → Br → Cl → F**:

- Electronegativity increases.
- Ionisation energy increases.
- Oxidising power increases.
- **Electron affinity does not increase continuously** because **chlorine has a more negative electron affinity than fluorine** due to less electron-electron repulsion in its larger 3p orbital.

Order of electron affinity: $\boxed{Cl > F > Br > I}$

Hence, electron affinity does **not** increase from iodine to fluorine.

**2. Answer: C)**

- $F + e^- \rightarrow F^-$ releases energy.
- $Cl + e^- \rightarrow Cl^-$ releases energy.
- $H + e^- \rightarrow H^-$ releases energy.
- To form $\mathbf{O^{2-}}$, oxygen must gain **two electrons**:
  - First electron gain is exothermic.
  - **Second electron gain is endothermic** because the second electron is added to an already negatively charged ion.

Therefore, the overall process requires absorption of energy.

**3. Answer: A**

**Lithium exhibits a diagonal relationship with magnesium.**

Representative elements are Groups **1, 2 and 13–17.**

In Period 5, electrons fill **5s → 4d → 5p**.

General configuration of f-block is $(n-2)f^{1-14}(n-1)d^{0-1}ns^2$.

**4. Answer: C)**

Valency **does not increase down a group**.

It generally remains constant for elements within the same group.

**5. Answer: A)**

Hydration energy increases with

- Higher ionic charge
- Smaller ionic radius

Order: $\boxed{Al^{3+} > Mg^{2+} > Na^+ > K^+ > Ba^{2+}}$

Therefore,

$Mg^{2+}$ has lower hydration energy than $Al^{3+}$.

**6. Answer: B)**

Valency in hydrides: Hence, $\boxed{1,2,3,4,3,2,1}$

**7. Answer: A)**

Lanthanide contraction causes Hf to have nearly the same atomic size as Zr because the 4f electrons shield the nuclear charge poorly.

Thus,

Zr ≈ Hf in size.

**8. Answer: A)**

4f electrons shield the increasing nuclear charge very poorly.

As a result,

- Effective nuclear charge increases.
- Atomic radius decreases gradually.

This is called **lanthanide contraction**.

**9. Answer: B)**

$O^{2-}$ and $Si^{4+}$ each have 10 electrons.

For isoelectronic species,

- Higher nuclear charge ⇒ smaller ionic radius.

Nuclear charges:

- O = 8
- Si = 14

Therefore, $Si^{4+}$ is much smaller than $O^{2-}$.

If $O^{2-}$ = 1 Å, so, 0.41 Å is the answer

**10. Answer: D)**

There is a **very large jump** between $IE_3$ and $IE_4$.

$IE_3 = 181$ eV, $IE_4 = 2000$ eV

This indicates that the element has **3 valence electrons**.

Hence, it belongs to **Group 13**.

Examples:

- B
- Al
- Ga

Properties:

- Usually metallic (especially Al, Ga, In, Tl).
- Oxide formula: $A_2O_3$

Example:$Al_2O_3$

which is **amphoteric**.

Hence all statements are correct.

**11. Answer: A)**

**Electronic configurations**

Cu :$[Ar]\,3d^{10}4s^1$

Zn :$[Ar]\,3d^{10}4s^2$

**First ionization energy**

Removing one electron:

- Cu loses $4s^1$ and attains stable $d^{10}$ configuration.
- Zn loses one 4s electron but still has $4s^1$.

Therefore,$IE_1(Cu) < IE_1(Zn)$

**Second ionization energy**

After first ionization:

$Cu^+$ :$3d^{10}$

$Zn^+$ :$3d^{10}4s^1$

Removing another electron:

- $Cu^+$ loses a 3d electron (very difficult)
- $Zn^+$ loses the remaining 4s electron (easier)

Hence,$IE_2(Cu) > IE_2(Zn)$

**12. Answer: D)**

**Ionization Energy** $\mathrm{IP_F} > \mathrm{IP_{Cl}}$

**Electron Affinity** $\mathrm{EA_{Cl}} > \mathrm{EA_F}$

**Electronegativity** $\mathrm{EN_F} > \mathrm{EN_{Cl}}$

**13. Answer: C**

Formation of $O^{2-}$, $S^{2-}$

requires addition of **second electron**.

The second electron is added to an already negatively charged ion.

Hence,

strong **electron-electron repulsion** must be overcome.

This repulsion is greater than the attraction by the nucleus.

Therefore,the second electron affinity is **positive (endothermic).**

**14. Answer: A)**

According to **Mulliken's definition**,

$$\boxed{\chi = \frac{IP + EA}{2}}$$

Rearranging,$2\chi = IP + EA$

or $\boxed{2\chi - IP - EA = 0}$

**15. Answer: A)**

Processes:

I)$Cl + e^- \rightarrow Cl^-$(Electron affinity)

II)$Cl^- \rightarrow Cl + e^-$(Removal of electron from chloride)

III)$Cl \rightarrow Cl^+ + e^-$(First ionization energy)

IV)$Cl^+ \rightarrow Cl^{2+} + e^-$(Second ionization energy)

**16. Answer: D**

Using the relationships:

- **Electron affinity of X = Ionization energy of $X^-$**

  Therefore,$IP(A^-) = EA(A) = 70$

  Option **B is correct**.
- **Electron affinity of $X^+$ = Ionization energy of X**

  Therefore,$EA(B^+) = IP(B) = 250$

  Option **C is correct**.
- Since **EA(B) > EA(A)**,

  $B^-$ is more stable than $A^-$, so statement **A** is acceptable.
- Comparing **EA($A^+$)** and **EA($B^+$)**:

  $EA(A^+) = IP(A) = 350$

  $EA(B^+) = IP(B) = 250$

  Hence,$EA(A^+) > EA(B^+)$

  Therefore statement **D** is **incorrect**.

**17. Answer: D)**

Solubility of Group 2 sulphates decreases down the group:

$MgSO_4 > CaSO_4 > SrSO_4 > BaSO_4$

- $MgSO_4$ is highly soluble.
- $Na_2SO_4$ is also soluble.
- $BaSO_4$ is almost insoluble in water.

**18. Answer: A)**

According to **Fajan's Rules**, polarizing power increases with:

- Higher positive charge
- Smaller ionic radius

Approximate ionic radii:

- $Li^+$ ≈ 0.76 Å
- $Mg^{2+}$ ≈ 0.72 Å

Both are **small ions with high charge density**, so they have similar polarizing power. This also explains the **diagonal relationship** between Li and Mg.

**19. Answer: A)**

Nature of oxides:

- **CaO** → Strongly basic oxide (least acidic)
- **CuO** → Basic oxide (less basic than CaO)
- **$H_2O$** → Neutral oxide
- **$CO_2$** → Acidic oxide (most acidic)

Thus, increasing acidic strength: $\boxed{CaO < CuO < H_2O < CO_2}$

**20. Answer: C)**

Down Group 15, the nature of oxides changes from acidic to basic: Therefore, **$Sb_2O_3$** is the amphoteric oxide

**One or more than one option correct type:**

**21. Correct Answers: A, B and C**

**A) $B(OH)_3$ – acidic**

Boric acid is a **weak Lewis acid** because it accepts $OH^-$ ions. Correct

**B) SnO, PbO – amphoteric**

Both react with **acids as well as bases**.

Example: $SnO + 2HCl \rightarrow SnCl_2 + H_2O$

$SnO + 2NaOH + H_2O \rightarrow Na_2[Sn(OH)_4]$

Correct

**C) $GeO_2$ – acidic**

Germanium dioxide is predominantly an **acidic oxide** (though it may show weak amphoteric behavior under some conditions). In school-level chemistry it is generally classified as **acidic**.

**22. Answer: B)**

Lutetium (Lu, atomic number 71) is classified as an **f-block element**

**23. Answer: C)**

**I .**Due to the **inert pair effect**, $Bi^{3+}$ is more stable than $Bi^{5+}$.

**II** Mn exhibits **+7** commonly, and **+8** in rare species such as gaseous **$MnO_4$** under special conditions; advanced chemistry texts recognize +8 as the highest oxidation state of Mn.

Accepted.

**III** Maximum oxidation state generally equals the group number for representative and transition elements.

**24. Answer: B)**

Successive ionisation energies always increase.

Therefore $IE_2 > IE_1$

for every element.

**25. Answer: D)**

Actual order$F < O$

because oxygen is to the left of fluorine in the same period.

Correct order is$F < O < O^{2-} < F^{-}$

Hence D is incorrect.

**26. Correct Answers: A, B and D**

**A)** Pauling EN(F) > Mulliken EN(F) (True)

**B)** $Cl_2O_7$ > $Cl_2O_5$ (EN of Cl)

Higher oxidation state increases electronegativity.

**C)** $CH_4$ > $CO_2$ (EN of C) **(correct)**

Carbon has higher oxidation state in $CO_2$.

Hence EN(C) is greater in $CO_2$.

**D)**$Cu^{2+}$ > $Cu^{+}$

Higher oxidation state means higher effective nuclear charge

**27. Correct Answers: A, C and D**

In general,higher ionisation energy → higher electronegativity.

B) Electronegativity increases mean metallic character increases.

Metallic character actually decreases.

C)Lower ionisation energy → stronger reducing property.

D)Cl has higher electron affinity than F.

Electron affinity order:$Cl > F > Br > I$

**28. Answer: A**

**Assertion: True**

Across a period, electron gain enthalpy generally becomes more negative, indicating that atoms release more energy when they gain an electron.

Example (Period 3):Na < Mg < Al < Si < P < S < Cl

(Exceptions include Be, Mg, N and noble gases due to stable electronic configurations.)

**Reason: True**

Across a period:

- Number of protons increases.
- Shielding effect remains nearly constant.
- Therefore, effective nuclear charge ($Z_e$ff) increases.

As a result, the nucleus attracts the incoming electron more strongly, making electron gain enthalpy more negative.

Thus, the reason correctly explains the assertion.

**29**. The large jump in successive ionisation energies indicates the number of valence electrons.

- **P:** Large jump after 1st IE → 1 valence electron ($ns^1$)
- R: Large jump after 2nd IE → 2 valence electrons ($ns^2$)
- Q: Large jump after 3rd IE → 3 valence electrons ($ns^2np^1$)
- S: Large jump after 4th IE → 4 valence electrons ($ns^2np^2$)

**30. Answer: C**

Second-period elements (Li, Be, B, C, N, O, F) differ from the other members of their groups because they have:

- Very small atomic size
- High charge-to-radius ratio
- High electronegativity

- No vacant d-orbitals

These factors give them unique properties.

**31. Answer: B**

Second-period elements have only 2s and 2p orbitals in their valence shell.
Since they do not have vacant d-orbitals, they cannot expand their octet.
Therefore, their maximum covalency is **4**.
Examples:

- Carbon → $CH_4$ (covalency = 4)
- Nitrogen → $NH_4^+$ (maximum covalency = 4)

**32. Answer: C**

First members of the p-block (C, N, O) are very small and have effective overlap of p-orbitals.
Therefore, they readily form:

- C=C
- C≡C
- N≡N
- C=O

Thus, **multiple bond formation** is a characteristic property of the first members of the p-block.
They cannot expand their valence shell because they lack vacant d-orbitals.

## Previous Year Questions

**1**. C. Ionization enthalpy generally increases across a period, there are two distinct "peaks" in Period 2. These two distinct peaks corresponds to Be and N.

**2**. B. X (-349 kJ/mol) = This is a very high (exothermic) value, characteristic of Chlorine, which has the highest electron affinity in the periodic table. Z (-295 kJ/mol) = This value is lower than Chlorine but still significant, fitting Iodine (which is further down Group 17). Y (-200 kJ/mol): This is the lowest magnitude in the set, fitting Sulfur (Group 16), which generally has a lower electron affinity than the halogens of similar periods.

**3**. C. Atomic radius follows two primary trends: it decreases across a period (left to right) and increases down a group.

**4**. B. Electron gain enthalpy is generally negative (exothermic) for most elements because they release energy when gaining an electron.

**5**. A. These elements are all in Period 3. While ionization enthalpy generally increases from left to right, there is an anomaly between Magnesium (Mg) and Aluminum (Al).

**6**. D. A diagonal relationship occurs between certain elements in the second and third periods that are positioned diagonally to each other.

**7**. C. To determine the group of a p-block element, we add 10 to the number of valence electrons (ns + np).

**8**. C. These species are isoelectronic, meaning they have the same number of electrons (10 electrons each).

**9**. D. According to Fajans' Rules, covalent character increases with a decrease in cation size and an increase in cation charge (high polarizing power).

**10**. A. As you move down a group in the periodic table, the atomic radius increases.

# 3. Acids, Bases and Salts

## LEVEL – I

**1. Correct Option: C) Sour**

Explanation:

*Acidus* is a Latin adjective meaning "sour, tart, or acidic." It's the root of the English word "acid."

**2. Correct Option: D) Ascorbic acid**

Explanation:

Ascorbic acid is also known as Vitamin C. Lemons are famous for their Vitamin C content, not the other acids listed.

**3. Correct Option: B) Sulphuric acid**

Explanation:

Sulphuric acid is a strong acid. It dissociates completely in water, producing a high concentration of hydrogen ions, making it a strong electrolyte.

**4. Correct Option: B) Acetic acid**

Explanation:

Acetic acid is the weak acid among the options. Sulphuric, nitric, and hydrochloric acids are all strong acids, meaning they virtually dissociate completely in water, whereas acetic acid only partially dissociates, leading to a lower concentration of hydrogen ions and weaker acidic character.

**5. Correct Option: D) HCl**

Explanation:

Mineral acids hail from inorganic sources like salts. Palmitic and acetic are organic (living things), carbonic can be both.

**6. Correct Option: B) Red**

Explanation:

In response to acidic conditions, blue litmus undergoes a chromic shift, resulting in a distinct red coloration.

**7. Correct Option: B) Red**

Explanation:

In an acidic medium, methyl orange exhibits a pronounced chromic shift, transitioning from its original orange hue to a distinct red colour.

**8. Correct Option: B) basicity of an acid**

Explanation:

The basicity of an acid refers to the number of hydrogen ions ($H^+$) it can furnish when it dissolves in water.

A) Acidity of base refers to $H^+$ concentration in a basic solution.

C) & D) classify acids by source, not $H^+$ donation capacity.

**9. Correct Option: C) Organic acid**

Explanation:

They're naturally found in living things and contain carbon, unlike mineral acids from inorganic sources. Citrusy citric acid or tangy lactic acid for examples.

**10. Correct Option: C) Sodium hydroxide**

Explanation:

A) $Na_2CO_3$ (washing soda): While exhibiting limited basicity under specific conditions, the carbonate ($CO_3^{2-}$) lacks the inherent electron density of strong bases for readily accepting $H^+$ ions.

B) $CH_3COOH$ (acetic acid): Primarily exhibiting acidic properties due to its ability to donate $H^+$ through the carboxylic acid group (-COOH), it lacks the structural features for significant base behaviour.

C) NaOH (sodium hydroxide): The presence of the hydroxide ($OH^-$) group, a potent electron donor, readily facilitates $H^+$ acceptance, classifying it as a strong base due to its high proton affinity.

D) HCl (hydrochloric acid): Possessing a readily available $H^+$ for donation, its structure lacks the characteristics necessary for accepting $H^+$ ions, solidifying its acidic nature.

**11. Correct Option: C) Sodium hydroxide**

Explanation:

A) NaOH (Strong Base): $NaOH \rightarrow Na^+(aq) + OH^-(aq)$
All $OH^-$ ions are deployed, ready for action.

B) Ammonium hydroxide ($NH_4OH$) (Weak Base): $NH_4OH + H_2O \rightleftharpoons NH_4^+(aq) + OH^-(aq)$
Only partial dissociation, resulting in a smaller $OH^-$ contingent.

C) Calcium hydroxide ($Ca(OH)_2$) (Moderately Strong Base): $Ca(OH)_2 \rightleftharpoons Ca^{2+}(aq) + 2OH^-(aq)$
Not all $Ca(OH)_2$ molecules dissociate, limiting the $OH^-$ presence.

D) Nitric acid ($HNO_3$) (Strong Acid): $HNO_3 \rightarrow H^+(aq) + NO_3^-(aq)$
$HNO_3$ releases $H^+$ ions, not $OH^-$, making it an acidic adversary.

**12. Correct Option: C) $NH_4OH$**

Explanation:

While strong bases like NaOH and KOH readily dissociate into their constituent ions ($Na^+/K^+$ and $OH^-$), $NH_4OH$ undergoes a partial equilibrium reaction with water:
$NH_4OH + H_2O \rightleftharpoons NH_4^+ + OH^-$
This equilibrium favours the undissociated $NH_4OH$ molecules, resulting in a lower concentration of free $OH^-$ ions compared to strong bases.

**13. Correct Option: D) Pink**

Explanation:

Phenolphthalein acts as a pH indicator, remaining colourless in acidic solutions and magically transforming to a vibrant pink in basic environments

**14. Correct Option: A) 2**

Explanation:

$H_2SO_4$, also known as sulphuric acid, is a dibasic acid, meaning it can donate two hydrogen ions ($H^+$) per molecule when dissolved in water. Basicity refers to the number of $H^+$ ions an acid donates, not its overall strength.

**15. Correct Option: C) 3**

Explanation:

When $Al(OH)_3$ reacts with any acid, $Al(OH)_3 + 3HCl \rightarrow AlCl_3 + 3H_2O$
It requires 3 acidic $H^+$ ions to neutralize therefore its acidity is 3.

**16. Correct Option: A) NaOH**

Explanation:

Monoacidic bases can accept only one hydrogen ion ($H^+$) per molecule.
NaOH readily accepts one $H^+$ ion from its dissociation in water,
$NaOH + H_2O \rightarrow Na^+ + OH^-$.
$Ba(OH)_2$, $Mg(OH)_2$, and $Al(OH)_3$ are diacidic or polyacidic bases, meaning they can accept two or more $H^+$ ions per molecule.
For example, $Ba(OH)_2$ can accept two $H^+$ ions:
$Ba(OH)_2 + 2H^+ \rightarrow Ba^{2+} + 2H_2O$.

**17. Correct Option: A) $Ba(OH)_2$**

Explanation:

Diacidic bases can accept two hydrogen ions ($H^+$) per molecule.

$Ba(OH)_2$ readily accepts two $H^+$ ions through its dissociation in water,

$Ba(OH)_2 + 2H^+ \rightarrow Ba^{2+} + 2H_2O$.

$NH_4OH$ and KOH are monoacidic bases like NaOH, accepting only one $H^+$ ion each.

$Al(OH)_3$ is a triacidic base, capable of accepting three $H^+$ ions due to its three $OH^-$ groups.

**18. Correct Option: D) $HNO_3$**

Explanation:

Arrhenius acids dissociate in water to release hydrogen ions ($H^+$).

$HNO_3 + H_2O \rightarrow H^+ + NO_3^-$.

$CO_2$ and $SO_3$ are acid anhydrides, reacting with water to form acids but not directly releasing $H^+$ ions.

$FeCl_3$ is a salt, not an acid, as it doesn't dissociate to release $H^+$ ions.

**19. Correct Option: C) NaCl**

Explanation:

Ionization refers to the separation of a molecule into charged ions in solution.

While HCl, $CH_3COOH$, and HBr are polar covalent molecules that dissociate into ions in water (e.g., $HCl \rightarrow H^+ + Cl^-$), NaCl is an ionic compound already composed of $Na^+$ and $Cl^-$ ions. Thus, it doesn't need further ionization in water.

**20. Correct Option: B) $H_2PO_2^-$**

Explanation:

$H_2PO_2^-$ (hydrogen phosphite ion) lacks a conjugate base. It's already donated its two acidic protons, leaving no remaining $H^+$ to accept and form a conjugate pair. Unlike $H_2PO_4^-$ (dihydrogen phosphate) and $H_2PO_3^-$ (hydrogen phosphite), which can accept protons ($H_2PO_4^- + H^+ \rightleftharpoons H_3PO_4$, $H_2PO_3^- + H^+ \rightleftharpoons H_3PO_3$), $H_2PO_2^-$ remains proton-less.

**21. Correct Option: D) Both Lewis and Bronsted neutralization**

Explanation:

This reaction exhibits characteristics of both:

1. Bronsted: $H_3O^+$ donates a proton ($H^+$) to $OH^-$, fulfilling the Bronsted acid-base definition.
2. Lewis: $H_3O^+$ acts as a Lewis acid (electron acceptor) by accepting a lone pair from $OH^-$, a Lewis base (electron donor).

**22. Correct Option: D) $H_2PO_4^-, PO_4^{-3}$**

Explanation:

A conjugate acid-base pair differs by one proton ($H^+$).

In A, B, and C, the difference between the two species is indeed one $H^+$:

A) $HSO_4^-$ gains $H^+$ to become $SO_4^{2-}$.

B) $H_2PO_4^-$ gains $H^+$ to become $HPO_4^{2-}$.

C) $H_2PO_4^-$ loses $H^+$ to become $H_3PO_4$.

However, in D, $H_2PO_4^-$ and $PO_4^{3-}$ differ by two protons. $H_2PO_4^-$ would need to lose two $H^+$ to reach $PO_4^{3-}$, not just one.

**23. Correct Option: A) $OH^-$**

Explanation:

The conjugate acid of a base differs by one proton ($H^+$).

The oxide ion ($O^{2-}$) accepts a proton to become the hydroxide ion ($OH^-$): $O^{2-} + H^+ \rightarrow OH^-$.

**24. Correct Option: D) $HN_3$**

Explanation:

Bases accept protons ($H^+$). While the other options readily accept protons due to their lone pairs ($N_2H_4$, $NH_2OH$, $(CH_3)_3N$), $HN_3$ acts as an acid itself, containing a readily donatable proton on the nitrogen atom.

**25. Correct Option: C) $BF_3$**

Explanation:

1. Brønsted-Lowry acids accept lone pairs and donate protons ($H^+$). $BF_3$ readily accepts a lone pair from a Lewis base like $F^-$ ($F^- + BF_3 \rightarrow BF_4^-$).
2. Arrhenius acids dissociate in water to release $H^+$ ions. $BF_3$ doesn't have any $H^+$ to release and doesn't dissociate in water, preventing it from exhibiting Arrhenius behaviour.

**26. Correct Option: C) Arrhenius theory**

Explanation:

According to Arrhenius theory, an acid is a substance that dissociates in water to release hydrogen ions ($H^+$). Sulphur trioxide ($SO_3$) readily dissolves in water and dissociates into $H^+$ and $SO_4^{2-}$ ions, making it an acid based on the Arrhenius definition.

The other options:

A) Lewis theory: While $SO_3$ can act as a Lewis acid by accepting electron pairs, it doesn't directly release $H^+$ in water, so it wouldn't be classified as an acid.

B) Usanovich theory: This lesser-known theory focuses on the interaction of protons and electrons, but it's not as widely used as the other options and wouldn't be the primary explanation for $SO_3$ being an acid.

D) Bronsted-Lowry theory: This theory expands on Arrhenius by stating that an acid is a proton donor. $SO_3$ can indeed donate a proton to water, making it an acid according to Bronsted-Lowry, but the Arrhenius definition is more specific about $H^+$ release in water.

**27. Correct Option: D) Acetylene**

Explanation:

The strongest conjugate base results from acetylene since it is the weakest acid among formic acid, benzoic acid, acetic acid, and acetylene.

The order of strength of acids is given below:

Acetylene < Acetic Acid < Formic Acid < Benzoic Acid

**28. Correct Option: A) $HNO_3$**

Explanation:

Amphoteric substances can act as both acids and bases, depending on the solution they're in. They have the ability to both donate and accept protons ($H^+$).

A) $HCO_3^-$ (bicarbonate ion) can act as a base by accepting protons ($HCO_3^- + H^+ \rightleftharpoons H_2CO_3$), or as an acid by donating its own proton ($HCO_3^- \rightleftharpoons CO_2 + OH^-$)

B) $H_2O$ (water) can act as an acid (donating $H^+$ to strong bases) and a base (accepting $H^+$ from strong acids).

C) $NH_3$ (ammonia) can act as a base by accepting $H^+$ (forming $NH_4^+$), but under specific conditions, it can also donate $H^+$ to even stronger bases like lithium amide.

$HNO_3$, however, is a strong acid and doesn't readily accept protons due to its highly electronegative nitrate group ($NO_3^-$) already possessing a strong hold on its protons. Therefore, it primarily remains an acid regardless of the solution it's in.

**29. Correct Option: A) Weak**

Explanation:

Strong electrolytes readily ionize in water, completely dissociating into ions.

Acetic acid is a weak acid compared to water. While it can still conduct electricity due to some dissociation, it does so to a much lesser extent compared to water.

When a strong electrolyte dissolves in acetic acid, the weakness of the solvent hinders its ability to fully ionize the electrolyte. The strong electrolyte experiences partial or incomplete dissociation, leading to a weaker overall ionization compared to its behaviour in water.

**30. Correct Option: C) Benzene**

Explanation:

In water, ammonia, and aqueous NaOH, HCl readily donates its proton ($H^+$), showcasing its acidic prowess. It forms $H_3O^+$ in water, $NH_4^+$ in ammonia, and $Cl^-$ and $Na^+$ in aqueous NaOH.

Benzene, however, is a relatively non-polar molecule with weak electron-accepting abilities, unlike the other options. Therefore, HCl remains stubbornly intact in benzene, failing to exhibit its acidic nature.

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## LEVEL – II

## SINGLE OPTION CORRECT TYPE

**1. Correct Option: B) Bitter**

Explanation:

Bases generally taste bitter, though not all and intensity varies. Strong ones have a soapy/astringent bitterness, while weak ones might be subtle or even slightly sweet

**2. Correct Option: A) Neutralization**

Explanation:

Neutralization refers to the chemical reaction between an acid and a base, where they react completely to form a salt and water. The resulting solution becomes neutral (neither acidic nor basic) due to the complete consumption of both the acidic and basic properties.

**3. Correct Option: C) Citric acid**

Explanation:

Citric acid is used for making sour drinks such as Limca.

**4. Correct Option: C) Inorganic acids**

Explanation:

A) Organic acids: These acids come from living organisms or their remains, such as citric acid (found in fruits) and acetic acid (vinegar).

B) Strong acids: This refers to the strength of an acid in terms of its ability to dissociate in water and donate protons. While some inorganic acids like sulfuric acid are strong, not all are, and strength isn't directly related to their origin.

C) Inorganic acids: These are acids derived from mineral sources, meaning they originate from non-living materials found in the Earth's crust. Examples include sulfuric acid ($H_2SO_4$), hydrochloric acid (HCl), and nitric acid ($HNO_3$).

D) Weak acids: Similar to strong acids, this categorizes acids based on their dissociation degree. Again, some inorganic acids are weak, but it's not a defining characteristic based on their source.

**5. Correct Option: C) Sulphuric acid**

Explanation:

Sulphuric acid is often referred to as the "king of chemicals" because of its widespread industrial use and importance in various chemical processes.

**6. Correct Option:**

Explanation:

A triacidic base is a base that can accept three protons ($H^+$ ions). This means it has three hydroxide ($OH^-$) groups available for donation.

NaOH (sodium hydroxide) and $Ca(OH)_2$ (calcium hydroxide) are monobasic bases, meaning they have only one $OH^-$ group and can accept only one $H^+$ ion.

$Al(OH)_3$ (aluminium hydroxide), however, has three $OH^-$ groups and can therefore accept three $H^+$ ions, qualifying it as a triacidic base.

$Mg(OH)_2$ (magnesium hydroxide) is a dibasic base, meaning it has two $OH^-$ groups and can accept two $H^+$ ions.

**7. Correct** Option: A) HCl

Explanation:

A conjugate base is formed when an acid donates a proton ($H^+$). In this case, the $Cl^-$ ion is formed when HCl loses its $H^+$:

$HCl\ (aq) \rightarrow H^+\ (aq) + Cl^-\ (aq)$

**8. Correct Option: C) $Na_2O$**

Explanation:

A) $CO_2$ (carbon dioxide): Although not a strong acid, $CO_2$ readily dissolves in water to form carbonic acid ($H_2CO_3$), making it acidic.

$CO_2(g) + H_2O(l) \rightleftharpoons H_2CO_3(aq)$

B) $SO_3$ (sulphur trioxide): Similar to $CO_2$, $SO_3$ reacts vigorously with water to form sulphurous acid ($H_2SO_3$), exhibiting acidic properties.

$SO_3(g) + H_2O(l) \rightarrow H_2SO_3(aq)$

C) $Na_2O$ (sodium oxide): Unlike the other options, $Na_2O$ reacts with water to form sodium hydroxide (NaOH), a strong base. This reaction consumes $H^+$ ions from water, making the resulting solution slightly alkaline, not acidic.

$Na_2O(s) + H_2O(l) \rightarrow 2NaOH(aq) + H^+(aq)$

E) $N_2O_5$ (dinitrogen pentoxide): This potent oxidizing agent also undergoes hydrolysis in water, forming nitric acid ($HNO_3$) and establishing its acidic character.

$N_2O_5(s) + H_2O(l) \rightarrow 2HNO_3(aq)$

**9. Correct Option: A) $ClO^-$**

Explanation:

Brønsted basicity is inversely proportional to the stability of the conjugate acid. In other words, a weaker conjugate acid indicates a stronger Brønsted base.

In this case, all the given anions are derived from the same acid, $HClO_4$ (perchloric acid). When they accept a proton ($H^+$), they form their respective conjugate acids:

- $ClO^- + H^+ \rightarrow HClO$ (hypochlorous acid)
- $ClO_2^- + H^+ \rightarrow HClO_2$ (chlorous acid)
- $ClO_3^- + H^+ \rightarrow HClO_3$ (chloric acid)
- $ClO_4^- + H^+ \rightarrow HClO_4$ (perchloric acid)

Among these conjugate acids, hypochlorous acid (HClO) is the weakest, thereby making $ClO^-$ the strongest Bronsted base.

**10. Correct Option: D) I < iii < ii < iv**

Explanation:

$HSO_3F$ is a super acid. Its acidic strength is greater than any given species.

The $pK_a$ value of other species are

$HCO_3^- \rightarrow$ 10.25

$H_3O^+ \rightarrow$ -1.74

$HSO_4^- \rightarrow$ 1.92

Lesser the $pK_a$ value higher will be its acidic strength. Hence, sequence of acidic strength will be $HSO_3F > H_3O^+ > HSO_4^- > HCO_3^-$

**11. Correct Option: B) Lewis acid**

Explanation:

A Lewis acid is a species that accepts a lone pair of electrons.

Reaction analysis: In this reaction, $AlCl_3$ accepts a lone pair of electrons from the chloride ion ($Cl^-$) to form the $AlCl_4^-$ complex ion. The empty 3p orbital on the aluminium atom acts as the electron pair acceptor.

Other options:

A) Lewis base: $Cl^-$ donates the lone pair, making it the Lewis base in this scenario.

A) Lewis salt: There's no formation of a salt in this reaction, which typically involves the complete transfer of electrons to form ions.

B) Arrhenius acid: $AlCl_3$ doesn't release $H^+$ in aqueous solution, so it doesn't fit the Arrhenius definition of an acid.

**12. Correct Option: C) Dative bond**

Explanation:

Dative bond: This type of bond involves one atom donating a lone pair of electrons to another atom, without sharing electrons equally. In a Lewis acid-base reaction, the base (electron donor) donates its lone pair to the empty orbital of the acid (electron acceptor), forming a dative bond.

Other options:

A) Covalent bond: While some covalent character may exist due to electron sharing, the emphasis is on the lone pair donation from the base, making it primarily a dative bond.

B) Ionic bond: This typically involves complete electron transfer, which isn't the case in Lewis
acid-base interactions.

D) H-bond: While hydrogen bonds can be present in the resulting molecule, they wouldn't be the primary bond formed in the actual neutralization reaction.

**13. Correct Option: C) $N_2$**

Explanation:

Lewis acids are electron acceptors, seeking to fill empty orbitals with lone pairs from other molecules.

$N_2$ possesses an empty π* antibonding orbital, making it eager to accept electron pairs. This empty orbital can readily interact with lone pairs on other molecules, engaging in electron pair sharing.

O, F, and $O_2$ primarily act as Lewis bases due to their lone pairs. They are electron donors, not acceptors.

**14**. **B) Lowry-Bronsted theory**.

**Lowry-Bronsted Theory Advantage:** The Lowry-Bronsted theory defines an acid as a **proton ($H^+$) donor** and a base as a **proton ($H^+$) acceptor**. This definition does not require a solvent or an aqueous medium.

**15. Correct Option: A) The presence of three oxygen atoms, $e^-$ density will increase on sulphur atom in the $SO_3$**

Explanation:

Lewis acid strength is directly related to the electron-accepting ability of the central atom. The easier it is for the atom to accept a lone pair of electrons, the stronger the Lewis acid.

In $SO_3$, the three oxygen atoms surrounding the sulphur atom are highly electronegative. They draw electron density towards themselves, creating a partial positive charge on the sulphur atom. This increases the sulphur's ability to accept lone pairs from other molecules, making $SO_3$ a stronger Lewis acid compared to $SO_2$. $SO_2$ has only two oxygen atoms, resulting in a less positive charge on the sulphur atom and a weaker electron-accepting ability compared to $SO_3$.

Therefore, the presence of three oxygen atoms in $SO_3$ plays a crucial role in enhancing its Lewis acidic strength by increasing the electron density deficit on the sulphur atom.

**16. Correct Option: C) $NBr_3$ has more acidic strength than $NI_3$**

Explanation:

In the context of Lewis acids, the electronegativity of the surrounding atoms plays a crucial role in determining their acidic strength. The more electronegative the surrounding atoms, the greater their ability to pull electron density away from the central N atom, creating a partial positive charge that makes it more receptive to accepting lone pairs (acting as a Lewis acid).

A) $NI_3$ vs. $NF_3$: Fluorine is more electronegative than iodine, so $NF_3$ will have a higher partial positive charge on the N atom, making it more acidic than $NI_3$.

B) $NI_3$ vs. $NBr_3$: Bromine is more electronegative than iodine, so $NBr_3$ will have a higher partial positive charge on the N atom, making it more acidic than $NI_3$.

D) $NCl_3$ vs. $NBr_3$: Chlorine is more electronegative than bromine, so $NCl_3$ will have a higher partial positive charge on the N atom, making it more acidic than $NBr_3$.

Therefore, based on the electronegativity trend, option C) $NBr_3$ has more acidic strength than $NI_3$ is the correct choice.

**17. Correct Option: D) It does not acts as Lewis acid**

Explanation:

A) While $BaCl_2$ is indeed an ionic compound, it doesn't directly influence its Lewis acid behaviour.

B) Barium (Ba) in $BaCl_2$ possesses filled electronic orbitals in its outermost shell ($6s^2$). There are no readily available empty orbitals for accepting lone pairs, a key characteristic of Lewis acids.

C) This statement refers to Beryllium (Be), not Barium. Additionally, while back bonding can occur in some Be compounds, it's not relevant to the Lewis acid nature of $BaCl_2$.

D) Both Ba and Cl in $BaCl_2$ have complete octets in their stable ionic configuration. They neither lack electron pairs nor possess empty orbitals to readily accept them, eliminating the possibility of Lewis acid behaviour.

**18. Correct Option: D) Acetic acid**

Explanation:

Amphiprotic solvents are those which can donate and accept a proton.

A) $2H_2O \rightleftharpoons H_3O^+ + OH^-$. Here one water molecule is accepting a proton and the other is donating, so water is amphiprotic.

C) Alcohol also can accept and donate a proton.

D) Acetic acid can only donate a proton but can't accept.

**19. Correct Option: B) $NH_3$**

Explanation:

Brønsted-Lowry theory defines acids as proton ($H^+$) donors.

A) $HSO_4^- \rightleftharpoons H^+ + SO_4^{2-}$ (sulphate ion readily donates a proton)

C) $BCl_3$ acts as a Lewis acid, accepting a lone pair from $Cl^-$ to form $BCl_4^-$ (indirectly involving proton transfer)

D) $HNO_3 \rightleftharpoons H^+ + NO_3^-$ (nitric acid readily dissociates)

However, $NH_3$ prefers the role of a Brønsted-Lowry base, readily accepting protons to form the ammonium ion: $NH_3 + H^+ \rightleftharpoons NH_4^+$.

**20. Correct Option: D) Lewis theory cannot explain relative strength of acids and bases**

Explanation:

Lewis theory: While it effectively defines acids and bases based on electron accepting/donating abilities, it lacks a quantitative measure for comparing their strengths. It tells you "who's who" in the acid-base world but doesn't provide a scale for "how strong" they are.

Arrhenius and Bronsted-Lowry theories: Both offer frameworks for comparing acid/base strengths. Arrhenius uses the degree of dissociation in water, while Bronsted-Lowry considers the pKa value (proton dissociation constant).

**Assertion and reason type:**

**21. Correct Option: A)** Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

Explanation:

This phenomenon is known as the leveling effect of water. Strong acids like HCl, $HNO_3$, and $H_2SO_4$ ionize completely (100%) in water. Because they all donate their protons entirely to water molecules, they all form hydronium ions ($H_3O^+$):

$HCl + H_2O \rightarrow H_3O^+ + Cl^-$

As a result, no strong acid can appear stronger than $H_3O^+$ in an aqueous solution, making it impossible to distinguish their relative strengths in water. To differentiate them, a less basic solvent like pure acetic acid must be used.

**Match the following:**

**22.** (a) → Q, (b) → S, (c) → P, (d) → R

**(a) Muriatic acid → (Q)** HCl (Hydrochloric acid historically carries this name).

**(b) King of chemicals → (S)** $H_2SO_4$ (Sulfuric acid is highly vital to industrial chemistry).

**(c) Aqua fortis → (P)** $HNO_3$ (Nitric acid translates from Latin as "strong water").

**(d) Phosphoric acid → (R)** $H_3PO_4$

**Comprehension Type Questions**

**21. Correct Option: B)** HCl

- **Explanation:** Hydrochloric acid (HCl) is a mineral acid that dissociates completely into its constituent ions in water. The others ($CH_3COOH$, $H_2CO_3$, and $H_3PO_3$) are weak acids that only partially ionize.

**22. Correct Option: C) Degree of ionization**

- **Explanation:** Strong acids possess a degree of ionization ($\alpha$) approximately equal to 1 (100% ionization), whereas weak acids have a degree of ionization much less than 1 ($\alpha \ll 1$), meaning they remain mostly intact as molecules.

**23. Correct Option: C)** $CH_3COOH$

- **Explanation:** Acetic acid ($CH_3COOH$) is an organic carboxylic acid that does not break down fully in water. Conversely, $HNO_3$, HCl, and $H_2SO_4$ are well-known strong mineral acids.

**Integer Type Questions**

**24. Answer: 2**

- **Explanation:** Basicity refers to the number of replaceable $H^+$ ions per molecule of an acid. Even though phosphorous acid contains three hydrogen atoms, its structural framework only binds two of them directly to highly electronegative oxygen atoms (P-OH bonds). The third hydrogen connects directly to phosphorus (P-H bond) and cannot be released as an ion.

**25. Answer: 3**

- **Explanation:** Acidity represents the number of replaceable hydroxide ($OH^-$) ions a base can yield in solution or the number of $H^+$ ions it can neutralize. Since one formula unit of $Al(OH)_3$ contains three hydroxide groups, its acidity is 3.

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**Salts:**

**1. Correct Option: D)** $\sqrt{K_W}$

Explanation:

At any temperature, the proton concentration of pure water is approximately $\sqrt{K_W} \approx 10^{-7}$ M, reflecting its equilibrium relationship with hydroxide ions via the ionic product of water ($K_w$).

**2. Correct Option: B) 0**

Explanation:

In a 1 M HCl solution, the complete dissociation of HCl occurs as

$HCl\ (aq) \rightarrow H^+(aq) + Cl^-(aq)$

$pH = -\log[H^+]$

$= -\log[1]$

$pH = 0$

**3. Correct Option: D) Decreased by 10 times**

Explanation:

1. Initial $H^+$ concentration:

$pH = -\log_{10}[H^+]$

$4 = -\log_{10}[H^+]$

$[H^+] = 10^{-4}$ M

2. Desired $H^+$ concentration for pH 5.0:

$pH\ 5.0 = -\log_{10}[H^+]_{new}$

$[H^+]_{new} = 10^{-5}$ M

3. Change in $H^+$ concentration:

$\frac{[H^+]_{new}}{[H^+]_{initial}} = \frac{10^{-5}\,M}{10^{-4}\,M} = \frac{1}{10}$

**4. Correct Option: D) Decreased by 100 times**

Explanation:

Initial $[H^+] = 10^{-2}$ M (pH 2). Doubling pH requires two one-unit increases, leading to: Final $[H^+] = 10^{-2} \times 10^{-1} \times 10^{-1} = 10^{-4}$ M (decreased by 100 times).

**5. Correct Option: A) 9**

Explanation:

Initial pH = 6, meaning $[H^+] = 10^{-6}$ M (because, $pH = -\log_{10}[H^+]$).

$H^+$ concentration decreases by 1000 times, so the new $[H^+] = 10^{-6}$ M / 1000 = $10^{-9}$ M.

new $pH = -\log_{10}(10^{-9}) = 9$.

**6. Correct Option: A) $FeSO_4$. $(NH_4)2SO_4$. $6H_2O$**

Explanation:

A mixed salt contains two different metal cations or two different non-metal anions.

A) $FeSO_4$. $(NH_4)2SO_4$. $6H_2O$ contains both $Fe^{2+}$ and $NH_4^+$ cations, along with $SO_4^{2-}$ anions.

B) $CH_3COONH_4$ (ammonium acetate) is a single salt, not a mixed salt. It has $NH_4^+$ as the cation and $CH_3COO^-$ as the anion.

C) $Na_2KPO_4$ (sodium potassium phosphate) is also a single salt. It has $Na^+$ and $K^+$ cations, along with $PO_4^{3-}$ anions.

D) $K_4[Fe(CN)_6]$ (potassium hexacyanoferrate) is a complex salt, not a mixed salt. It has the complex anion $[Fe(CN)_6]^{4-}$ and $K^+$ cations.

**7. correct option is B) Slaked lime.**

**Explanation**

Bleaching powder, chemically known as calcium oxychloride ($CaOCl_2$), is commercially prepared by passing chlorine gas ($Cl_2$) over dry **slaked lime** ($Ca(OH)_2$).

The balanced chemical equation for the reaction is:

$Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O$

**8. Correct Option: B) POP has the property of setting into a hard mass on wetting with water**

Explanation:

A) Normal salts are neutral in nature. So, A is wrong.

C) Salt is an electrovalent or ionic compound formed by the partial or complete replacement of the replaceable (ionisable) hydrogen atoms (i.e.,H+ ions) of an acid by a metallic ion or ammonium ion. So, C is wrong.

**9. The correct option is B) $Na_2SO_4 \cdot 10H_2O$**

Explanation:

A phenomenon where a compound loses its water of crystallization on exposure to dry air is called efflorescency. Substances showing efflorescency are called efflorescent substances.

$Na_2SO_4 \cdot 10H_2O$ (Glauber's salt or sodium sulfate decahydrate) is a classic example of an efflorescent substance. When left in the air, it spontaneously loses its water molecules to form anhydrous sodium sulfate ($Na_2SO_4$).

$Na_2SO_4 \cdot 10H_2O \xrightarrow{\text{exposed to air}} Na_2SO_4 + 10H_2O$

**10. The correct option is B) It produces carbon dioxide gas.**

**Explanation**

When baking soda (sodium bicarbonate, $NaHCO_3$) is added to cake batter and heated during the baking process, it undergoes a chemical reaction called **thermal decomposition**.

The heat causes the baking soda to break down, releasing **carbon dioxide gas ($CO_2$)** according to the following equation:

$2NaHCO_3(s) \xrightarrow{\Delta} Na_2CO_3(s) + H_2O(g) + CO_2(g)$

**11. Correct Option: C) Sodium hydroxide**

Explanation:

A) Silver chloride (AgCl):

$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$, equilibrium heavily favours the solid

B) Calcium sulphite ($CaSO_3$): It exhibits limited solubility, forming a slightly cloudy solution

$CaSO_3(s) \rightleftharpoons Ca^{2+}(aq) + SO_3^{2-}(aq)$, equilibrium favours the solid

C) Sodium hydroxide (NaOH):

$NaOH(s) \rightarrow Na^+(aq) + OH^-(aq)$

D) Sodium aluminate ($NaAlO_2$): This complex salt forms a colloidal suspension rather than a true solution

$NaAlO_2(s) + H_2O \rightleftharpoons Na^+(aq) + AlO_2(OH)^-(aq) + H_2O$

**12. Correct Option: D) $CaOCl_2$**

Explanation:

The reaction refers to the chlorination of calcium hydroxide ($Ca(OH)_2$) to form calcium oxychloride ($CaOCl_2$), also known as bleaching powder.

$2Ca(OH)_2(s) + Cl_2(g) \rightarrow CaOCl_2(s) + CaCl_2(s) + H_2O(l)$

**13. Correct Option: C) Neutral salt**

Explanation:

Strong acids and strong bases completely dissociate in water, meaning their ions readily separate and remain free in the solution. When a strong acid reacts with a strong base, the resulting salt is formed by the combination of their respective ions. Since both strong acids and strong bases have no inherent acidic or basic properties due to complete dissociation, their conjugate ions formed in the salt also lack these properties. Therefore, the resulting salt exhibits neither acidic nor basic character, making it a neutral salt.

**14. Correct Option: B) Basic salt**

Explanation:

Washing soda, also known as sodium carbonate ($Na_2CO_3$), is formed from the reaction of a strong base (sodium hydroxide, NaOH) with a weak acid (carbonic acid, $H_2CO_3$).

$2NaOH(aq) + H_2CO_3(aq) \rightarrow Na_2CO_3(aq) + 2H_2O(l)$

In this reaction, the strong base completely dissociates in water, while the weak acid partially dissociates. The resulting salt, $Na_2CO_3$, contains the conjugate base of the weak acid ($CO_3^{2-}$) which retains some of its basic properties.

**15. Correct Option: D) Carbon dioxide**

Explanation:

When you mix baking powder with a liquid and an acidic ingredient, the following reaction occurs:

$2NaHCO_3(aq) + H^+(aq) \rightarrow Na_2CO_3(aq) + CO_2(g) + H_2O(l)$

**16. Correct Option: C) 1/ 2**

Explanation:

Plaster of Paris, also known as calcium sulphate hemihydrate ($CaSO_4 \cdot ½H_2O$), contains one molecule of water of crystallization per two formula units of calcium sulphate. This translates to half a water molecule per formula unit, making option C the correct choice.

**17. Correct Option: D) Calcium oxide and carbon dioxide**

Explanation:

When calcium carbonate ($CaCO_3$) is heated, it undergoes a decomposition reaction:

$CaCO_3 (s) \rightarrow CaO (s) + CO_2 (g)$

This reaction essentially breaks down the calcium carbonate molecule, releasing calcium oxide (CaO, also known as lime) as a solid and carbon dioxide ($CO_2$) as a gas. So, both products, calcium oxide and carbon dioxide, are formed during the heating process.

**18. Correct Option: B) Washing soda**

**Explanation:**

Permanent hardness of water is caused by the presence of soluble chlorides and sulfates of calcium and magnesium ($CaCl_2$, $MgCl_2$, $CaSO_4$, $MgSO_4$).

When **washing soda** (sodium carbonate, $Na_2CO_3$) is added to hard water, it reacts with these soluble salts to form **insoluble precipitates** of calcium carbonate ($CaCO_3$) and magnesium carbonate ($MgCO_3$), which can then be easily filtered out.

$CaCl_2 + Na_2CO_3 \rightarrow CaCO_3 \downarrow + 2NaCl$

$MgSO_4 + Na_2CO_3 \rightarrow MgCO_3 \downarrow + Na_2SO_4$

**19. Correct Option: B) Fixed number of water molecules present in one formula unit of the salt**

Water of crystallization is the fixed number of water molecules that are chemically combined into the crystalline framework of a salt unit. These molecules are essential for maintaining the crystal's specific shape and, in many cases, its color.

- **Examples:** * Copper sulfate pentahydrate ($CuSO_4 \cdot 5H_2O$) contains 5 molecules of water of crystallization per formula unit, giving it a bright blue color.
  - Washing soda ($Na_2CO_3 \cdot 10H_2O$) contains 10 molecules of water of crystallization.

**20. Correct Option: A) $O_2$**

Explanation:

The chlor-alkali process produces chlorine gas ($Cl_2$), hydrogen gas ($H_2$), and sodium hydroxide (NaOH), but it does not yield oxygen gas ($O_2$) as a product.

$2NaCl(aq) + 2H_2O(l) \rightarrow Cl_2(g) + H_2(g) + 2NaOH(aq$

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**Level-II**

**1. Correct Option: C) $C_5H_5NH_2$ ($P^{Kb}$ = 3.2)**

Explanation:

$P^{Kb}$ is the negative logarithm of the base dissociation constant (Kb). Higher $P^{Kb}$ values indicate weaker bases because they dissociate less readily in water.

Comparing the $P^{Kb}$ values:

- $C_5H_5NH_2$ ($P^{Kb}$ = 3.2) has the highest $P^{Kb}$ value, making it the weakest base.
- $NH_4OH$ ($P^{Kb}$ = 4.8) has a lower $P^{Kb}$ value than $C_5H_5NH_2$, indicating it's a stronger base.
- $C_6H_5NH_2$ ($P^{Kb}$ = 9.4) and $C_9H_7N$ ($P^{Kb}$ = 9.2) have even higher $P^{Kb}$ values, making them even weaker bases than $C_5H_5NH_2$.

**2. Correct Option: C) 1 and 4 are correct**

Explanation:

1) In a neutral medium, $[H^+]$ = $[OH^-]$ = 1 x $10^{-7}$ mol $L^{-1}$ at $25^oC$. So, pH = -log $[H^+]$ = 7 at $25^oC$ is correct.

As the temperature increases, $K_w$ increases. For example, pH range is from 0 to 14 at $25^oC$ and $K_w$ = $10^{-14}$ and neutral pH point for pure water is 7.

But pH range is 0 to 13 at $80^oC$ and $K_w$ = $10^{-13}$ and neutral pH is 6.5 for pure water at this temperature. So, pH of water decreases with rise in temperature but will still remain neutral.

2) When base is added $[OH^-]$ of water increases and pH increases. So, the given option is wrong.

3) When acid is added $[H^+]$ of water increases and pH decreases. So, the given option is wrong.

4) pH + pOH = p$K_w$,

Where $K_w$ = -log ($K_w$) and $K_w$ is the ion product constant of water. For pure water at $25^oC$, $K_w$ is approximately 1 x $10^{-14}$, and at this temperature, the pH of pure water is 7.

So, at any temperature, the sum of the pH and pOH in water is constant and equal to the p$K_w$ value for that temperature.

**3. Correct Option: A) Temperature**

Explanation:

B) Nature of the solution: Obviously, an acidic solution will have a different pH than a basic or neutral one.

C) Degree of dissociation of acid (or) base: The extent to which an acid or base releases its ions affects the concentration of $H^+$ and $OH^-$ ions, directly impacting pH.

D) Volume of the solution: While not as significant as other factors, changing the volume of a solution (without altering the amount of acid/base) can slightly affect the concentration of ions and consequently, the pH.

**4. Correct Option:**

Explanation:

$K_a$, the acid dissociation constant, measures the extent of an acid's dissociation in water. A low $K_a$ value indicates limited dissociation, meaning only a small fraction of the acid molecules ionize into $H^+$ and conjugate base. Conversely, a high $K_a$ value signifies strong dissociation, with a significant portion of the acid molecules releasing protons.

Therefore, a weak acid is characterized by a low $K_a$, reflecting its limited tendency to dissociate and release $H^+$ ions.

Examining the other options

A) A very high $K_a$ indicates a strong acid, not a weak one.

B) $P^{Ka}$ (the negative logarithm of $K_a$) is inversely related to $K_a$. So, a low $K_a$ translates to a high $P^{Ka}$, not a very low one.

C) Alpha (α) represents the degree of dissociation, which is typically low for weak acids due to their limited ionization.

**5. Correct Option: D) $(NH_4)_2CO_3$**

Explanation:

A) $Na_2CO_3$ (sodium carbonate):

This salt is derived from a weak acid ($H_2CO_3$) and a strong base (NaOH). However, the carbonate ion ($CO_3^{2-}$) has a weak ability to pull protons from water, meaning the solution remains slightly alkaline due to excess $OH^-$ ions.

B) $AlCl_3$ (aluminium chloride):

This salt is formed from a strong acid (HCl) and a strong base ($Al(OH)_3$). Therefore, the solution has no excess $H^+$ or $OH^-$ ions and will be neutral (pH ≈ 7).

C) NaCl (sodium chloride):

This salt is derived from strong acid (HCl) and strong base (NaOH). Similar to $AlCl_3$, it has no excess ions and will have a neutral pH.

D) $(NH_4)_2CO_3$ (ammonium carbonate):

This salt is formed from a weak acid ($H_2CO_3$) and a weak base ($NH_4OH$). While the carbonate ion contributes some alkalinity, the ammonium ion ($NH_4^+$) can react with water to release $H^+$ ions. This reaction, known as hydrolysis, can tip the balance slightly towards acidity, potentially resulting in a pH below 7.

**6. Correct Option: C) salt of strong base and strong acid**

Explanation:

Strong acids and bases completely dissociate in water, meaning they break down entirely into their respective ions. For example, HCl (strong acid) dissociates into $H^+$ and $Cl^-$, and NaOH (strong base) dissociates into $Na^+$ and $OH^-$. When a salt is formed from a strong acid and a strong base, both components are already completely dissociated. Therefore, the resulting salt simply exists as separate ions in the solution, achieving 100% ionization.

Examining the other options

A) Salt of strong acid and weak base: The weak base only partially dissociates in water, meaning some of its molecules remain intact. This incomplete dissociation translates to an incomplete ionization of the salt as well.

B) Salt of strong base and weak acid: Similar to option A, the weak acid's partial dissociation hinders the complete ionization of the salt.

D) Both A & B: As explained, neither salts of strong acids with weak bases nor salts of strong bases with weak acids achieve complete ionization due to the incomplete dissociation of one of the components.

**7. Correct Option: B) Sodium carbonate decahydrate**

Explanation:

The chemical name of washing soda is sodium carbonate decahydrate, often represented as $Na_2CO_3 \cdot 10H_2O$

**8. Correct Option: D) Both (B) & (C)**

Explanation:

A) Dehydration of NaCl solution: This doesn't directly produce sodium hydroxide. Dehydration focuses on removing water molecules, and while a concentrated NaCl solution can be dehydrated to form anhydrous NaCl, it wouldn't yield NaOH.

B) Electrolysis of aqueous NaCl solution: This is one method for producing sodium hydroxide, known as the chlor-alkali process. The overall reaction involves the electrolytic decomposition of brine (NaCl solution):

$2NaCl(aq) + 2H_2O(l) \rightarrow Cl_2(g) + H_2(g) + 2NaOH(aq)$

As you can see, this process generates chlorine gas ($Cl_2$), hydrogen gas ($H_2$), and the desired sodium hydroxide (NaOH) solution.

C) Chlor-alkali process: This encompasses option B and refers specifically to the electrolytic production of chlorine, hydrogen, and sodium hydroxide from brine.

D) Both (B) & (C): This is the correct answer. Electrolysis of aqueous NaCl solution, particularly via the chlor-alkali process, is the primary method for industrial production of sodium hydroxide

**9. Correct Option: A) 2**

Explanation:

Gypsum, also known as calcium sulphate dihydrate, has the chemical formula $CaSO_4 \cdot 2H_2O$. This indicates that one molecule of calcium sulphate ($CaSO_4$) combines with two molecules of water ($2H_2O$) as water of crystallization.

**10. Correct Option: C) Hydrogen**

Explanation:

Active metals react with dilute mineral acids to undergo a single displacement reaction, displacing hydrogen from the acid to liberate hydrogen gas ($H_2$).

Metal + Dilute Acid → Salt + Hydrogen gas ($H_2 \uparrow$)

*Example:* $Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$

**11. Correct Option: B) Salt and water**

Explanation:

Metal oxides are basic in nature. When a basic metal oxide reacts with an acid, it undergoes a neutralization reaction to yield salt and water.

Metal Oxide (Basic) + Acid → Salt + Water

*Example:* $CuO(s) + 2HCl(aq) \rightarrow CuCl_2(aq) + H_2O(l)$

**12. Correct Option: B) 0.16 N**

Explanation:

When mixing two solutions of the same solute, use the mixing formula:

$$N_{\text{result}} = \frac{N_1V_1 + N_2V_2}{V_1 + V_2}$$

- Given values: $V_1 = 20$ mL, $N_1 = 0.1$ N, $V_2 = 30$ mL, $N_2 = 0.2$ N
- Total Volume $(V_1 + V_2) = 20 + 30 = 50$ mL

Substitute the values:

$$N_{\text{result}} = \frac{(0.1 \times 20) + (0.2 \times 30)}{50} = \frac{2 + 6}{50} = \frac{8}{50} = 0.16 \text{ N}$$

**13. Correct Option: A) 0.05 N**

Explanation:

1. Find milliequivalents (meq) of base (NaOH):
   meq of NaOH = $N_1 \times V_1 = 0.2$ N × 25 mL = 5.0 meq
2. Find milliequivalents (meq) of acid (HCl):
   meq of HCl = $N_2 \times V_2 = 0.1$ N × 25 mL = 2.5 meq
3. Determine excess components:
   Since meq of NaOH > meq of HCl, the resulting solution is basic.
   Excess meq of $OH^-$ = 5.0 − 2.5 = 2.5 meq
4. Calculate excess ion concentration (Normality of excess ions):
   Total Volume $(V_1 + V_2) = 25 + 25 = 50$ mL

$$\text{Excess } [OH^-] = \frac{\text{Excess meq}}{\text{Total Volume (mL)}} = \frac{2.5}{50} = 0.05 \text{ N}$$

**One or More Than One Option Correct Type**

**14. Correct Options: A, B, and C**

Explanation:

A mixed salt is a salt that contains more than one basic radical (cation other than $H^+$) or more than one acidic radical (anion other than $OH^-$).

- $NaKCO_3$ contains two different cations: $Na^+$ and $K^+$. (Mixed Salt)
- $CaOCl_2$ (Bleaching powder) contains two different anions: $Cl^-$ and $OCl^-$. (Mixed Salt)
- $Na_2KPO_4$ contains two different cations: $Na^+$ and $K^+$. (Mixed Salt)
- $K_4[Fe(CN)_6]$ contains a simple cation and a complex anion, classifying it as a complex salt, not a mixed salt.

**15. Correct Options: A, B, and D**

Explanation:

Bleaching powder ($CaOCl_2$) acts as a strong disinfectant due to the release of chlorine, it bleaches organic coloration in fabrics via oxidation, and serves as a powerful oxidizing agent in chemical processes. Option C is incorrect because baking soda ($NaHCO_3$) is used for baking cakes, not bleaching powder.

Comprehension Type Questions

**16. Correct Option: B) Due to presence of ions**

Explanation:

Salts are ionic compounds. When dissolved in water, the crystal lattice breaks apart, releasing free-moving cations and anions that serve as charge carriers to conduct electric current.

**17. Correct Option: C) Crystalline solids**

Explanation:

Salts possess an orderly, repeating three-dimensional arrangement of constituent ions held tightly by strong electrostatic forces, forming a defined crystalline structure at room temperature.

**18. Correct Option: B) Polar solvents**

Explanation:

According to the rule "like dissolves like," ionic compounds like salts are highly soluble in polar solvents (such as water) because polar solvent molecules effectively stabilize and separate the individual ions.

Assertion and Reason Type Questions

**19. Correct Option: A)** Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

Explanation:

Pure water undergoes self-ionization to a tiny degree ($K_w = 1.0 \times 10^{-14}$ at 25°C). Because only a minuscule fraction of water molecules break down into free ions ($H_3O^+$ and $OH^-$), it conducts electricity poorly, categorizing it perfectly as a weak electrolyte.

**Matrix Match Type Questions**

**20. Correct Matching Summary:(a) → Q, (b) → P, (c) → R, (d) → S**

- (a) Sodium hydroxide → (Q) Used in the manufacture of paper and artificial fibers (Rayon process).
- (b) Bleaching powder → (P) Used as an oxidizing agent in chemical industries.
- (c) Baking soda → (R) Used as a core ingredient in antacids ($NaHCO_3$ neutralizes stomach acid).

- (d) Washing soda → (S) Used in the manufacture of glass ($Na_2CO_3$ acts as a silica flux).

**Integer Type Questions**

**21.** **Answer:** It depends directly on the concentration. If the exponent value given in the blank space is $10^{-n}$, the value of $x$ is simply that integer magnitude $n$, because $\text{pH} = -\log[H^+]$.

**22.** **Answer: 0.18**

Explanation:

Use the mixing formula:

$$N_{\text{result}} = \frac{N_1V_1 + N_2V_2}{V_1 + V_2}$$

$$N_{\text{result}} = \frac{(0.5 \times 10) + (0.1 \times 40)}{10 + 40} = \frac{5 + 4}{50} = \frac{9}{50} = 0.18 \text{ N}$$

Therefore, the value of $x$ is $0.18$.

## Previous Year Questions

**1**. B. Hydroxide ion concentration is logarithmic to pOH, which is inversely related to pH (pOH = 14 - pH); thus, the solution with the highest pH (P) has the highest $OH^-$.

**2**. C. The dehydration of Gypsum (Y) to Plaster of Paris (X) occurs at a critical temperature of 373 K (100°C) to yield the hemihydrate; exceeding this temperature leads to "dead burnt plaster" ($CaSO_4$).

**3**. A. Only Experiments I (Slaked lime, a base) and III (Vinegar, an acid) show correct color transitions for all four indicators; Experiment II is incorrect (Methyl Orange turns red in acid), and Experiment IV is incorrect (Phenolphthalein turns pink in base).

**4**. B. The bulb illuminates only when solution X is an electrolyte (acid or base) that provides mobile charge carriers; glucose is a non-electrolyte, and solid NaCl lacks ion mobility.

**5**. C. In the Chlor-alkali process, chlorine evolves at the anode (X) and hydrogen at the cathode (Y); NaOH is formed near the cathode, not the anode, and the process produces caustic soda, not baking soda.

**6**. B. Thermolysis of Sodium hydrogen carbonate ($NaHCO_3$) produces Sodium carbonate (a washing powder constituent) and $CO_2$ gas (which precipitates $CaCO_3$ in lime water).

**7**. C. These pH values match the characteristic ranges of the listed biological and household fluids, where lemon juice is strongly acidic and milk of magnesia is moderately basic.

**8**. C. The reaction describes the protonation of the bicarbonate anion, acting as a Brønsted-Lowry base, to form carbonic acid which subsequently undergoes rapid dehydration to evolve $CO_2$ gas (Y); this endothermic decomposition and gas evolution mechanism is the primary basis for its application in soda-acid fire extinguishers and systemic antacids.

**9**. D. This flowchart delineates the Solvay process where the carbonation of ammoniacal brine (i) yields sparingly soluble sodium bicarbonate (ii) due to the common-ion effect; subsequent thermal calcination produces sodium carbonate (iii) and regenerates $CO_2$ (iv), which is analytically confirmed via the formation of a $CaCO_3$ precipitate (V) in a saturated $Ca(OH)_2$ solution.

**10**. C. Potassium acetate ($CH_3COOK$) is the salt of a weak acid ($CH_3COOH$) and a strong base (KOH), leading to significant anionic hydrolysis where the acetate ion acts as a conjugate base to generate $OH^-$ ions, thereby driving the equilibrium pOH lower and the pH above 7.

# 4. CHEMICAL BONDING

## LEVEL – I

**1. Correct Option: A) Chemical bond**

**Explanation**:

Chemical bond encompasses the various forces (covalent, ionic, metallic) that bind atoms in molecules, crystals, and other structures

B) Valence: While valence refers to the number of electrons in an atom's outermost shell that can participate in bonding, it's not the force itself but rather a property that influences bonding behaviour.

C) Dative bond: This is a specific type of covalent bond where one atom donates both electrons in the bond, but it doesn't encompass all types of chemical bonds.

D) Hydrogen bond: This is a type of intermolecular force, not a true chemical bond, and it primarily occurs between molecules with hydrogen atoms bonded to highly electronegative atoms like oxygen or nitrogen.

**2. Correct Option: B) Exothermic**

**Explanation**:

When atoms form chemical bonds, they share electrons, creating a more stable configuration compared to their individual states. This process usually releases energy, making the overall reaction exothermic. The released energy can be in the form of heat, light, or sound

**3. Correct Option: A) Potential energy of system decreases**

**Explanation**:

When atoms form chemical bonds, they typically achieve a more stable electronic configuration by sharing electrons. This creates a lower energy state compared to the individual atoms, resulting in a decrease in the system's potential energy.

**4. Correct Option: A) Kessel and Lewis**

**Explanation**:

The significance of the stable octet in the formation of chemical bonds was first recognized by Gilbert N. Lewis and Kossel independently.

**5. Correct Option: A) Valence electrons**

**Explanation**:

B) Core electrons: These are electrons located closer to the nucleus in inner energy levels and are tightly bound to the nucleus. They are not directly involved in chemical bonding.

C) Lone pair electrons: These are valence electrons that are not involved in bonding with other atoms. While they are part of the valence shell, they are not the defining characteristic of all valence electrons.

D) Bonding electrons: These are valence electrons that are involved in forming chemical bonds with other atoms. Again, while they are a subset of valence electrons, they don't encompass all of them.

**6. Correct Option: B) Core electrons**

**Explanation**:

A) Valence electrons: These are the electrons present in the outermost energy level, not the inner one. They are crucial for chemical bonding and determining an atom's reactivity.

C) Lone pair electrons: These are valence electrons that are not involved in bonding with other atoms. While they are part of the valence shell, they are not the defining characteristic of all valence electrons.

D) Bonding electrons: These are valence electrons that are involved in forming chemical bonds with other atoms. Again, while they are a subset of valence electrons, they don't encompass all of them.

**7. Correct Option: C) $Cl^-$ ion**

**Explanation**:

A) Cl atom: A neutral chlorine atom has 7 valence electrons. While it can achieve an octet by sharing electrons with other atoms, it wouldn't have a complete octet in its isolated state as a Lewis dot structure.

B) $Cl^+$ ion: This ion loses an electron, resulting in 6 valence electrons. An octet is not achieved, making it unstable as a Lewis dot structure.

C) $Cl^-$ ion: By gaining an electron, chlorine gains a total of 8 valence electrons (7 from the atom + 1 gained), achieving a stable octet configuration. This is represented by a chlorine atom surrounded by 8 dots (7 dots representing its own electrons and 1 dot representing the gained electron).

D) $Cl^{2+}$ ion: This ion loses an electron, leaving it with 6 valence electrons, similar to $Cl^+$ ion and lacking a stable octet.

**8. Correct Option: C) He**

**Explanation**:

A) Li: Loses 1 electron to achieve a stable octet configuration (filled outer shell), but lacks the exceptional stability of He's filled first energy level.

B) Mg: Similar to Li, loses 2 electrons for an octet configuration, gaining stability but not reaching He's unique level.

C) He: Despite only 2 electrons (not 8), its filled first energy level orbital provides unmatched stability due to optimal electron-proton ratio and minimal shielding.

D) Ne: Achieves a stable octet with 8 electrons, following the rule, but its stability doesn't surpass He's unique first-level configuration.

**9. Correct Option: B) 4, 6, 7**

**Explanation**:

Carbon (C): 4 - It's in Group 14 of the periodic table, meaning it has 4 electrons in its outermost shell.

Sulfur (S): 6 - Located in Group 16, so it has 6 valence electrons.

Fluorine (F): 7 - As the first element in Group 17 (the halogens), it has 7 valence electrons.

**10. Correct Option: D) Both A and B.**

**Explanation**:

A) $CH_4$: Carbon in $CH_4$ forms four covalent bonds with each hydrogen atom, achieving an octet configuration. So, it obeys the octet rule.

B) $CO_2$: Each oxygen atom in $CO_2$ forms a double bond with the central carbon, sharing two electrons each. Carbon forms four single bonds with each oxygen, sharing one electron each. This also satisfies the octet rule for all atoms.

C) $BF_3$: Boron in $BF_3$ forms three covalent bonds with each fluorine atom, sharing one electron each. This fulfills the octet rule for boron and fluorine.

D) Both A and B: As explained above, both $CH_4$ and $CO_2$ obey the octet rule.

**11. Correct Option: A) $BCl_3$**

**Explanation**:

A contracted octet molecule has fewer than 8 electrons around its central atom due to an inability to fulfill the usual octet rule.

A. $BCl_3$: Boron has 3 valence electrons, forms 3 bonds (sharing 1 each), resulting in 6 electrons (contracted octet).

B. $CH_4$: Carbon has 4, forms 4 bonds, gets 8 electrons (octet rule).

C. $SF_6$: Sulfur has 6, forms 6 bonds, gets 12 electrons (exceeds octet).

D. $PCl_5$: Phosphorus has 5, forms 5 bonds, gets 10 electrons (exceeds octet).

**12. Correct Option: D) $CH_4$**

**Explanation**:

A super octet molecule has more than 8 electrons around its central atom

A. $PCl_5$: Phosphorus (central atom) has 5 valence electrons, forms 5 bonds, acquires 10 electrons (super octet, violates octet rule).

B. $IF_7$: Iodine has 7, forms 7 bonds, gets 14 electrons (super octet).

C. $SF_6$: Sulfur has 6, forms 6 bonds, gets 12 electrons (super octet).

D. $CH_4$: Carbon has 4, forms 4 bonds, gets 8 electrons (octet rule satisfied).

**13. Correct Option: D) Formal charge**

**Explanation**:

The formal charge is the difference between the number of valence electrons in an isolated atom and the number of electrons assigned to that atom in the Lewis structure.

A) Atomic number: This signifies the total number of protons, not related to electron distribution in molecules.

B) Oxidation number: While related to charge, it focuses on electron gain/loss during reactions, not electron distribution within a molecule.

C) Valency: This indicates the combining capacity of an atom, not the specific electron allocation in a Lewis structure.

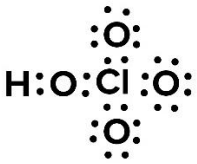
**14. Correct Option: 0 (please don't compare with options it's a printing mistake)**

**Explanation**:

Formal Charge = $V - N - \frac{B}{2}$

Where, V = the number of valence electrons present in the neutral atom of the compound. N = the number of non – bonding valence electrons in the same atom.

B = the total number of electrons present in the bonds that the atom under consideration forms with the other atoms in the molecule.



Formal Charge of Cl= $7 - 0 - \frac{14}{2} = 0$

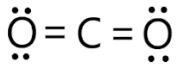
**15. Correct Option: A) 0**

**Explanation**:

Formal Charge = $V - N - \frac{B}{2}$

Where, V = the number of valence electrons present in the neutral atom of the compound. N = the number of non – bonding valence electrons in the same atom.

B = the total number of electrons present in the bonds that the atom under consideration forms with the other atoms in the molecule.



Formal Charge of C= 4 – 0 - $\frac{8}{2}$ = 0

**16. Correct Option: A) Valency**

**Explanation**:

Valency describes the combining capacity of an atom, meaning the number of bonds it can form with other atoms.

B) Chemical bond: This refers to the actual force connecting atoms, not the potential for forming bonds.

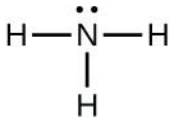
C) Oxidation state: This captures the hypothetical charge an atom would have if all its bonds were ionic, not its bonding capacity.

D) Oxidation number: Similar to oxidation state, it focuses on the theoretical charge, not the inherent tendency to form bonds.

**17. Correct Option: B) 3**

**Explanation**:

Nitrogen has 5 valence electrons ($2s^2 2p^3$). In $NH_3$, N forms covalent bonds with 3 hydrogen atoms. It shares one electron with each H, resulting in 3 shared electrons. This fulfils the octet rule (8 electrons for stability).

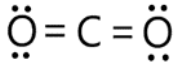


ammonia

**18. Correct Option: C) 4**

**Explanation**:

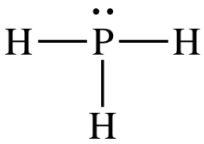
In the $CO_2$ molecule, carbon (C) forms two double bonds with oxygen (O). Each double bond involves the sharing of two electron pairs. Therefore, the valency of carbon in $CO_2$ is 4.



**19. Correct Option: D) 3**

**Explanation**:

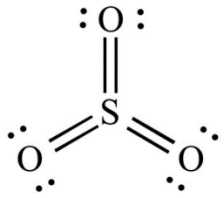
In the $PH_3$ molecule, phosphorus (P) forms three covalent bonds with hydrogen (H). Each covalent bond involves the sharing of one electron pair. Therefore, the valency of phosphorus in $PH_3$ is 3.



**20. Correct Option: B) 6**

**Explanation**:

In the $SO_3$ molecule, sulphur (S) forms six covalent bonds with oxygen (O). Each covalent bond involves the sharing of one electron pair. Therefore, the valency of sulphur in $SO_3$ is 6.



**LEVEL – II**

**SINGLE OPTION CORRECT TYPE**

**1. Correct Option: B) 16**

**Explanation**:

Element 20 (calcium) has 2 valence electrons and seeks stability through the octet rule (8 outer shell electrons).

A) 11 (Sodium, Na): Sodium typically loses one electron to achieve a stable electron configuration.

B) 16 (Sulphur, S): Sulphur typically gains two electrons to achieve a stable electron configuration.

C) 18 (Argon, Ar): Argon is a noble gas with a stable electron configuration.

D) 10 (Neon, Ne): Neon is a noble gas with a stable electron configuration.

Therefore, atom with atomic number 20 is most likely to combine chemically with the atom of atomic number 16.

**2. Correct Option: B) $F^+$**

**Explanation**:

F has 7 electrons in its valence shell. Thus, to attain stability, it should have lost one electron.

**3. Correct Option: A) $XY_2$**

**Explanation**:

Given, X....[Ar] $4s^2$

X is a metal and forms $X^{+2}$ ion

Y....[Ne] $3s^23p^5$

Y is a non-metal and forms $Y^-$ ion

$X^{+2}$ $Y^-$

Applying criss-cross method the formula is written as $XY_2$

**4. Correct Option: C) $H_2O$, $NH_3$, CO, $CH_4$**

**Explanation**:

A: $PCl_5$ violates octet rule (P has 10 electrons).

B: $BH_3$ violates octet rule (B has 6 electrons).

C: All atoms in $H_2O$, $NH_3$, CO, and $CH_4$ achieve 8 electrons (octet rule).

D: $CO_2$ and $AlCl_3$ violate octet rule (O with 7, Al with 6 electrons).

**5. Correct Option: D) $IF_7$**

**Explanation:**

A. Carbon (C) in $CH_4$: 4 valence electrons + 4 shared with H (1 each) = 8 electrons (octet rule satisfied).

B. Carbon (C) in $CO_2$: 4 valence electrons + 4 shared with O (2 each) = 8 electrons (octet rule satisfied).

C. Nitrogen (N) in $NF_3$: 5 valence electrons + 3 shared with F (1 each) = 8 electrons (octet rule satisfied).

D. Iodine (I) in $IF_7$: 7 valence electrons + 7 shared with F (1 each) = 14 electrons (significantly exceeds octet rule).

**6. Correct Option: C) 14**

**Explanation:**

Xenon (Xe): It has 8 valence electrons in the outermost shell ($5s^25p^6$).

Fluorine (F): Each F atom has 7 valence electrons ($1s^22s^22p^5$).

1. Shared electrons: $XeF_4$ has 4 F atoms, and each forms a single bond with Xe (1 shared electron/bond). So, Xe shares 4 electrons (4 x 1).

2. Remaining valence electrons in Xe: 8 valence electrons - 4 shared electrons = 4 electrons.

3. Lone pairs in Xe: Divide remaining electrons by 2 (as each lone pair has 2 electrons): 4 electrons / 2 = 2 lone pairs.

4. Lone pairs in F: Each F has 3 lone pairs (7 valence electrons - 1 shared electron = 6 electrons / 2 = 3 lone pairs).

Total lone pairs: 2 lone pairs (Xe) + 3 lone pairs x 4(4 x F) = 14 lone pairs.

**ONE OR MORE THAN ONE OPTION CORRECT TYPE**

**7. Correct Option: A, B , C**

**Explanation:**

Formal Charge = V – N - B/2

Where, V = the number of valence electrons present in the neutral atom of the compound. N = the number of non – bonding valence electrons in the same atom.

B = the total number of electrons present in the bonds that the atom under consideration forms with the other atoms in the molecule.

Number 2 O atom: FC = 6 – 4 – 4/2 = 0

Number 1 O atom: FC = 6 – 2 – 6/2 = 1

Number 3 O atom: FC = 6 – 6 – 2/2 = -1

**8. Correct Option: A, B, C**

**Explanation:**

Expanded octet refers to a central atom having more than 8 electrons around it due to exceptions to the octet rule. Among the given options:

A) $SF_6$: Sulphur (central atom) has 6 valence electrons and shares 6 electrons with fluorine (1 each), resulting in 12 electrons around it (expanded octet).

B) $PCl_5$: Phosphorus has 5 valence electrons and shares 1 with each chlorine, leading to 10 electrons around it (expanded octet).

C) $IF_7$: Iodine has 7 valence electrons and shares 1 with each fluorine, giving it 14 electrons around it (expanded octet).

D) $BeCl_2$: Beryllium only has 2 valence electrons and shares 1 with each chlorine, totaling 4 electrons (expanded octet rule not satisfied).

**9. Correct Option: C, D**

**Explanation:**

Duplet configuration means having only two valence electrons in the outermost shell.

A) $H^-$: Hydrogen gains one electron, resulting in two (duplet) around it.

B) $H_2$: Each hydrogen shares one electron with the other, both achieving two (duplet) electrons.

C) $Li^{2+}$: Lithium typically loses 1 electron to achieve a duplet configuration. In this case, $Li^{2+}$ has lost 2 valence electrons, leaving it with one valence electron.

D) $Be^{3+}$: Beryllium loses three electrons, leaving it with no valence electrons (not a duplet).

**10. Correct Option: B) By sharing of electrons**

**Explanation:**

Octet rule states that atoms tend to have 8 electrons in their outermost shell (valence shell) to achieve stability.

A. Electron transfer: While transferring electrons can sometimes lead to an octet configuration (e.g., ionic bonding), it's not the primary path due to potential energy imbalances created.

C. Gaining electrons: This can contribute to octet configuration, but it depends on the initial electron count and may not always be applicable.

D. Not sharing electrons: This wouldn't fulfill the octet rule, as electrons remain isolated and don't contribute to filling the outer shell.

Assertion and reason type:

**11. Correct Option: A) Both A and R are true, and R is the correct explanation of A**

**Explanation:**

- **Assertion (A) is true:** Lewis dot structures are fundamental diagrams used in chemistry to visually represent how atoms connect and bond with one another to form molecules.
- **Reason (R) is true:** Lewis structures achieve this by specifically mapping out the valence electrons (the electrons in the outermost shell) as dots around the chemical symbols of the elements.
- **Is it the correct explanation?** Yes. Because chemical bonding entirely depends on how outermost valence electrons are shared (covalent bonds) or transferred (ionic bonds), showing the arrangement of these outermost electrons is precisely *why* Lewis structures allow us to understand molecular bonding.

**12. Correct Option: A) Both A and R are true, and R is the correct explanation of A**

**Explanation:**

- **Assertion (A) is true:** Valency is defined as the combining capacity of an atom. A hydrogen atom consistently exhibits a valency of 1.
- **Reason (R) is true:** A neutral hydrogen atom has only 1 electron in its lone valence shell (the K shell). To reach a stable, filled outer shell configuration (a stable duplet resembling Helium), it requires exactly 1 more electron. It achieves this stability either by sharing its single electron (forming a single covalent bond, like in $H_2O$ or $H_2$) or by completely losing it (forming an $H^+$ ion, like in acids).
- **Is it the correct explanation?** Yes. The fundamental definition of valency is the number of electrons an atom must gain, lose, or share to reach a stable electronic configuration. Therefore, the fact that hydrogen shares or loses exactly 1 electron to become stable directly dictates and explains why its valency is 1.

**MATRIX MATCHING TYPE**

**13.** **(a) → R, (b) → S, (c) → Q, (d) → P**

**(a) Magnesium:** * **Atomic Number:** 12

- **Electronic Configuration:** 2, 8, 2
- **Valence Electrons:** 2
- **Matching Dot Symbol:** Mg surrounded by **2 dots → (R)**

**(b) Chlorine:** * **Atomic Number:** 17

- **Electronic Configuration:** 2, 8, 7
- **Valence Electrons:** 7
- **Matching Dot Symbol:** Cl surrounded by **7 dots → (S)**

**(c) Sodium:** * **Atomic Number:** 11

- **Electronic Configuration:** 2, 8, 1
- **Valence Electrons:** 1
- **Matching Dot Symbol:** Na with **1 dot → (Q)**

**(d) Sulphur:** * **Atomic Number:** 16

- **Electronic Configuration:** 2, 8, 6
- **Valence Electrons:** 6
- **Matching Dot Symbol:** S surrounded by **6 dots → (P)**

**14. Correct Option: a)-R), b)-S), c)-Q), d)-P)**

**Explanation**:

a) Boron (B): 3

- It has the electron configuration: $1s^2 2s^2 2p^1$.
- Valence electrons are those in the outermost shell, so in this case, the single electron in the 2p orbital is considered the valence electron.

b) Calcium (Ca): 2

- Its electron configuration is: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$.
- The outermost shell (4s) contains two valence electrons.

c) Bromine (Br): 7

- Its configuration is: $1s^2 2s^2 2p^6 3s^2 3p^5$.
- Similar to boron, the seven valence electrons reside in the outermost shell (3p), with five electrons contributing to the valence count.

d) Krypton (Kr): 8

- Its configuration is: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6$.
- Although it has electrons in the 4d orbital, the valence electrons are considered those in the outermost 4p orbital, which has a full eight electrons, making Kr a noble gas.

**COMPREHENSION TYPE**

**Passage-I:**

**15. Correct Option: B) 0, 3, 2, 0**

**Explanation**:

**Helium (He):** Helium has a full outer shell with 2 electrons ($1s^2$). It doesn't need to gain or lose electrons to achieve stability, so its valency is 0.

**Phosphorus (P):** Phosphorus has 5 valence electrons ($3s^2 3p^3$). It can gain 3 electrons to achieve an octet configuration ($P^{3-}$), resulting in a valency of 3. Alternatively, it can share 5 electrons with other atoms ($3s^2 3p^2$), also achieving an octet, leading to a valency of 5.

**Sulphur (S):** Sulfur has 6 valence electrons ($3s^2 3p^4$). It can gain 2 electrons ($S^{2-}$) or share 4 electrons ($3s^2 3p^2$) to reach an octet, resulting in valencies of 2 or 4, respectively. Additionally, it can participate in hypervalent bonding, sharing 6 electrons ($3s^2 3d^2$), leading to a valency of 6.

**Neon (Ne):** Similar to helium, neon has a full outer shell with 8 electrons ($1s^2 2s^2 2p^6$). It doesn't need to gain or lose electrons, so its valency is 0.

**16. Correct Option: B) 1, 7**

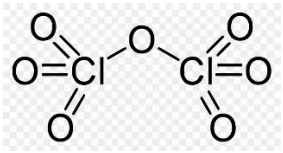
**Explanation**:

A) Cl in HCl:

In HCl, chlorine shares one electron with hydrogen to form a covalent bond. Its valence electrons are $3s^2 3p^5$, and sharing one electron allows it to achieve a stable octet configuration. Therefore, valency of Cl in HCl is 1

B) Cl in $Cl_2O_7$:

In the $Cl_2O_7$ molecule, each chlorine (Cl) forms 3 double bonds and 1 single bond with oxygen (O). Valency of Cl in $Cl_2O_7$ = 3 x 2 + 1 = 7



**17. Correct Option: C) 3**

**Explanation**:

Al has 3 valence electrons ($3s^2 3p$). It shares 1 electron each with 3 oxygen atoms in $Al_2O_3$. Sharing 3 electrons allows Al to reach a stable octet (8 total electrons).

**Passage-II:**

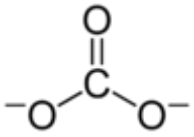
**18. Correct Option: D) 0**

**Explanation**:

Formal Charge = $V - N - \frac{B}{2}$

Where, V = the number of valence electrons present in the neutral atom of the compound. N = the number of non – bonding valence electrons in the same atom.

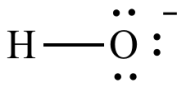
B = the total number of electrons present in the bonds that the atom under consideration forms with the other atoms in the molecule.



Formal Charge of C = $4 - 0 - \frac{8}{2} = 0$

**19. Correct Option: C) -1**

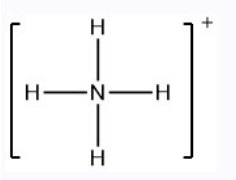
**Explanation**:



Formal Charge of O = $6 - 6 - \frac{2}{2} = -1$

**20. Correct Option: A) 1**

**Explanation**:



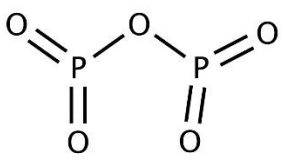
Formal Charge of N = $5 - 0 - \frac{8}{2} = 1$

**INTEGER TYPE**

**21. Correct Option: 5**

**Explanation**:

In the $P_2O_5$ molecule, each P forms 2 double bonds and 1 single bond with O. Each double bond involves the sharing of two electron pairs. And one single bond involves the sharing of 1 electron pair. Valency of P in $P_2O_5 = 2 \times 2 + 1 = 5$



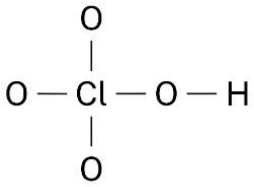
**22. Correct Option: 3**

Explanation:

Formal Charge = V – N - $\frac{B}{2}$

Where, V = the number of valence electrons present in the neutral atom of the compound. N = the number of non – bonding valence electrons in the same atom.

B = the total number of electrons present in the bonds that the atom under consideration forms with the other atoms in the molecule.



Formal Charge of Cl = 7 – 0 – $\frac{8}{2}$ = 3

## Ionic Bond

### LEVEL – I

**1. Correct Option: A) Transfer of electron**

**Explanation**:

Ionic bonds are formed by the complete transfer of electrons between atoms. One atom loses electrons and becomes a positively charged ion (cation), while another atom gains electrons and becomes a negatively charged ion (anion).

B) Sharing of electrons defines covalent bonds, where atoms share electrons to achieve a stable octet configuration.

C) While loss of electrons is involved in ionic bond formation, it's not the complete picture. Both loss and gain of electrons are necessary for the ionic bond to form.

D) Gain of electrons alone can occur in other types of bonding, not just ionic.

**2. Correct Option: B) Electrostatic force**

**Explanation**:

Ionic compounds consist of oppositely charged ions held together by strong electrostatic forces of attraction. These forces arise due to the Coulombic interaction between positive and negative charges.

A) Van der Waals forces are much weaker intermolecular forces that exist between neutral atoms or molecules due to temporary fluctuations in electron distribution. While they exist to some extent in ionic compounds, they are not the primary force holding them together.

C) Hydrogen bonding involves specific interactions between polar molecules with hydrogen atoms bonded to highly electronegative elements like oxygen, nitrogen, or fluorine. It's not relevant in ionic compounds, which lack these functionalities.

D) Repulsive forces occur between like charges and cannot form a stable structure like an ionic compound.

**3. Correct Option: A) Electro negative element**

**Explanation**:

A) Electropositive elements: These tend to lose electrons easily and become positively charged ions (cations). They have low electronegativity, meaning they don't attract electrons strongly.

B) Electronegative elements: These tend to gain electrons readily and become negatively charged ions (anions). They have high electronegativity, meaning they have a strong pull towards electrons.

C) Non-metals: While most non-metals are electronegative and gain electrons, some can exhibit varying behavior depending on the context. For example, hydrogen can sometimes lose an electron, and halogens may share electrons in covalent bonds.

**4. Correct Option: B) Electropositive elements**

**Explanation**:

A) Electropositive elements: These tend to lose electrons easily and become positively charged ions (cations). They have low electronegativity, meaning they don't attract electrons strongly.

B) Electronegative elements: These tend to gain electrons readily and become negatively charged ions (anions). They have high electronegativity, meaning they have a strong pull towards electrons.

C) Non-metals: While most non-metals are electronegative and gain electrons, some can exhibit varying behavior depending on the context. For example, hydrogen can sometimes lose an electron, and halogens may share electrons in covalent bonds

**5. Correct Option: B) loss of 2 electrons**

**Explanation**:

Magnesium (Mg) is an electropositive element, meaning it tends to lose electrons to achieve stability. It has 2 valence electrons in its outermost shell ($3s^2$), and losing these 2 electrons would give it the stable electronic configuration of a noble gas (neon) with a full outer shell (8 electrons).

A) & C) Gaining electrons wouldn't be favorable for Mg as it already has 2 valence electrons and wouldn't achieve a full outer shell with 3 or 4 electrons.

D) Losing 1 electron wouldn't result in a stable configuration either. Mg would have 1 remaining valence electron, leaving it positively charged but not achieving the same stability as losing both and resembling a noble gas.

**6. Correct Option: D) Ionic bond**

**Explanation**:

Potassium (K) is an electropositive element, readily losing its single valence electron to become $K^+$. Chlorine (Cl) is an electronegative element, readily gaining an electron to become $Cl^-$. This complete transfer of electrons from K to Cl defines an ionic bond, not a covalent or dative bond.

**7. Correct Option: C) Electro valency**

**Explanation**:

A. Valency: This refers to the combining power of an atom, indicating how many bonds it can form. It doesn't necessarily specify whether electron transfer or sharing occurs.

B. Covalency: This term describes the formation of covalent bonds through electron sharing, not applicable to ionic bonds.

C. Electro valency: This specifically refers to the number of electrons an atom gains or loses when forming an ionic bond. This transfer of electrons allows each atom to achieve a stable outer shell configuration.

D. Atomic number: This is the number of protons in an atom's nucleus and doesn't change during bond formation.

**8. Correct Option: A) +1**

**Explanation**:

Potassium (K) is an electropositive element with one valence electron in its outermost shell ($4s^1$). In KCl, K atoms readily lose their single valence electron to achieve a stable octet configuration similar to a noble gas (argon) with a full outer shell (8 electrons). Losing one electron results in a positively charged potassium ion ($K^+$).

Electrovalency represents the number of electrons lost or gained. Since K loses one electron, its electrovalency in KCl is +1.

**9. Correct Option: B) K and $O_2$**

**Explanation**:

A. Na and Ca: Both are metals and tend to lose electrons, making them less likely to form an ionic bond where one loses and the other gains.

B. K and $O_2$: Potassium (K) is a highly electropositive metal with a low ionization energy, readily losing one electron. Oxygen (O) is an electronegative non-metal with a high electron affinity, readily gaining two electrons. This electron transfer scenario creates oppositely charged ions, leading to ionic bonding.

C. $O_2$ and $Cl_2$: Both are non-metals, so sharing electrons in a covalent bond is more likely than complete transfer.

D. $Br_2$ and $I_2$: Both are halogens, forming covalent bonds with each other.

**10. Correct Option: B) High charge on ions, large cation and small anion**

**Explanation**:

High charge on ions: Greater charge difference leads to stronger electrostatic attraction, making the ionic bond more stable.

Large cation and small anion: This size difference allows for closer packing of the ions, maximizing electrostatic attraction and minimizing repulsive forces.

**11. Correct Option: B) Non direction**

**Explanation**:

Ionic bonds are formed due to the electrostatic attraction between oppositely charged ions.

This attraction exists in all directions around each ion, not just in specific directions like covalent bonds.

**12. Correct Option: D) They exhibit space isomerism**

**Explanation**:

A. High melting and boiling points: Ionic compounds have strong electrostatic forces between oppositely charged ions, requiring high energy to overcome, leading to high melting and boiling points.

B. Good conductors in molten state: When molten, ionic compounds dissociate into free ions, enabling them to conduct electricity.

C. Exist as solids: Ionic compounds typically form stable crystal lattices due to the strong electrostatic forces, resulting in solid states.

D. Space isomerism: This type of isomerism arises from different spatial arrangements of atoms within a molecule. However, ionic compounds lack individual molecules and instead have repeating crystal structures. The arrangement of ions in these structures is determined by the minimization of repulsion and maximization of attraction, not by specific spatial relationships like in molecules.

**13. Correct Option: B) Fused NaCl**

**Explanation**:

A. Crystalline NaCl: Despite having charged ions arranged in a regular lattice, these ions are fixed in their positions and cannot move freely. This lack of mobile charges prevents conduction in solid NaCl.

B. Fused NaCl: When heated to a high temperature, NaCl melts, forming a molten state. In this state, the ions become free to move, allowing them to carry current and making the solution a good conductor.

C. Solid KCl: Similar to crystalline NaCl, the ions in solid KCl are fixed in their positions, leading to no electrical conductivity.

D. Diamond: Diamond is a covalent network solid where electrons are tightly shared between carbon atoms. This lack of free mobile charges also prevents electrical conduction.

**14. Correct Option: A) Good conductors**

**Explanation**:

Ionic compounds in their solid state are not good conductors of electricity. This is because the ions are fixed in their crystal lattice and cannot move freely. However, when ionic compounds are fused (melted) or dissolved in water (aqueous state), the situation changes. In these states, the ionic compounds dissociate into free ions. These ions are no longer fixed in place and can move freely. The freely moving ions can carry electric current, making the fused or aqueous solution a good conductor.

**15. Correct Option: C) Solution state**

**Explanation**:

Ionic compounds are typically solids in their pure state. In this solid state, the ions are fixed in a crystal lattice and cannot move freely. For a reaction to occur, the reactants need to collide with enough energy (activation energy) for the reaction to proceed. In a solid state, with fixed ions, collisions are less frequent and the activation energy is high, making reactions less likely. However, when ionic compounds are dissolved in water, they dissociate into free ions. These free ions can move around freely in the solution, increasing the frequency of collisions and lowering the activation energy. This makes ionic reactions much more likely to occur in solution

**16. Correct Option: D) RbCl**

**Explanation**:

- Non-directional bonds: These bonds lack a specific directionality and exist equally in all directions around the atom.
- RbCl: This is an ionic compound formed by the complete transfer of electrons from rubidium (Rb) to chlorine (Cl). The resulting oppositely charged ions ($Rb^+$ and $Cl^-$) attract each other due to electrostatic forces, which act in all directions around each ion.
- $NCl_3$, $BeCl_2$, and $BCl_3$: These are covalent compounds where atoms share electrons to achieve stable configurations. The sharing of electrons occurs in specific orbitals and leads to directional bonds with well-defined geometries (trigonal pyramidal for $NCl_3$, linear for $BeCl_2$, and trigonal planar for $BCl_3$).

**17. Correct Option: A) Lattice energy/enthalpy**

**Explanation**:

A. Lattice energy: This term precisely refers to the energy needed to break all the ionic bonds in one mole of a solid ionic compound, separating it into individual gaseous ions.

B. Electrostatic force: While electrostatic forces are the primary attraction between ions in an ionic compound, "electrostatic force" doesn't directly describe the energy required for separation.

C. Nuclear energy: This refers to energy within the nucleus of an atom, unrelated to the separation of ions in a compound.

D. Ionic energy: While this term might seem relevant, it's not a widely used or recognized term for lattice energy.

**18. Correct Option: D) Charge and size of the ions**

**Explanation**:

A. Charge on the ions: Higher charges lead to stronger electrostatic attraction between ions, resulting in higher lattice energy.

B. Size of the ions: Smaller ions can pack closer together, minimizing repulsion and maximizing attraction, leading to higher lattice energy.

C. Packing of the ions: The arrangement of ions in the crystal lattice also impacts the overall attraction and repulsion forces, influencing lattice energy.

**19. Correct Option: C) Oppositely charged ions**

**Explanation**:

Electrovalent compounds are formed by the complete transfer of electrons between atoms. This transfer results in the formation of positively charged cations and negatively charged anions. These oppositely charged ions are attracted to each other by strong electrostatic forces.

**20. Correct Option: D) $Al_2O_3$ > CaO > $MgBr_2$ > NaCl**

**Explanation**:

Lattice energy: Increases with higher charge on ions and smaller ion size.

$Al_2O_3$: Highly charged $Al^{3+}$ cation and small $O^{2-}$ anion lead to the highest lattice energy.

CaO: Smaller $Ca^{2+}$ compared to $Mg^{2+}$ and the same $O^{2-}$ contribute to a higher lattice energy than $MgBr_2$.

$MgBr_2$: Larger $Br^-$ anion reduces the lattice energy compared to CaO.

NaCl: Largest anion ($Cl^-$) among the options results in the lowest lattice energy.

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**LEVEL – II**

**SINGLE OPTION CORRECT TYPE**

**1. Correct Option: B) Fluorine**

**Explanation**:

Potassium is an electropositive element with one valence electron it tends to lose easily. Among the options, fluorine is the most electronegative element, readily gaining an electron to achieve a stable configuration. This strong difference in electronegativity facilitates the complete transfer of the electron from potassium to fluorine, forming an ionic bond. While other halogens (chlorine, bromine, iodine) are also electronegative, their larger sizes compared to fluorine lead to slightly weaker attractions with potassium and potentially some covalent character in the bond.

**2. Correct Option: A) LiCl > $BeCl_2$ > $BCl_3$ > $CCl_4$**

**Explanation:**

Ionic character: Increases with decreasing electronegativity difference between the elements involved.

- LiCl: Li has the lowest electronegativity, leading to the largest difference with Cl and the most ionic character.
- $BeCl_2$: Be is slightly more electronegative than Li, resulting in a slightly smaller difference and less ionic character compared to LiCl.
- $BCl_3$: B is even more electronegative than Be, leading to a further decrease in electronegativity difference and a trend towards more covalent character.
- $CCl_4$: C is the most electronegative among the cations, resulting in the smallest difference with Cl and the least ionic character, bordering on covalent.

**3. Correct Option: D) $CaI_2$**

**Explanation**:

Ionic character decreases with increasing size and decreasing electronegativity of the anion.

Iodine (I) in $CaI_2$ is the largest anion among the options, resulting in weaker electrostatic attraction with the $Ca^{2+}$ cation. The electronegativity of I is also lower compared to F, Br, and Cl. This smaller electronegativity difference reduces the driving force for complete electron transfer, leading to a more covalent character in the bond.

**4. Correct Option: D) Anhydrous and aqueous**

**Explanation**:

- Anhydrous HCl: In this state, HCl exists as a gas. Gases generally lack sufficient mobile ions to conduct electricity effectively. Therefore, anhydrous HCl is a bad conductor.
- Aqueous HCl: When dissolved in water, HCl dissociates into $H^+$ and $Cl^-$ ions. These freely moving ions can carry electric current, making aqueous HCl a good conductor.

**5. Correct Option: A) Li**

**Explanation**:

Electrovalent compounds are formed due to the complete transfer of electrons.

Lithium (Li) has the lowest ionization energy among the options, meaning it requires the least energy to lose its single valence electron. Losing this electron readily gives it a stable configuration similar to helium, making it highly favorable for Li to form ionic bonds.

While Na, Be, and Mg also tend to lose electrons, their higher ionization energies compared to Li make them slightly less inclined to form purely ionic bonds

**6. Correct Option: A) $X^+Y^-$**

**Explanation**:

The compound formed between a strongly electropositive element (X) and a strongly electronegative element (Y) with both being univalent (having a charge of +1 and -1 respectively) would be an ionic compound. In an ionic compound, electrons are transferred from the electropositive element to the electronegative element.

So, the correct representation for the compound formed would be $X^+Y^-$

**7. Correct Option: A) $A_2B_3$**

**Explanation**:

Element A needs to lose 3 electrons to achieve a stable octet configuration (similar to noble gases).

Element B needs to gain 2 electrons to achieve a stable octet configuration.

To minimize the number of ions formed, A loses 3 electrons while B gains 2 electrons per molecule. This leads to the formation of 2 A cations ($A^{2+}$) and 3 B anions ($B^-$) per molecule.

Combining these ions in the simplest ratio gives the formula $A_2B_3$.

**8. Correct Option: A) Ionic**

**Explanation**:

Cations are positively charged ions, and anions are negatively charged ions. These oppositely charged ions are attracted to each other due to electrostatic forces. When these ions pack together in a crystal lattice, they form an ionic compound.

Metallic compounds are composed of metal atoms, covalent compounds involve sharing electrons between atoms, and dipole-dipole interactions occur between neutral molecules.

**ONE OR MORE THAN ONE OPTION CORRECT TYPE**

**9. Correct Option: A & D**

**Explanation**:

Ionic bonds form between atoms when one atom (electron acceptor) gains one or more electrons from another atom (electron donor), resulting in oppositely charged ions attracted to each other by electrostatic forces.

A) High electron affinity: Electron acceptors need high electron affinity to readily attract and stabilize the donated electrons.

B) Less size of anion: Anion size has little bearing on ionic bond formation as long as the charges can interact effectively.

C) Low charge of anion: A low charge on the anion would reduce the electrostatic attraction with the cation, weakening the bond.

D) High electronegativity: High electronegativity reflects a strong attraction for electrons, making the atom a good electron acceptor.

**10. Correct Option: A& B**

**Explanation**:

Electrovalency refers to the number of electrons gained or lost by an atom to form an ion.

In ionic compounds, metals lose electrons (positive electrovalency) and non-metals gain electrons (negative electrovalency).

A. KCl: Potassium (K) loses 1 electron (electrovalency +1), and Chlorine (Cl) gains 1 electron (electrovalency -1).

B. NaCl: Sodium (Na) loses 1 electron (electrovalency +1), and Chlorine (Cl) gains 1 electron (electrovalency -1).

C. MgO: Magnesium (Mg) loses 2 electrons (electrovalency +2), and Oxygen (O) gains 2 electrons (electrovalency -2).

D. CaO: Calcium (Ca) loses 2 electrons (electrovalency +2), and Oxygen (O) gains 2 electrons (electrovalency -2).

**11. Correct Option: A, B, C & D**

**Explanation**:

A. They have high melting point and boiling point: Correct. The strong electrostatic forces between oppositely charged ions require a lot of energy to overcome, resulting in high melting and boiling points.

B. Ionic bonds are non-directional: Correct. Ionic bonds arise from the attraction between opposite charges, not specific orbital overlaps, making them non-directional.

C. Ionic compounds are hard and brittle: Correct. The rigid structure of ionic crystals and strong ionic bonds lead to hardness, but also make them susceptible to shattering rather than bending, making them brittle.

D. Ionic compounds exist as solids: Correct. The strong ionic forces favor tightly packed arrangements, resulting in solid-state structures.

**12. Correct Option: A, B, C, D**

**Explanation**:

Each statement represents the ionization process of a metal atom, where it loses electrons to achieve a stable, filled outer electron shell configuration. This electron loss results in the formation of a positively charged ion (cation).

**Assertion and reason type:**

**13. Correct Option: A)** Both A and R are true, and R is the correct explanation of A
**Explanation**:

- Assertion (A) is true: Lattice enthalpy is the amount of energy released when one mole of an ionic crystalline solid is formed from its constituent gaseous ions. Conversely, it is also the energy required to completely separate one mole of a solid ionic compound into its gaseous ions. A higher value means the ions are bound together tightly, indicating a stronger, more stable ionic compound.
- Reason (R) is true: By definition, a stronger chemical bond requires more energy to overcome or break.
- Is it the correct explanation? Yes. The stability and strength of the ionic compound (Assertion) is directly measured by how much energy is needed to break those bonds (Reason). Therefore, R is the correct explanation of A.

**14. Correct Option: A)** Both A and R are true, and R is the correct explanation of A
**Explanation**:

- Assertion (A) is true: Ionic compounds, such as sodium chloride (NaCl), consistently exhibit very high melting and boiling points.
- Reason (R) is true: Inside an ionic crystal lattice, millions of positively charged cations and negatively charged anions are locked in a rigid three-dimensional arrangement, held together by incredibly powerful, non-directional electrostatic forces of attraction (Coulombic forces).
- Is it the correct explanation? Yes. To melt a solid ionic compound, you must provide enough thermal energy to overcome these strong forces and allow the ions to move freely as a liquid. Because these attractive forces (Reason) are so strong, a vast amount of heat energy is required to disrupt the lattice, which directly results in a high melting point (Assertion).

**Matching:**

**15**. **P → a, Q → b, R → c, S → d**

**P. Ionic bond → a. Electrostatic attraction**

- **Explanation:** An ionic bond is formed by the complete transfer of electrons from a metal to a non-metal, resulting in the creation of oppositely charged ions. These ions are held together firmly by strong **electrostatic forces of attraction**.

**Q. Lattice enthalpy → b. Energy required to separate ions**

- **Explanation:** Lattice enthalpy is defined as the amount of **energy required to completely separate** one mole of a solid ionic compound into its constituent gaseous ions.

**R. Cation → c. Positively charged ion**

- **Explanation:** When an atom loses one or more electrons, it gains a net positive charge. This **positively charged ion** is called a cation.

**S. Anion → d. Negatively charged ion**

- **Explanation:** When an atom gains one or more electrons, it acquires a net negative charge. This **negatively charged ion** is called an anion.

**COMPREHENSION TYPE**

**Passage – I:**

**16. Correct Option: D) $X_2Y$**
**Explanation**:
Since element X is univalent (loses one electron) and element Y is bivalent (gains two electrons), two atoms of X will combine with one atom of Y to balance the charges.

This gives us the formula $X_2Y$, where the subscripts indicate the number of atoms of each element present in the compound.

**17. Correct Option: C) 35**

**Explanation**:

A) 10: This is neon (Ne), a noble gas with a full outer shell and very low electronegativity. It won't readily accept an electron from sodium, making an ionic bond unlikely.

B) 34: This is selenium (Se), which has some electronegativity but not enough to form a strong ionic bond with sodium. It's more likely to form a covalent bond.

C) 35: This is chlorine (Cl), which has a high electronegativity and readily accepts an electron from sodium. This forms the strong ionic compound sodium chloride (NaCl).

D) 37: This is rubidium (Rb), another metal with lower electronegativity than chlorine. While it can form an ionic compound with sodium, it wouldn't be as strong as with chlorine.

**18. Correct Option: C) Higher electronegativity**

**Explanation**:

In an ionic bond, one atom loses electrons and becomes a positively charged cation. This atom typically has lower electronegativity as it readily gives up electrons.

The other atom gains electrons and becomes a negatively charged anion. This atom has higher electronegativity as it attracts and holds electrons strongly.

**Passage – II:**

**19. Correct Option: C) II < IV < III < I**

**Explanation**:

The ionic character of a bond depends on the electronegativity difference between the participating atoms. Higher the difference, greater the ionic character.

- LiF: Lithium (Li) has low electronegativity (0.98), while Fluorine (F) has the highest (3.98). This large difference (3) leads to significant electron transfer, making LiF highly ionic.
- $ClF_3$: Chlorine (Cl) has moderate electronegativity (3.19), but the three Fluorines still create a considerable difference (around 2.81). $ClF_3$ is relatively ionic but less than LiF.
- $K_2O$: Potassium (K) has low electronegativity (0.82), and Oxygen (O) has moderate (3.44). The difference (2.62) suggests moderate ionic character.
- $SO_2$: Sulfur (S) has moderate electronegativity (2.58), and Oxygen (O) also has moderate (3.44). The smaller difference (around 0.86) makes $SO_2$ less ionic than the others.

**20. Correct Option: D) II, III and IV**

**Explanation**:

I) AgBr: Silver bromide (AgBr) is generally considered insoluble in water due to its strong ionic bond and high lattice energy.

II) $BaSO_4$: Barium sulfate ($BaSO_4$) is also highly insoluble in water due to its strong ionic bond and large size of the sulfate ion.

III) KOH: Potassium hydroxide (KOH) is a highly soluble ionic compound that readily dissolves in water to form potassium and hydroxide ions.

IV) $Ca_3(PO_4)_2$: Calcium phosphate ($Ca_3(PO_4)_2$) has low solubility in water due to its strong ionic bonds and large size of the phosphate ion.

**21. Correct Option: D) Both (b) and (c)**

**Explanation**:

For solubility in water, the trend should be decreasing as we move down a group (from Ca to Sr to B) or across a period (from F to Cl to Br). This is because the larger size of the ions or increased attraction between the ions and the smaller radius of the $F^-$ ion make it harder for the ionic compound to dissolve in water.

The correct order for solubility in water is:

$BF_2 > SrF_2 > CaF_2$

KF > NaF > LiF

**INTEGER TYPE**

**22. Correct Option: 4**

**Explanation**:

In ionic bonds, the maximum electrovalence is determined by the number of valence electrons the atom can lose to achieve a stable, noble gas configuration. While some elements like aluminum can lose 3 electrons, the maximum electrovalence is typically +4, achieved by elements like beryllium and titanium.

**23. Correct Option: 1**

**Explanation**:

Chlorine initially has 7 valence electrons (in its outer shell). For stability, it aims to achieve a full outer shell like a noble gas, which typically requires 8 electrons. In NaCl, sodium loses 1 electron, which is then gained by chlorine, resulting in both sodium ($Na^+$) and chlorine ($Cl^-$) having stable 8-electron outer shells.

## Covalent Bond and Coordinate Covalent Bond

**LEVEL – I**

**1. Correct Option: B) Covalent bond**

**Explanation**:

A) Ionic bond: Ionic bonds involve the complete transfer of electrons from one atom to another, not equal sharing.

B) Covalent bond: Covalent bonds are formed by the sharing of electron pairs between atoms, usually resulting in a stable electron configuration for both.

C) Dative bond: This is a type of covalent bond where one atom donates both electrons in the shared pair. While related, it's not the general definition of a bond formed by equal sharing.

D) Coordinate covalent bond: This is another term for a dative bond.

**2. Correct Option: C) $H_2O$**

**Explanation:**

NaCl, KCl, and $MgCl_2$: These compounds are all ionic compounds. They form through the transfer of electrons, not sharing, resulting in positively charged cations ($Na^+$, $K^+$, $Mg^{2+}$) and negatively charged anions ($Cl^-$).

$H_2O$: This is a covalent compound. Each hydrogen atom shares one electron with the oxygen atom, forming a stable configuration for all three atoms.

**3. Correct Option: D) The number of electrons shared by an atom.**

**Explanation**:

A) The number of electrons are accepted by an atom: This describes electronegativity, not covalency.

B) The number of electrons are donated by an atom: This applies to dative covalent bonds, but not the broader concept of covalency.

C) The number of electrons transferred by an atom: This definition applies to ionic bonds, not covalent bonds.

D) The number of electrons shared by an atom: This accurately describes the essence of covalency, where atoms pool their electrons to form stable configurations.

**4. Correct Option: C) 4**

**Explanation**:

Covalent is defined as the number of bonds an element forms with other elements in a molecule.

In a Carbon dioxide molecule, each Oxygen shares electrons with carbon, forming two bonds each with Carbon. As the total number of bonds formed with carbon is , hence, the covalency of carbon in carbon dioxide is 4.

**5. Correct Option: C) 2**

**Explanation**:

An oxygen molecule ($O_2$) consists of two oxygen atoms joined by a double bond. A double bond is a type of covalent bond where two pairs of electrons are shared between the atoms.

**6. Correct Option: B) 3**

**Explanation**:

Nitrogen molecules ($N_2$) are held together by a triple covalent bond. This means that each nitrogen atom shares three pairs of electrons with the other nitrogen atom.

**7. Correct Option: C) 3**

**Explanation**:

Elements in Group IVA (also known as Group 14) have 4 valence electrons in their outermost shell. To achieve a stable octet configuration (8 valence electrons), they can share electrons with other atoms. However, sharing electrons comes with a stability limit. Sharing more than 3 electrons with another atom can lead to repulsion between the electron clouds, making the bond unstable. Therefore, for two IVA group elements to bond with each other, the maximum number of shared electron pairs is 3, resulting in a triple bond.

**8. Correct Option: D) F-F**

**Explanation**:

A nonpolar covalent bond occurs when the electronegativity difference between the two atoms is very small or nonexistent. In this case, the electrons are shared equally between the atoms.

Among the options provided, the bond between two fluorine atoms (F-F) is nonpolar because fluorine atoms have similar electronegativities, so the electron pair is shared equally.

**9. Correct Option: B) Low melting point**

**Explanation**:

A) High solubility in water: This depends on the polarity of the covalent compound. Polar covalent compounds generally dissolve in water, while nonpolar ones do not.

B) Low melting point: Covalent bonds are weaker than ionic bonds due to the sharing of electrons instead of complete transfer. This results in lower melting and boiling points for covalent compounds.

C) High electrical conductivity: Covalent compounds generally do not have free ions and are poor conductors of electricity.

D) High boiling point: Similar to melting point, covalent compounds tend to have lower boiling points due to weaker bonds compared to ionic compounds.

**10. Correct Option: C) They show ionic reactions**

**Explanation**:

A) They may exhibit isomerism: Isomerism is the phenomenon where different molecules have the same chemical formula but different structures. This can occur in covalent compounds due to the way the atoms are arranged.

B) They have low melting and boiling points: As mentioned earlier, covalent bonds are weaker than ionic bonds, leading to lower melting and boiling points for covalent compounds.

D) They show molecular reactions: Covalent compounds react by breaking and forming new covalent bonds between molecules. This is known as a molecular reaction.

**11. Correct Option: B) Absence of free electrons**

**Explanation**:

Diamond is a bad conductor of electricity because it lacks free electrons. In diamond, each carbon atom forms strong covalent bonds with four neighboring carbon atoms, creating a rigid three-dimensional network structure known as a tetrahedral arrangement. This arrangement locks the valence electrons in place, preventing them from moving freely and conducting electricity.

**12. Correct Option: A) Small cation and large anion**

**Explanation**:

Fajans' rule: ionic bonds favor small cations and large anions. This maximizes charge density and polarization, strengthening electrostatic attraction and electron transfer.

B) Small cation and small anion: Repulsion

C) Large cation and large anion: Weaker attraction

D) Large cation and small anion: Less effective electron transfer

**13. Correct Option: B) $SiCl_4$**

**Explanation**:

Factors affecting covalent character:

1. Electronegativity difference: Smaller difference indicates more sharing of electrons, leading to more covalent character.
2. Size of cation: Smaller cation leads to higher charge density and greater polarization of the anion, increasing covalent character.

A. NaCl: High electronegativity difference and large $Na^+$ cation suggest strong ionic character.

B. $SiCl_4$: Smaller electronegativity difference and smaller $Si^{4+}$ cation compared to $Na^+$ imply more covalent character.

C. $AlCl_3$: Smaller electronegativity difference than NaCl but larger $Al^{3+}$ cation than $Si^{4+}$ leads to less covalent character than $SiCl_4$.

D. $AgCl_2$: High electronegativity difference and large $Ag^+$ cation indicate strong ionic character.

**14. Correct Option: A) $KCl < CaCl_2 < AlCl_3 < SnCl_4$**

**Explanation**:

Covalent character generally increases with:

1. Decreasing electronegativity difference: Smaller difference between the electronegativity of the cation and anion leads to more sharing of electrons and a more covalent bond.
2. Smaller cation size: Smaller cation creates a more concentrated positive charge, attracting the shared electrons more strongly and increasing covalent character.

**15. Correct Option: A) $Ni^{3+}$**

**Explanation**:

Polarizing power of an ion is determined by its charge-to-radius ratio. Higher the ratio, greater the ability of the ion to distort the electron cloud of surrounding anions, leading to greater polarizability.

**16. Correct Option: A) Maximum polarization is done by a cation of high charge.**

**Explanation**:

Maximum polarization is done by a cation of high charge

According to Fajans rules, polarization will be favoured by small size of cation, large size of anion and high charge of cation and anion.

**17. Correct Option: C) $PbCl_2$**

**Explanation**:

Factors affecting covalent character:

1. Electronegativity difference: Smaller difference indicates more sharing of electrons and a more covalent bond.
2. Size of cation: Smaller cation creates a more concentrated positive charge, attracting the shared electrons more strongly and increasing covalent character.

**18. Correct Option: C) Coordinate covalent bond**

**Explanation**:

A. Ionic bond: This type of bond involves the complete transfer of electrons from one atom to another, not just sharing.

B. Covalent bond: This is the sharing of electrons equally between two atoms.

C. Metallic bond: This is a type of bonding where electrons are delocalized across a metal lattice, not shared between specific atoms.

D. Coordinate covalent bond (dative covalent bond): This type of bond involves the sharing of an electron pair donated by only one atom, with the other atom providing an empty orbital.

**19. Correct Option: D) Coordinate Covalent bond**

**Explanation**:

- $NH_3$ (ammonia) acts as a Lewis base: It has a lone pair of electrons on the nitrogen atom that it can donate.
- $BF_3$ (boron trifluoride) acts as a Lewis acid: It has an empty orbital on the boron atom that can accept electron density.

In the formation of their adduct, $NH_3$ donates its lone pair to the empty orbital of $BF_3$, resulting in a coordinate covalent bond. This bond forms readily due to the strong attraction between the electron-rich $NH_3$ and the electron-deficient $BF_3$.

**20. Correct Option: B) Electrons of the atom are shared between two atoms**

**Explanation**:

A) Electrons are equally shared by the atoms: This is a characteristic of a regular covalent bond, not a coordinate covalent bond.

C) Hydrogen bond is formed: Hydrogen bonds can be present alongside coordinate covalent bonds, but they are not directly related to the electron sharing mechanism.

D) Electrons are not shared by the atoms: This is incorrect. The definition of a coordinate covalent bond involves the sharing of an electron pair, even though both electrons in the pair come from one atom.

**LEVEL – II**

**SINGLE OPTION CORRECT TYPE**

**1. Correct Option: D) $BeCl_2$**

**Explanation**:

- Electronegativity difference: Smaller difference indicates less complete electron transfer, leading to more covalent character. Be has high electronegativity (1.57), closer to Cl (3.16) compared to Mg, Sr, and S.
- Cation size: Smaller cations have higher effective nuclear charge, attracting electrons more tightly and reducing ionic character. $Be^{2}+$ is the smallest cation among the options.

Therefore, the combination of smaller electronegativity difference and smaller cation size makes $BeCl_2$ the least ionic among the given compounds.

**2. Correct Option: D) $CaI_2$**

**Explanation:**

As per Fajan's rule as the size of anion increases covalent character increase.

Order of anions size: $F^- < Cl^- < Br^- < I^-$

So, order of covalent character is $CaF_2 < CaCl_2 < CaBr_2 < CaI_2$

**3. Correct Option: B) Fluorine**

**Explanation**:

As per Fajan's rule as the size of anion increases covalent character increase. Since fluorine is smaller among all, so the answer is fluorine.

**4. Correct Option: D) MgS**

**Explanation**:

Factors affecting solubility:

1. Hydration energy: This is the energy released when ions are surrounded by water molecules. Larger ions interact more favourably with water, leading to higher hydration energy and generally better solubility.
2. Lattice energy: This is the energy required to break apart the crystal lattice of a solid ionic compound. It depends on the charges of the ions and the distance between them. Higher charges and smaller distances lead to higher lattice energy and lower solubility.

A. NaCl: $Na^+$ is larger than $Mg^{2+}$, and $Cl^-$ is smaller than $S^{2-}$. This gives $Na^+$ a potential advantage in hydration energy but also means a tighter lattice in NaCl due to smaller ions.

B. $Na_2S$: Both $Na^+$ and $S^{2-}$ are larger than $Cl^-$, potentially favoring hydration energy, but the lattice has two $S^{2-}$ interacting with smaller $Na^+$, again suggesting a tighter structure.

C. $MgCl_2$: $Mg^{2+}$ is smaller than $Na^+$, which could hinder its hydration. $Cl^-$ is smaller than $S^{2-}$, potentially contributing to a tighter lattice.

D. MgS: Both $Mg^{2+}$ and $S^{2-}$ are larger ions, which could improve hydration energy. However, the interaction between a small $Mg^{2+}$ and a large $S^{2-}$ in the lattice can still be strong, potentially leading to higher lattice energy.

**5. Correct Option: D) 4**

**Explanation**:

Covalency refers to the number of covalent bonds an atom forms with other atoms to achieve a stable electron configuration.

**6. Correct Option: A) Covalent bond**

**Explanation**:

When two non-metals come together, they are more likely to share electrons rather than transfer them completely. This sharing of electrons results in a covalent bond.

B) Ionic bond: Ionic bonds typically form between a metal and a non-metal, involving the complete transfer of electrons from the metal to the non-metal.

C) Dative bond: While a type of covalent bond, it's a specific scenario where one atom donates both electrons in the shared pair. It's not the general term for bonds between non-metals.

D) Metallic bond: Metallic bonds form between metal atoms through a delocalized sea of electrons, not between individual atoms like in the case of non-metals.

**ONE OR MORE THAN ONE OPTION CORRECT TYPE**

**7. Correct Option: A, B, D**

**Explanation**:

A & B: Smaller $Li^+$ ion creates stronger polarization of $Cl^-$ leading to weaker ionic bonds in LiCl, resulting in lower melting point and higher solubility in organic solvents.

C: Smaller size and higher charge density of $Li^+$ make LiCl slightly more ionic than NaCl. However, the difference is minor.

D: Ionic conductivity relies on mobile ions. Smaller and more tightly packed $Li^+$ ions in LiCl make it less conductive than NaCl with larger and more loosely packed $Na^+$ ions.

**8. Correct Option: A, D**

**Explanation**:

A. $H_3O^+$ (hydronium ion): In this ion, a water molecule ($H_2O$) donates a lone pair of electrons from its oxygen atom to an empty orbital on a hydrogen ion ($H^+$), forming a coordinate covalent bond between the oxygen and the hydrogen.

B. $BF_3$ (boron trifluoride): This molecule does not contain any coordinate covalent bonds. It has three covalent bonds between boron and fluorine atoms.

C. $H_2O$ (water): This molecule only has regular covalent bonds between hydrogen and oxygen atoms.

D. $NH_4^+$ (ammonium ion): An ammonia molecule ($NH_3$) donates a lone pair of electrons from its nitrogen atom to an empty orbital on a hydrogen ion ($H^+$), forming a coordinate covalent bond between the nitrogen and the hydrogen.

**9. Correct Option: C) Lewis base**

**Explanation**:

A. Lewis acids: These are electron pair acceptors and tend to have empty orbitals. They do not donate electrons.

B. Donors: This is a generic term that can apply to any substance that donates something, including electrons. However, it doesn't specifically refer to electron pair donors in the context of Lewis theory.

C. Lewis bases: These are electron pair donors and have lone pairs of electrons or negative charges, making them capable of donating electrons to Lewis acids to form coordinate covalent bonds.

D. Acceptor: This term is similar to Lewis acid, but it's not as specific in the context of Lewis theory.

**10. Correct Option: A, B**

**Explanation**:

A. Ionic bond: The $NH_4^+$ (ammonium) ion and the $Cl^-$ (chloride) ion are held together by an ionic bond.

B. Covalent bonds: Within the $NH_4^+$ ion, covalent bonds exist between the nitrogen atom and each of the four hydrogen atoms. These bonds involve the sharing of electrons between the atoms.

**11. Correct Option: A, D**

**Explanation**:

A nonpolar bond is a bond in which the electrons are shared equally between the two atoms. This results in no positive or negative end to the bond.

Electronegativity is a measure of an atom's ability to attract electrons in a bond. The greater the difference in electronegativity between two atoms, the more polar the bond will be.

- $CCl_4$ All four atoms are identical (carbon and chlorine), so there is no difference in electronegativity and the electrons are shared equally.
- $P_4$ All four atoms are identical (phosphorus), so there is no difference in electronegativity and the electrons are shared equally.

**12. Correct Option: A) Lewis acid**

**Explanation**:

A. Lewis acids: These are electron pair acceptors and possess empty orbitals to accommodate electron pairs donated by Lewis bases. They do not donate electrons themselves.

B. Donar: This is a general term for any substance that gives something, including electrons. But it's not specific to the context of Lewis theory and electron pair donation.

C. Lewis base: These are electron pair donors with lone pairs of electrons or negative charges, capable of donating electrons to Lewis acids to form coordinate covalent bonds.

D. Acceptor: Similar to Lewis acid, but less specific within Lewis theory.

**13. Correct Option: D) Both b and C**

**Explanation**:

B) The ligand has no lone pair of electrons hence it is called a Lewis base: In a coordinate covalent bond, the ligand is the electron donor and must have a lone pair to donate. It cannot be a Lewis base without a lone pair.

C) The metal ion has empty orbitals hence it accepts a lone pair of electrons from the ligand: While some Lewis acids do have empty orbitals, it's not a strict requirement for accepting an electron pair. Lewis acids can also accept electrons into higher-energy orbitals or delocalize the donated electrons throughout their structure.

**14. Correct Answer: A)** Both A and R are true and R is the correct explanation of A**.**

**Explanation:**

- **Assertion is true:**
  **$AlCl_3$** has a much lower melting point (about **192.4°C**, sublimes) than **NaCl** (**801°C**).
- **Reason is true:**
  According to **Fajans' rules**, **$Al^{3+}$** is a small, highly charged cation with strong polarizing power. It distorts the electron cloud of **$Cl^-$**, making the **Al–Cl bond significantly covalent**.

- Since **covalent compounds** have **weak intermolecular forces** compared to the strong electrostatic forces in ionic compounds, they generally have **lower melting points**.

  Therefore, the reason correctly explains the assertion.

**15. Correct Answer: A)** Both A and R are true and R is the correct explanation of A.

**Explanation:**

- **Assertion is true:**

  Most covalent compounds (e.g., $CO_2$, $CH_4$, $Cl_2$) have **low melting and boiling points**.

- **Reason is true:**

  Although atoms within a covalent molecule are held together by **strong covalent bonds**, the **forces between molecules (Van der Waals forces)** are weak.

- Only a small amount of energy is needed to overcome these intermolecular forces, so these compounds melt and boil at relatively low temperatures.

  Hence, the reason correctly explains the assertion.

**MATRIX MATCHING TYPE**

**16. Correct Option: a)-Q), b)-P), c)-R), d)-S)**

**Explanation**:

a) Diamond is a pure covalent network solid where carbon atoms share electrons to form strong covalent bonds.

b) Table salt (NaCl) is an ionic compound where $Na^+$ and $Cl^-$ ions are held together by strong electrostatic forces due to complete electron transfer.

c) Ammonium ion is a polyatomic ion where N forms covalent bonds with H atoms. Additionally, N donates a lone pair to an $H^+$ ion, forming a dative covalent bond.

d) Ammonium chloride combines characteristics of the individual ions: ionic bond between $NH_4^+$ and $Cl^-$, covalent bonds within $NH_4^+$, and a dative bond between N and $H^+$.

**17. Correct Option: a)-Q), b)-S), c)-P), d)-R)**

**Explanation**:

a) $AlF_3$ is an ionic compound due to the high electronegativity difference between Al and F, resulting in complete electron transfer from Al to F and electrostatic attraction between ions.

b) $CCl_4$ is a covalent molecule as all atoms share electrons. Its tetrahedral structure and weak intermolecular forces lead to a liquid state at room temperature.

c) Graphite is a layered structure with weak van der Waals forces between sheets, making it slippery and a good lubricant.

d) $BCl_3$ has an empty orbital and can accept an electron pair from a Lewis base, acting as a Lewis acid in various reactions.

**COMPREHENSION TYPE**

**Passage – I:**

**18. Correct Option: B) HF > $H_2O$ > $SO_2$ > $NH_3$**

**Explanation**:

Polarity is a measure of the separation of electric charge within a molecule. A molecule is polar if it has a positive end and a negative end. The polarity of a molecule is determined by the difference in electronegativity between the atoms in the molecule. Electronegativity is a measure of an atom's ability to attract electrons towards itself in a bond.

The greater the difference in electronegativity between two atoms, the more polar the bond between them will be. And the more polar bonds a molecule has, the more polar the molecule will be.

In the given options, HF has the largest electronegativity difference, followed by $H_2O$, $SO_2$, and $NH_3$. Therefore, the correct order of decreasing polarity is HF > $H_2O$ > $SO_2$ > $NH_3$.

**19. Correct Option: D) Fajan's rule**

Explanation:

Fajan's rule states that smaller cations and larger anions favor covalent character due to increased polarization of the anion by the cation.

- In $AlCl_3$, the $Cl^-$ anion is larger than the $F^-$ anion in $AlF_3$. This makes the electron cloud of $Cl^-$ easier to polarize by the smaller $Al^{3+}$ cation in $AlCl_3$, leading to more covalent character.
- $AlF_3$ has both a smaller anion ($F^-$) and a smaller cation ($Al^{3+}$) compared to $AlCl_3$. This minimizes polarization and strengthens the ionic character of the bond.

**20. Correct Option: A) $AlF_3$**

Explanation:

- Electronegativity difference: Fluorine has the highest electronegativity among halogens, leading to the largest electronegativity difference with aluminium in $AlF_3$, promoting an ionic bond.
- Cation size: While all halides have larger anions than the $Al^{3+}$ cation, fluorine, being the smallest halogen, minimizes potential polarization effect in $AlF_3$, further strengthening its ionic character.

**INTEGER TYPE**

**21. Correct Option: 2**

**Explanation**:

In hydrogen peroxide, each oxygen atom is bonded to one hydrogen atom on one side and to another oxygen atom on the other side. Each bond between oxygen and hydrogen involves the sharing of one electron pair, meaning each oxygen atom participates in two covalent bonds. Since each oxygen atom has six valence electrons, sharing one electron pair in each bond fulfils the octet rule for both oxygen atoms, achieving stability

**22. Correct Option: 3**

**Explanation**:

Each nitrogen atom has five valence electrons in its outer shell. To achieve the stable octet configuration (eight electrons in the outer shell), each nitrogen atom shares three electrons with the other nitrogen atom. Each shared pair of electrons forms a covalent bond, resulting in a triple bond (N≡N).

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## Valence Bond Theory (VBT)

**LEVEL – I**

**1. Correct Option: B) Decreases the bond length**

**Explanation**:

When atoms form bonds, they share electrons in their outermost orbitals. This sharing attracts the positively charged nuclei of the atoms closer together. This attraction overcomes the opposing force of electron-electron repulsion between the shared electrons and the inner electrons of the atoms. As a result, the internuclear distance (bond length) decreases compared to the distance between the isolated atoms.

**2. Correct Option: A) P and P orbitals**

**Explanation**:

Stronger bonds result from greater overlap between the orbitals of bonding atoms. P orbitals have more complex shapes than S orbitals, allowing for more extensive overlap and stronger interactions. When P orbitals overlap end-to-end (head-on), the overlap is maximized, leading to a particularly strong bond.

**3. Correct Option: B) Orbitals overlap will not occur**

**Explanation**:

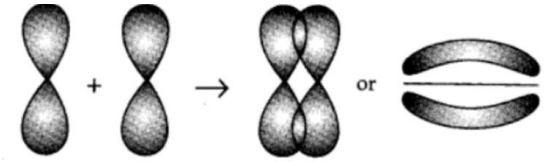
The Pauli Exclusion Principle states that two electrons within an atom cannot occupy the same quantum state, which includes having the same spin. In other words, electrons with the same spin cannot pair up.

For a diatomic molecule to form, orbitals containing unpaired electrons on individual atoms must overlap, allowing sharing of electron density and bonding. Since both atoms have electrons with the same spin, there are no unpaired electrons available for overlap, thus preventing bond formation.

**4. Correct Option: D) P-P orbitals side wise manner**

**Explanation**:

- Pi (π) bonds: Formed by the sideways or lateral overlapping of p orbitals. The p orbitals don't overlap end-to-end, but their lobes interact sideways, creating electron density above and below the plane of the atoms. This characterizes the pi bond.




- Sigma (σ) bonds: Formed by the axial or end-to-end overlapping of atomic orbitals. This overlap can occur between s-s orbitals, s-p orbitals, or p-p orbitals (where they overlap directly along their internuclear axis in end-to-end manner). This head-on overlap creates a cylindrical electron density distribution around the bond axis, which defines the sigma bond.

**5. Correct Option: B) sp - sp**

**Explanation**:

As we know that hybrid orbital always gives better overlap than pure orbitals because of the same size and equivalent energy. Out of various hybrid orbitals above the most efficient overlapping is shown by the sp-sp hybrid orbitals as there are 50% s and 50% p.

Since greater the s-character, smaller is the hybridized orbital and more is the tendency for overlapping. Also, more is the bond energy, smaller is the bond length.

**6. Correct Option: B) $BF_3$**

**Explanation**:

A. $BeCl_2$ (sp): Only 2 valence electrons, forms 2 bonds using unhybridized orbitals.

B. $BF_3$ ($sp^2$): 3 valence electrons, forms 3 bonds by combining 1s and 2p orbitals into trigonal planar $sp^2$ hybrids.

C. $NH_3$ ($sp^3$): 5 valence electrons, forms 3 bonds and has 1 lone pair, achieved by combining 1s and 3p orbitals into tetrahedral $sp^3$ hybrids.

D. $XeF_3$ ($sp^3d$): 8 valence electrons, forms 3 bonds and has 2 lone pairs, requiring the involvement of 1s, 3p, and 1d orbital in $sp^3d$ hybridization.

**7. Correct Option: D) $BF_3$**

**Explanation**:

$IF_3$ has bent-T shape

$PCl_3$ and $NH_3$ has pyramidal geometry

$BF_3$ has trigonal planar geometry. $BF_3$ does not have any lone pair, the three F atoms are arranged around central Boron at a bond angle of $120^o$

**8. Correct Option: B) Trigonal planar, $sp^2$**

**Explanation**:

$BF_3$ has trigonal planar geometry. $BF_3$ does not have any lone pair, the three F atoms are arranged around central Boron at a bond angle of $120^o$

$BF_3$ - Boron trifluoride ($BF_3$) undergoes $sp^2$ hybridization. Boron (B) has three valence electrons, and in $BF_3$, it forms three sigma bonds with three fluorine (F) atoms. To accommodate the three sigma bonds, boron undergoes $sp^2$ hybridization.

**9. Correct Option: B) $BF_3$, $AlF_3$**

Explanation:

A) $BF_3$, $NF_3$: Although both have three central atoms bonded to three other atoms, $NF_3$ has a pyramidal geometry due to the presence of a lone pair on the nitrogen atom.

B) $BF_3$, $AlF_3$: Both $BF_3$ and $AlF_3$ have three electron domains (three bonding atoms) around the central atom, leading to the same trigonal planar geometry without any lone pairs influencing the molecular shape.

C) $BeF_2$, $H_2O$: $BeF_2$ has a linear geometry with two fluorine atoms bonded to beryllium in a straight line. $H_2O$ has a bent or V-shaped geometry due to the presence of two lone pairs on the oxygen atom.

D) $BCl_3$, $PCl_3$: Both have a trigonal pyramidal geometry, but not the same as the trigonal planar geometry of $BF_3$ and $AlF_3$. This is because they have a lone pair on the central atom (B or P).

**10. Correct Option: A) 25% s- character**

**Explanation**:

$Sp^3$ hybridization involves combining one s orbital and three p orbitals to form four $sp^3$ hybrid orbitals. Each $sp^3$ hybrid orbital contains:

- 25% s character: This comes from the contribution of the single s orbital.
- 75% p character: This comes from the contribution of the three p orbitals.

**11. Correct Option: B) $SF_4$**

Explanation:

A. $SO_4^{2-}$: Sulphur in $SO_4^{2-}$ has four single bonds and one negative charge, indicating four electron domains (bonding atoms and lone pairs). This necessitates $sp^3$ hybridization for optimal orbital overlap and bonding.

B. $SF_4$: In $SF_4$, sulphur forms four single bonds with fluorine atoms. However, it also has one lone pair of electrons. Due to the presence of this lone pair, sulphur undergoes $sp^3d$ hybridization to accommodate the five electron domains (four bonds and one lone pair).

C. $SF_2$: Sulphur in $SF_2$ has two single bonds and two lone pairs, resulting in four electron domains. Therefore, $sp^3$ hybridization allows for efficient bonding with the fluorine atoms.

D. $SCl_2$: Similar to $SF_2$, sulphur in $SCl_2$ forms two single bonds and has two lone pairs, requiring $sp^3$ hybridization for the four electron domains.

**12. Correct Option: A) $NH_3$**

**Explanation**:

A. $NH_3$ (ammonia): Nitrogen has 5 valence electrons, forming 3 bonds with H atoms and having 1 lone pair. This leads to 4 electron domains (3 bonds + 1 lone pair) and necessitates $sp^3$ hybridization, resulting in a trigonal pyramidal shape.

B. $SiF_4$ (silicon tetrafluoride): Silicon has 4 valence electrons, forming 4 bonds with F atoms. This results in 4 electron domains (all bonds) and $sp^3$ hybridization, leading to a tetrahedral shape.

C. $H_2O$ (water): Oxygen has 6 valence electrons, forming 2 bonds with H atoms and having 2 lone pairs. This creates 4 electron domains (2 bonds + 2 lone pairs) and necessitates $sp^3$ hybridization, leading to a bent or V-shaped geometry.

D. $BF_3$ (boron trifluoride): Boron has 3 valence electrons, forming 3 bonds with F atoms. This results in 3 electron domains (all bonds) and $sp^2$ hybridization, leading to a trigonal planar shape.

**13. Correct Option: B) $sp^3 > sp^2 > sp$**

**Explanation**:

Hybrid orbital size is primarily determined by the contribution of s and p orbitals:

- s orbitals are smaller and more spherical than p orbitals.
- As the s character in a hybrid orbital increases, its size decreases.

In this specific case:

- $sp^3$ orbitals: Contain 25% s character and 75% p character. They are the largest due to the lowest s character contribution.
- $sp^2$ orbitals: Contain 33.3% s character and 66.6% p character. They are smaller than $sp^3$ orbitals due to the higher s character.
- sp orbitals: Contain 50% s character and 50% p character. They are the smallest due to the highest s character contribution.

**14. Correct Option: D) Increases progressively**

**Explanation**:

As hybridization changes, the shapes of the hybrid orbitals and the angles between them also change.

- $Sp^3$: This has tetrahedral geometry with four hybrid orbitals arranged at 109.5° angles for maximum orbital overlap and minimum repulsion.
- $Sp^2$: This has trigonal planar geometry with three hybrid orbitals arranged at 120° angles, a bit closer due to the influence of more directional p orbitals.
- Sp: This has linear geometry with two hybrid orbitals aligned at 180°, closest due to the dominance of p orbitals with minimal s character.

**15. Correct Option: A) $sp^3d$**

**Explanation**:

The number of sigma bonds formed by the phosphorus atom is five and no lone pair is attached to the central atom so the steric number of $PCl_5$ is 5

For steric number 5, the type of hybridization is $sp^3d$

**16. Correct Option: B) Octahedral**

**Explanation**:

- $sp^3d^2$ hybridization: This combines one s orbital, three p orbitals, and two d orbitals to form six hybrid orbitals.
- These six hybrid orbitals are arranged in an octahedral geometry, with bond angles of 90° between each orbital.
- Compounds like $SF_6$ and $XeF_6$ exhibit $sp^3d^2$ hybridization and an octahedral shape.

**17. Correct Option: A) $sp^3d^2$ hybridization**

**Explanation**:

A. $sp^3d^2$ hybridization: Involves six hybrid orbitals arranged in an octahedral geometry. This arrangement leads to bond angles of 90° between each pair of bonding orbitals.

B. $sp^3d$ hybridization: has four hybrid orbitals, but they form a trigonal prism geometry, resulting in bond angles of less than 90°.

C. $dsp^2$ hybridization: Has four hybrid orbitals in a trigonal planar geometry, leading to bond angles of 120°.

D. $dsp^3$ hybridization: Has five hybrid orbitals in a trigonal bipyramidal geometry, with bond angles ranging from 90° to 120°.

**18. Correct Option: D) Octahedral**

**Explanation**:

Hybridization = $\frac{1}{2}(s + v - q)$

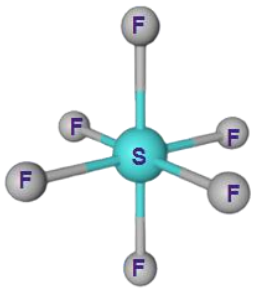
where, s = number of monovalent surrounding atoms

v = number of valence electrons on the central atom

q = charge (with sign)

In case of $SF_6$, s = 6, v = 6, q = 0 Hence, x = 6

∴ Hybridization becomes $sp^3d^2$ and geometry is octahedral.




**19. Correct Option: D) Pentagonal bipyramidal**

**Explanation**:

Hybridization (x) = $\frac{1}{2}(s + v - q)$

In $IF_7$,

v = Total number of valence electron pairs=7

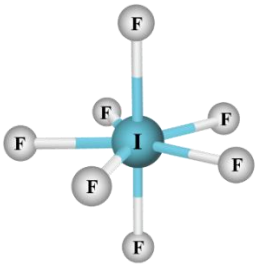
s = Total number of bonding electron pairs=7

q = Total number of lone pairs=0

x = 7

So hybridisation is $sp^3d^3$.

Shape is pentagonal bipyramidal




**20. Correct Option: D) 4,3**

Explanation:

Steric number = Number of bonded atoms + Number of lone pairs

$NH_4^+$:

- Steric number = 4 (bonding atoms) + 0 (lone pairs) = 4
- Since the steric number is 4, the hybridization is $sp^3$, which involves 4 hybrid orbitals.

$NO_3^-$:

- Steric number = 3 (bonding atoms) + 1 (lone pair) = 4
- However, in the case of $NO_3^-$, the presence of a lone pair doesn't contribute to the hybridization state. Only the bonding atoms determine the hybridization.
- Therefore, the hybridization is $sp^2$ (trigonal planar) using 3 hybrid orbitals.

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**LEVEL – II**

**SINGLE OPTION CORRECT TYPE**

**1. Correct Option: C) $SO_4^{2-}$ and $BF_4$**

**Explanation**:

Isostructural species have the same arrangement of atoms in space, even though they might differ in the types of atoms involved. Here's why the other options are not iso-structural:

A) $CH_3^-$ and $CH_3^+$: Both have tetrahedral geometry, but the extra electron in $CH_3^-$ changes the electron density distribution and bond lengths.

B) $NH_4^+$ and $NH_3$: $NH_4^+$ has a tetrahedral geometry while $NH_3$ is pyramidal due to the lone pair on nitrogen.

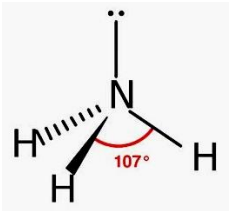
D) $NH_4^+$ and $BF_4^-$: $NH_4^+$ has a tetrahedral geometry while $BF_4^-$ is square planar due to different hybridization and no lone pairs.

**2. Correct Option: C) $NH_3$ & $H_2O$**

**Explanation**:

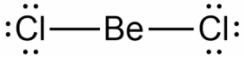
A. $NH_3$ (Ammonia):

- Has a trigonal pyramidal shape due to sp3 hybridization around the nitrogen atom with one lone pair.
- Non-linear due to the presence of the lone pair affecting the bond angles.



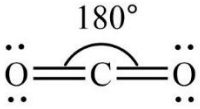

**B. $BeCl_2$ (Beryllium chloride):**

- Has a linear shape due to the absence of lone pairs and sp hybridization around the beryllium atom.
- Linear and does not deviate from a straight line.



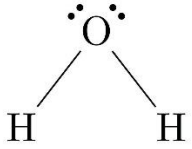

**C. $CO_2$ (Carbon dioxide):**

- Has a linear shape due to the absence of lone pairs and $sp^2$ hybridization around the carbon atom.
- Linear and does not deviate from a straight line.



**D. $H_2O$ (Water):**

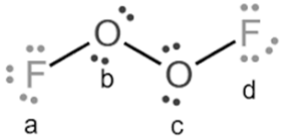
- Has a bent or V-shaped shape due to $sp^3$ hybridization around the oxygen atom with two lone pairs.
- Non-linear due to the presence of the lone pairs affecting the bond angles.



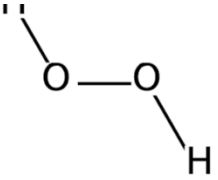

**3. Correct Option: B) $H_2O_2$**

**Explanation**:

$O_2F_2$: Both oxygen atoms have 2 lone pairs and 1 bonding pair each, leading to a bent or V-shaped structure with a bond angle of approximately 103°. This shape is known as an "open book" or "bent" structure.




$H_2O_2$: Both oxygen atoms have 2 lone pairs and 1 bonding pair each, resulting in a similar bent or V-shaped structure with a bond angle of around 105°. This shape also resembles an "open book" or "bent" configuration.

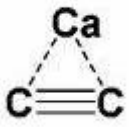



All the other options have a linear shape due to the absence of lone pairs and sp hybridization.

**4. Correct Option: B) One sigma and two pi bonds**

**Explanation**:

$CaC_2$ is an ionic compound, not a covalent molecule. It exists as $Ca^{2+}$ and $C_2^{-2}$ ions. Within the $C_2^{-2}$ ion, the carbon atoms form a triple bond. A triple bond consists of one sigma bond and two pi bonds.




**5. Correct Option: C) 109°**

**Explanation**:

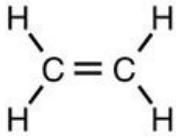
In $BH_4^-$, the boron atom (B) undergoes $sp^3$ hybridization to accommodate its four electron domains (one bond with each H and one lone pair). $sp^3$ hybridization involves mixing one 2s orbital and three 2p orbitals to form four tetrahedral hybrid orbitals. Due to the tetrahedral arrangement of these hybrid orbitals, the H-B-H bond angle in $BH_4^-$ is 109.5°, which is close to the ideal angle for $sp^3$ hybridization.

**6. Correct Option: B) $sp^2$ hybridization**

**Explanation**:

Each carbon atom forms three sigma (σ) bonds: two with hydrogen atoms and one with the other carbon atom. To accommodate these bonds, each carbon

undergoes $sp^2$ hybridization. This involves mixing one 2s orbital with two 2p orbitals, resulting in three sp2 hybrid orbitals arranged in a trigonal planar geometry (120° apart). The remaining 2p orbital on each carbon remains unhybridized and forms a pi (π) bond with the other carbon atom, contributing to the double bond character of ethylene.




**ONE OR MORE THAN ONE OPTION CORRECT TYPE**

**7. Correct Option: D) Shape of molecule depends upon type of hybridization**

**Explanation**:

A) Hybridization involves orbitals of comparable energies, not necessarily nearly equal. Small energy differences may exist but shouldn't significantly hinder mixing.

B) Hybrid orbitals are similar in energy but not identical in all respects. They inherit some characteristics from the original atomic orbitals they mix from.

C) Hybrid orbitals can form sigma (σ) and pi (π) bonds, not just sigma bonds. Sigma bonds arise from head-on overlap, while pi bonds result from sideways overlap of hybrid orbitals.

**8. Correct Option: D) $SnCl_2$**

**Explanation**:

Isostructural molecules have the same arrangement of atoms in space, even though they might differ in the types of atoms involved.

$CO_2$, $SO_2$, and $NO_2$: All three molecules have a linear geometry with a central atom (C, S, or N) bonded to two terminal atoms (O). This common arrangement makes them iso-structural.

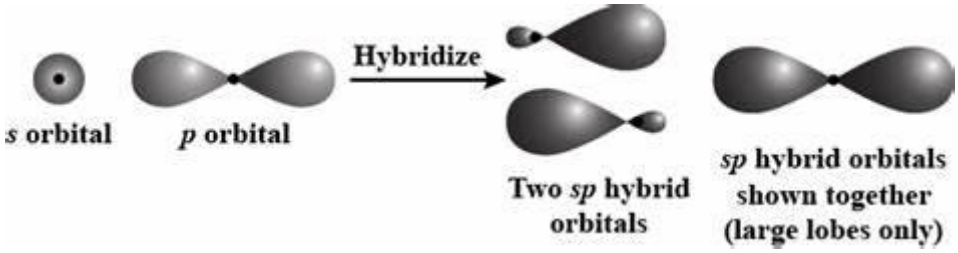
$C_2H_2$: This molecule has a linear geometry with a triple bond between the two carbon atoms. Each carbon is also bonded to a hydrogen atom, resulting in a different spatial arrangement compared to the other options.

$SnCl_2$: While $SnCl_2$ is linear, the central tin atom (Sn) has $sp^2$ hybridization, resulting in a bent (V-shaped) geometry due to the presence of one lone pair. This differs from the linear structure of $CO_2$, $SO_2$, and $NO_2$.

**9. Correct Option: B) Two equivalent hybrid orbitals at 180°**

**Explanation**:

Hybridizing one s orbital and one p orbital results in sp hybridization. This involves mixing one 2s orbital with one 2p orbital (usually the 2px orbital). The mixing creates two sp hybrid orbitals with equal contributions from the s and p orbitals. These sp orbitals are not mutually perpendicular (like in sp3 hybridization with three orbitals) or arranged in a plane (like in sp2 hybridization with three orbitals). Instead, they are oriented at 180° to each other, forming a linear geometry.



**10. Correct Option: B) $sp^2$**

**Explanation**:

Number of sigma bond around the Sulphur atom in $SO_2$ = 2 (one with each oxygen atom)

Sulphur atom has 6 valence electrons, out of which 4 of them are shared with two oxygen atoms in $SO_2$.

So, Number of lone pairs of electrons on a Sulphur atom in $SO_2 = \frac{6-4}{2} = 1$

Hence, steric number (X) for $SO_2$ = 2 + 1 = 3 and it is already known to you that for X = 3, hybridization is $sp^2$

**11. Correct Option: B) S-S sigma bond is stronger than P-P sigma bond**

**Explanation**:

A) Hybrid orbitals can form only sigma bonds: Hybrid orbitals can form both sigma (σ) and pi (π) bonds, depending on their overlap patterns. Sigma bonds result from head-on overlap, while pi bonds arise from sideways overlap.

C) Double bonds cause less repulsion than single bonds: Double bonds involve more electron density concentrated in the bonding region, leading to greater repulsion compared to single bonds where electron density is more spread out.

D) The number of hybrid orbitals formed are twice the number of orbitals participating in hybridization: Depending on the specific hybridization involved, the number of hybrid orbitals can be equal to or less than twice the number of participating orbitals. For example, sp hybridization involves one s and one p orbital, resulting in two sp hybrid orbitals.

**12. Correct Option: A) $Cl_2$**

**Explanation**:

P-P overlapping refers to the interaction between unpaired electrons in p orbitals of the same atom. This overlap leads to the formation of a sigma (σ) bond.

A. $Cl_2$: Both chlorine atoms have an unpaired electron in their 3p orbitals. These 3p orbitals overlap sideways to form a pi (π) bond, resulting in the diatomic chlorine molecule.

B. HCl: The hydrogen atom only has one 1s orbital, and the chlorine atom has an unpaired electron in its 3p orbital. They form a sigma (σ) bond by overlapping their s orbitals, not p orbitals.

C. $H_2O$: The oxygen atom has two unpaired electrons in its 2p orbitals, but they form sigma bonds with the hydrogen atoms (s-p overlap), not pi bonds.

D. $NH_3$: The nitrogen atom has an unpaired electron in a 2p orbital, but it forms sigma bonds with the hydrogen atoms (s-p overlap) and uses its remaining 2p orbitals to hold lone pairs, not for pi bonds.

**13. Correct Answer: A) Both A and R are true and R is the correct explanation of A.**

**Explanation:**

- **Assertion is true:**

  A **σ (sigma) bond** is generally **stronger** than a **π (pi) bond**.

- **Reason is true:**
  - **Sigma bond (σ):** Formed by **head-on (axial)** overlap of orbitals (s–s, s–p, p–p, or hybrid orbitals). This overlap is **maximum**, making the bond stronger.
  - **Pi bond (π):** Formed by **sideways (lateral)** overlap of two parallel p-orbitals. The overlap is **less effective**, so the bond is weaker.

  Since greater overlap results in a stronger bond, the reason correctly explains the assertion.

**14. Correct Answer: C) A is true but R is false.**

**Explanation:**

- **Assertion is true:**
  In **ethylene ($C_2H_4$)**, the carbon-carbon double bond consists of:
  - **One σ bond** (formed by head-on overlap of $sp^2$ hybrid orbitals)
  - **One π bond** (formed by sideways overlap of unhybridized p orbitals)
- **Reason is false:**
  Carbon atoms in ethylene are **$sp^2$ hybridized**, **not $sp^3$ hybridized**.
  - Each carbon forms **three σ bonds** using **$sp^2$ hybrid orbitals**.
  - One unhybridized p orbital on each carbon overlaps sideways to form the **π bond**.

Therefore, the assertion is true, but the reason is incorrect.

**MATRIX MATCHING TYPE**

**15. Correct Option: a-R, b-S, c-P, d-Q**

**Explanation**:

a. Ethane ($CH_3CH_3$): Each carbon forms 4 single bonds using $sp^3$ hybridization (1s orbital + 3p orbitals).

b. Ethylene ($CH_2=CH_2$): Each carbon forms 3 sigma bonds (using $sp^2$ hybridization) and a pi bond (using an unhybridized p orbital).

c. Acetylene (HC≡CH): Each carbon forms 1 sigma bond and 2 pi bonds, requiring sp hybridization (1s + 1p).

d. Benzene ($C_6H_6$): Each carbon forms 3 sigma bonds and 1 pi bond, employing $sp^2$ hybridization.

**16. Correct Option: a-R, b-P, c-S, d-Q**

**Explanation**:

a) $BCl_3$: Each Cl forms a single bond with B, totaling 3 electron domains (bonds).

b) $NH_3$: Each H forms a single bond with N, and N has 1 lone pair, totaling 4 electron domains.

c) $PCl_5$: Each Cl forms a single bond with P, totaling 5 electron domains.

d) $XeF_4$: Each F forms a single bond with Xe, and Xe has 2 lone pairs, totaling 6 electron domains.

**COMPREHENSION TYPE**

**Passage – I:**

**17. Correct Option: B) $sp^2$**

**Explanation**:

Hybridization (x) = $\frac{1}{2}(s + v - q)$

where, s = number of monovalent surrounding atoms

v = number of valence electrons on the central atom

q = charge (with sign)

In case of $BH_3$, s = 3, v = 3, q = 0 Hence, x = 3

∴ Hybridization becomes $sp^2$

**18. Correct Option: C) sp**

**Explanation**:

Hybridization (x) = $\frac{1}{2}(s + v - q)$

where, s = number of monovalent surrounding atoms

v = number of valence electrons on the central atom

q = charge (with sign)

In case of $NO_2^+$, s = 3, v = 2, q = -1 Hence, x = 2

∴ Hybridization becomes sp

**19. Correct Option: A) $sp^2$, sp**

**Explanation**:

Hybridization (x) = $\frac{1}{2}(s + v - q)$

where, s = number of monovalent surrounding atoms

v = number of valence electrons on the central atom

q = charge (with sign)

In case of $CO_3^{2-}$, s = 0, v = 4, q = -2 Hence, x = 3

∴ Hybridization becomes $sp^2$

In case of $CO_2$, s = 0, v = 4, q = 0 Hence, x = 2

∴ Hybridization becomes sp

**20. Correct Option: D) $sp^3d^3$**

**Explanation**:

A) $sp^3$

The angle between the sp3 hybrid orbitals is 109.28°

B) $sp^3d$

Three hybrid orbitals lie in the horizontal plane inclined at an angle of 120° to each other, known as the equatorial orbitals. The remaining two orbitals lie in the vertical plane at 90 degrees plane of the equatorial orbitals, known as axial orbitals.

C) $sp^3d^2$

They are inclined at an angle of 90 degrees to one another.

D) $sp^3d^3$

The angle between every five bonds in the X-Y plane is 72°. Whereas the angle between the bond along Z-axis and X-Y plane is 90°.

**INTEGER TYPE**

**21. Correct Option: 4**

**Explanation**:

Each carbon atom in acetylene undergoes sp hybridization. This involves mixing one 2s and one 2p orbital from each carbon atom, resulting in two sp hybrid orbitals per carbon atom. Since there are two carbon atoms in acetylene, the total number of hybrid orbitals formed is 4 (2 sp orbitals per carbon atom * 2 carbon atoms).

**22. Correct Option: 3**

**Explanation**:

In $SO_2$, the sulphur atom (S) uses a total of 3 of its orbitals for bonding:

- 1 s orbital: This orbital hybridizes with 3 p orbitals to form four $sp^3$ hybrid orbitals.
- 2 p orbitals: Two of the four $sp^3$ hybrid orbitals participate in sigma bonding with the oxygen atoms.

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# VSEPR Theory & Bond Parameters

**LEVEL – I**

**1. Correct Option: B) B**

**Explanation**:

Given that the compound X has a bond angle of 120 degrees, it suggests a trigonal planar geometry, which indeed requires a central atom with three valence electrons. Among the options provided, boron (B) is the element with three valence electrons, consistent with the trigonal planar geometry.

**2. Correct Option: C) 1:3**

**Explanation**:

A bond angle of 109.28° corresponds to sp3 hybridization. In sp3 hybridization, one 2s orbital and three 2p orbitals from the same atom hybridize to form four sp3 hybrid orbitals.

The contribution of each orbital to the hybrid orbital is not equal. The s orbital contributes 25% to the sp3 hybrid orbital, while each p orbital contributes 75%. Therefore, the ratio of s character to p character is:

25% s character / 75% p character = 1:3

**3. Correct Option: D) $ICl_3$**

**Explanation**:

A. $XeO_3$: The central xenon atom has 1 lone pairs and 3 bonding pairs (double bonds with oxygen)

B. $SF_4$: The central sulfur atom has 1 lone pair and 4 bonding pairs (single bonds with fluorine).

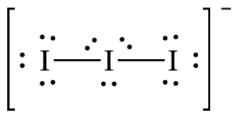
C. $PCl_3$: The central phosphorus atom has 1 lone pair and 3 bonding pairs (single bonds with chlorine).

D. $ICl_3$: The central iodine atom has 2 lone pairs and 3 bonding pairs (single bonds with chlorine).

**4. Correct Option: A) Linear**

**Explanation**:

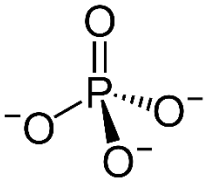
Each iodine atom (I) forms a single covalent bond with the central iodine atom, resulting in two I-I bonds. The central iodine atom also has one negative charge, which corresponds to a lone pair of electrons. To minimize electron-electron repulsion, the lone pair occupies an equatorial position, forming a linear arrangement of the three iodine atoms: I - I - I.



**5. Correct Option: B) Tetrahedral**

**Explanation**:

The central phosphorus atom (P) forms four single covalent bonds with four oxygen atoms (O). No lone pairs are present on the phosphorus atom. Based on the VSEPR theory (Valence Shell Electron Pair Repulsion), the four electron domains (four bonding pairs) around the phosphorus atom arrange themselves in a tetrahedral geometry to minimize electron-electron repulsion.



**6. Correct Option: D) $NH_3$, $PH_3$**

**Explanation**:

A) $BF_3$, $NH_3$:

- $BF_3$ has a trigonal planar geometry due to $sp^2$ hybridization and three bonding domains (three fluorine atoms).
- $NH_3$ has a trigonal pyramidal geometry due to $sp^3$ hybridization and four electron domains (three hydrogen atoms and one lone pair).

B) $H_2O$, $C_2H_2$:

- $H_2O$ has a bent geometry due to $sp^3$ hybridization and two lone pairs.
- $C_2H_2$ has a linear geometry due to sp hybridization and one sigma and two pi bonds.

C) $CO_2$, $SO_2$:

- $CO_2$ has a linear geometry due to sp hybridization and two double bonds.
- $SO_2$ has a bent geometry due to $sp^2$ hybridization and one lone pair.

D) $NH_3$, $PH_3$:

- Both $NH_3$ and $PH_3$ have trigonal pyramidal geometry due to $sp^3$ hybridization and four electron domains (three hydrogen atoms and one lone pair).

**7. Correct Option: D) $SO_3$**

**Explanation**:

$SO_2$: Sulfur atom has $sp^2$ hybridization, using one s and two p orbitals to form three $sp^2$ hybrid orbitals. One lone pair remains on the sulfur atom.

$BF_3$: Boron atom has $sp^2$ hybridization, using one s and two p orbitals to form three $sp^2$ hybrid orbitals. No lone pairs are present.

$B_2H_6$: Each boron atom has $sp^3$ hybridization, using one s and three p orbitals to form four $sp^3$ hybrid orbitals. Each boron atom is bonded to two hydrogen atoms.

$SO_3$: Sulfur atom has $sp^3d$ hybridization, using one s, three p orbitals, and one d orbital to form five $sp^3d$ hybrid orbitals. One lone pair remains on the sulfur atom.

**8. Correct Option: $CO_2$**

**Explanation**:

A. $BF_3$: This molecule has a trigonal planar geometry with $sp^2$ hybridization around the central boron atom. This results in bond angles of **120°** between the boron and fluorine atoms.

B. $CH_4$: This molecule has a tetrahedral geometry with $sp^3$ hybridization around the central carbon atom. This results in bond angles of **109.5°** between the carbon and hydrogen atoms.

C. $CO_2$: This molecule has a linear geometry with sp hybridization around the central carbon atom. This results in bond angles of **180°** between the carbon and oxygen atoms.

D. $SO_2$: This molecule has a bent geometry with $sp^2$ hybridization around the central sulphur atom. This results in bond angles of **~119°** between the sulphur and oxygen atoms.

**9. Correct Option: B) 1**

**Explanation**:

1. Central Atom: Xe (Xenon)
2. Bonding:

- One double bond with oxygen (O) forming Xe-O bond.
- Four bonds with fluorine (F) forming Xe-F single bonds.

3. Total valence electrons in Xe**:** 8
4. Electrons used in bonding: 6 (2 with O + 4 with F)
5. Remaining valence electrons: 2
6. Lone pairs: Since Xe needs to satisfy the octet rule, it forms one lone pair with the remaining 2 electrons.

**10. Correct Option: B) Pyramidal**

**Explanation**:

According to the VSEPR model, molecules arrange their electron domains (bonding pairs and lone pairs) to minimize electron-electron repulsion. With 4 electron domains, the most stable geometry for $ClO_3^-$ is trigonal pyramidal, where the three

Cl-O bonds form a triangle and the lone pair occupies one corner above the plane, creating a pyramidal shape.

**11. Correct Option: D) 4, 0**

**Explanation**:

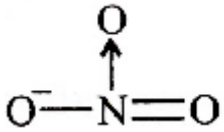
Due to the presence of one negative charge, number of valence electrons= 5+ 1=6

One *O*-atom forms two bonds (= bond) and two *O*-atom are shared with two electrons of *N*-atom

Thus, 3*O*-atoms are shared with 8 electrons of *N*-atom.

Number of bond pairs (or shared pairs) =4

Number of lone pairs =0



**12. Correct Option: D) Different with 1,0 and 2 lone pair of electrons respectively**

**Explanation**:

$SF_4$: Sulfur (S) in $SF_4$ has 1 lone pair and 4 bonding pairs, leading to an $sp^3d$ hybridization and a see-saw shape.

$CF_4$: Carbon (C) in $CF_4$ has 4 bonding pairs and no lone pairs, resulting in $sp^3$ hybridization and a tetrahedral shape.

$XeF_4$: Xenon (Xe) in $XeF_4$ has 2 lone pairs and 4 bonding pairs, leading to $sp^3d^2$ hybridization and a square planar shape.

**13. Correct Option: D) $C_2H_2$**

**Explanation**:

Factors influencing C-C bond length:

1. Bond order: A higher bond order corresponds to a shorter bond length due to stronger electron-nucleus attraction.
2. Hybridization: sp hybridization creates shorter bonds compared to $sp^2$ and $sp^3$.

A. $C_2H_6$ (ethane): All C-C bonds are single (bond order 1) and $sp^3$ hybridized.

B. $C_2H_4$ (ethylene): One C-C bond is double (bond order 2) and $sp^2$ hybridized, while the other is single.

C. $C_6H_6$ (benzene): All C-C bonds are part of a delocalized pi system with resonance, effectively increasing bond order. They are $sp^2$ hybridized.

D. $C_2H_2$ (acetylene): Both C-C bonds are triple (bond order 3) and sp hybridized.

**14. Correct Option: D) $CO < CO_2 < CO_3^{-2}$**

**Explanation**:

The bond length in a molecule depends primarily on two factors:

1. Bond order: Higher bond order generally corresponds to shorter bond length due to stronger electron-nucleus attraction.
2. Negative charge: The presence of a negative charge delocalizes electron density across the molecule, effectively increasing the size of the electron cloud and leading to longer bonds.

- CO: This molecule has a triple bond between C and O, resulting in the highest bond order (3) and the shortest bond length among the three options.
- $CO_2$: This molecule has a double bond between C and each O, leading to a bond order of 2 and a bond length longer than CO but shorter than $CO_3^{-2}$.
- $CO_3^{-2}$: This ion has a single bond between C and each O, with a negative charge delocalized across the molecule. The combination of lower bond order and negative charge leads to the longest bond length among the three.

**15. Correct Option:**

**16. Correct Option: D) HI**

**Explanation**:

As the size of halogen atom(x) increases, the H-x bond length increases, and hence H-X bond strength decreases.

H-x bond energy decreases from H-F to H-I

H-I has least bond energy...

Bond energy order: H-F > H-Cl > H-Br > H-I

**17. Correct Option: C) sp - s**

**Explanation**:

The highest bond energy arises from the smallest overlap, which occurs with sp hybridization.

- smaller orbital size: sp orbitals are more concentrated compared to $sp^2$, $sp^3$, or $sp^3d$, leading to greater overlap with a standard s orbital.
- better directional overlap: sp orbitals have a more linear shape, maximizing overlap with an s orbital along the internuclear axis.

**18. Correct Option: A) $F_2 > Cl_2 > Br_2 > I_2$**

**Explanation**:

Bond energy typically increases with increasing electronegativity difference between the bonded atoms. Fluorine has the highest electronegativity among halogens, leading to the strongest F-F bond in $F_2$. As we move down the halogen group, electronegativity decreases, resulting in a weaker bond. Additionally, atomic size increases, creating greater orbital overlap and consequently lower bond energy. Therefore, $Cl_2$ will have a weaker bond than $F_2$, $Br_2$ weaker than $Cl_2$, and $I_2$ the weakest among them

**19. Correct Option: B) O-H**

**Explanation**:

Electronegativity difference: Oxygen (O) is significantly more electronegative than carbon (C), sulfur (S), and chlorine (Cl). This larger difference leads to stronger ionic character and a stronger bond.

Bond polarity: Due to the high electronegativity difference, O-H is the most polar bond, with stronger electrostatic attraction between the atoms.

**20. Correct Option: A) $NO_2^+$**

**Explanation**:

1. Hybridization: $NO_2^+$ has sp hybridization on the central nitrogen, while $NO_2$, $NO_2^-$, and $NO_3^-$ have $sp^2$ hybridization. sp orbitals are more linear, leading to larger bond angles.
2. Lone pairs: $NO_2^-$ and $NO_3^-$ have lone pairs on the central atom, repelling bonding pairs and decreasing the bond angle. $NO_2^+$ has no lone pairs, further increasing the bond angle potential.

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**LEVEL – II**

**SINGLE OPTION CORRECT TYPE**

**1. Correct Option: D) $CO_2$**

**Explanation**:

A. $CH_4$: The carbon atom in $CH_4$ is $sp^3$ hybridized, leading to a tetrahedral geometry with bond angles of 109.5°.

B. $H_2O$: Oxygen in $H_2O$ is $sp^3$ hybridized, but two lone pairs repel bonding electron pairs, resulting in a bent shape and bond angles of 104.5°.

C. $NH_3$: Nitrogen in $NH_3$ is $sp^3$ hybridized with one lone pair. The lone pair repulsion reduces the bond angle to 107° compared to $CH_4$.

D. $CO_2$: Carbon in $CO_2$ is $sp^2$ hybridized, forming a linear molecule with a 180° bond angle. No lone pairs are present to cause repulsion.

**2. Correct Option: A) $NH_3$**

**Explanation**:

Key factors for a pyramidal shape:

1. Central atom with three bonded atoms: All the listed molecules have three bonded atoms.
2. Presence of a lone pair: This is crucial for the pyramidal shape. In $NH_3$, the nitrogen atom has one lone pair along with three hydrogen atoms attached.

Only $NH_3$ has the combination of a lone pair and $sp^3$ hybridization that generates the characteristic pyramidal shape.

**3. Correct Option: A) S - S**

**Explanation**:

Factors affecting bond strength:

- Atom size: Smaller atoms can form closer and stronger bonds due to less electron-electron repulsion. However, this is not the sole factor.
- Electronegativity: Higher electronegativity difference between atoms leads to greater attraction and a stronger bond.

While oxygen is indeed smaller than sulphur, its significantly higher electronegativity outweighs the slight size advantage in terms of overall bond strength. As we move down the group to Se and Te, their larger sizes weaken the bond energy due to the reduced orbital overlap and weaker electrostatic interactions.

**4. Correct Option: D) $XeO_4$**

**Explanation**:

For a molecule to be planar, it needs to fulfil two key requirements:

1. Central atom hybridization: The central atom in the molecule must be hybridized in a way that generates orbitals capable of forming planar bonds. This typically means $sp^2$ or $dsp^2$ hybridization, where the hybridized orbitals lie in the same plane.
2. No lone pairs around the central atom: Lone pairs occupy space and repel bonding pairs, pushing them out of the plane and disrupting planarity. Therefore, the central atom should not have any lone pairs for a truly planar structure.

| A) $XeF_4$: | C) $XeO_2F_2$ |
|---|---|
| Valence electrons of Xe= 8<br>Bounded electrons = 4<br>Lone pairs = (8-4)/2 = 2 | Valence electrons of Xe= 8<br>Bounded electrons = 6<br>Lone pairs = (8-6)/2 = 1 |
| B) $XeOF_2$ | D) $XeO_4$ |
| Valence electrons of Xe= 8<br>Bounded electrons = 4<br>Lone pairs = (8-4)/2 = 2 | Valence electrons of Xe= 8<br>Bounded electrons = 8<br>Lone pairs = (8-8)/2 = 0 |

**5. Correct Option: A) $XeF_2$, $IF_2^-$**

**Explanation**:

Hybridization = $\frac{\text{Valence e}- + \text{Monovalent atoms} + \text{Anion} - \text{Cation}}{2}$

A) $XeF_2$, $IF_2^-$

$XeF_2$ Hybridization = (8+2)/2 = 5 ($sp^3d$)
$IF_2^-$ Hybridization = (7+2+1)/2 = 5 ($sp^3d$)
B) $NH_3$, $BF_3$
$NH_3$ Hybridization = (5+3)/2 = 4 ($sp^3$)
$BF_3$ Hybridization = (3+3)/2 = 3 ($sp^2$)
C) $CO_3^{2-}$, $SO_3^2$
$CO_3^{2-}$ Hybridization = (4+0+2)/2 = 3 ($sp^2$)
$SO_3^{2-}$ Hybridization = (6+0+2)/2 = 4 ($sp^3$)
D) $PCl_5$, $ICl_5$
$PCl_5$ Hybridization = (5+5)/2 = 5 ($sp^3d$)
$ICl_5$ Hybridization = (7+5)/2 = 6 ($sp^3d^2$)

**6. Correct Option: A) Three**

**Explanation**:

In an octahedral molecule, the central atom M is surrounded by six identical atoms X arranged at the corners of a cube. Each X atom forms a bond with the central atom M at an angle of 90°. However, there are three specific X-M-X bond angles that are exactly 180°. These angles lie along the axes of the cube, where two X atoms are directly opposite each other on the same axis.

**ONE OR MORE THAN ONE ANSWER TYPE**

**7. Correct Option: C, D**

**Explanation**:

C) Slightly more than $NH_3$: This is because both molecules have a trigonal pyramidal geometry due to $sp^3$ hybridization of the central atom (P in $PH_3$ and N in $NH_3$). However, the larger size of the phosphorus atom in $PH_3$ compared to nitrogen in $NH_3$ leads to slightly less repulsion between the hydrogen atoms, resulting in a slightly larger bond angle in $PH_3$ (around 93.3°) compared to $NH_3$ (around 107°).

D) Much more than $PF_3$: This is due to the difference in hybridization between the central atoms. $PH_3$ has $sp^3$ hybridization, while $PF_3$ has $sp^3d^2$ hybridization. The inclusion of d orbitals in $PF_3$ expands the valence shell, leading to a larger bond angle (around 101.3°) compared to $PH_3$.

**8. Correct Option: None are always true**

**Explanation**:

A) EN of central atom:

Decreasing EN generally leads to increased bond angle, but not always. Lower EN means less attraction to shared electrons, potentially reducing repulsion and allowing wider angles. However, other factors like size and hybridization play a role.

B) Size of central atom:

Increasing size can sometimes lead to increased bond angle, but again, not always. Larger atoms have more space to accommodate electron density, potentially reducing repulsion and allowing wider angles. However, hybridization and the nature of atom B also influence bond angles.

C) EN of atom B:

The effect of decreasing EN of atom B on bond angle is not consistent. Lower EN in B means less attraction towards shared electrons, potentially leading to less repulsion and possibly wider bond angles. However, this can be mitigated by other factors like the EN difference between A and B and the hybridization of the central atom.

D) EN of atom B:

Similar to C, the effect of decreasing EN of B is not always consistent on bond angle.

**9. Correct Option: B, C, D**

**Explanation**:

A) $XeF_4$: While $XeF_4$ has a central Xe atom surrounded by four atoms, it adopts a square planar geometry with bond angles of 90° due to $sp^3d^2$ hybridization and the absence of lone pairs.

B) $XeF_2$: With only two fluorine atoms, $XeF_2$ adopts a linear geometry with a bond angle of 180°. Square planar geometry requires four atoms around the central atom.

C) $XeOF_4$: This molecule has a tetrahedral geometry due to $sp^3$ hybridization on the Xe atom and the presence of four lone pairs (two lone pairs on Xe and two on the oxygen atoms). Square planar geometry requires no lone pairs.

D) $CH_3Cl$: The carbon atom in $CH_3Cl$ has $sp^3$ hybridization, resulting in a tetrahedral geometry with bond angles of approximately 109.5°. Square planar geometry requires a central atom with $sp^2d^2$ hybridization.

**10. Correct Option: A, B**

**Explanation**:

Nitrogen remains $sp^3$ hybridized even after forming the adduct. Boron changes from $sp^2$ in $BF_3$ to $sp^3$ in the adduct due to accepting a lone pair from $NH_3$. This changes its shape from trigonal planar to tetrahedral.

**Assertion and reason type:**

**11. Correct Answer: A) Both A and R are true and R is the correct explanation of A.**

**Explanation:**

- **Assertion is true:**

  According to **VSEPR (Valence Shell Electron Pair Repulsion) theory**, when there are **two electron pairs** (bond pairs only) around the central atom, they arrange themselves as far apart as possible, giving a **linear geometry**.

- **Reason is true:**

  Two electron pairs experience mutual repulsion. To **minimize repulsion**, they remain **180° apart**, which produces a linear shape.

  Thus, the reason correctly explains the assertion.

**12. Correct Answer: A) Both A and R are true and R is the correct explanation of A.**

**Explanation:**

- **Assertion is true:**
  - **$NH_3$:** Bond angle = **107°**
  - **$H_2O$:** Bond angle = **104.5°**

  Therefore, the bond angle in water is smaller.

- **Reason is true:**
  - **$NH_3$:** 3 bond pairs + **1 lone pair**
  - **$H_2O$:** 2 bond pairs + **2 lone pairs**

  Lone pair–lone pair and lone pair–bond pair repulsions are stronger than bond pair–bond pair repulsions. Since **$H_2O$ has two lone pairs**, the bond pairs are pushed closer together, reducing the bond angle from **107° to 104.5°**.

  Hence, the reason correctly explains the assertion.

**Order of Repulsions (VSEPR Theory)**

| Lone pair–Lone pair > Lone pair–Bond pair > Bond pair–Bond pair |
|---|

Because water has more lone pairs, it experiences greater repulsion, leading to a smaller H–O–H bond angle.

**MATRIX MATCHING TYPE**

**13. Correct Option: a-Q, b-P, c-R, d-S**

**Explanation**:

$BrF_3$: The central bromine atom (Br) in $BrF_3$ has one lone pair and $sp^3d$ hybridization, resulting in a T-shaped geometry.

$[PCl_4]^+$: The phosphorus atom (P) in the positively charged tetrahedral ion $[PCl_4]+$ lacks lone pairs and has $sp^3$ hybridization, leading to a square planar geometry.

$XeF_4$: The xenon atom (Xe) in $XeF_4$ has no lone pairs and $sp^3d^2$ hybridization, resulting in a regular tetrahedral geometry.

$XeO_3$: The xenon atom (Xe) in $XeO_3$ has three lone pairs and $sp^3d$ hybridization, causing the molecule to adopt a pyramidal shape with one lone pair occupying the apex.

**14. Correct Option: a-R, b-Q, c-P, d-S**

**Explanation**:

a) $ClF_3$: Chlorine has one lone pair, resulting in $sp^3d$ hybridization and a T-shaped geometry.

b) $NF_3$: Nitrogen has one lone pair, leading to $sp^3$ hybridization and a pyramidal shape.

c) $BF_3$: Boron has no lone pairs, and $sp^2$ hybridization gives it a trigonal planar geometry.

d) $SO_2$: Sulfur has one lone pair, resulting in $sp^2$ hybridization and a bent (V-shaped) geometry.

**COMPREHENSION TYPE**

**Passage – I:**

**15. Correct Option: B) IV<III<II<I**

**Explanation**:

As the electronegativity of central atom increases the bond angle increases due to repulsion between bond pair and bond pair as bond pairs are more close to the central atom.

**16. Correct Option: D) IV<II<III<I**

**Explanation**:

I ($NH_3$): Nitrogen is $sp^3$ hybridized with one lone pair, resulting in a trigonal pyramidal geometry and a bond angle of ~107°.

II ($PH_3$): Phosphorus is $sp^3$ hybridized with one lone pair, similar to $NH_3$, but the larger P atom leads to less repulsion and a slightly wider bond angle of ~93.3°.

III ($AsH_3$): Arsenic is $sp^3$ hybridized with one lone pair, similar to $NH_3$ and $PH_3$, but the even larger As atom results in an even wider bond angle of ~91.8°.

IV ($SbH_3$): Antimony is $sp^3$ hybridized with one lone pair, but the significantly larger Sb atom causes the widest bond angle of ~90°.

**17. Correct Option: A) $OSF_2$**

**Explanation**:

Factors affecting the X-O-X bond angle:

1. Size of the halogen atom (X): Smaller size allows for closer packing and less steric hindrance, leading to a smaller bond angle.
2. Lone pairs: Their absence minimizes repulsion between electron pairs, allowing for a smaller bond angle.

- $OSF_2$: Fluorine is the smallest halogen, and there are no lone pairs on oxygen, leading to the smallest bond angle.

- $OSCl_2$, $OSBr_2$, $OSI_2$: All three have larger halogens than fluorine, increasing the bond angle. Additionally, $OSBr_2$ and $OSI_2$ have lone pairs on oxygen, further widening the bond angle.

**Passage – II:**

**18. Correct Option: D) $PI_3$**

**Explanation**:

Larger halogens like iodine (I) create more steric hindrance compared to smaller ones like fluorine (F), leading to wider bond angles. Therefore, as the halogen size increases down the group from F to I, the repulsion between the lone pairs and bonding pairs increases, resulting in the largest bond angle in $PI_3$.

**19. Correct Option: C) $XeF_4$**

**Explanation**:

a) $BrF_3$ has T-shape but perfect T but little distorted due to repulsion between lone pair and bond pair.

b) $ClF_3$ has T-shape but perfect T but little distorted due to repulsion between lone pair and bond pair.

c) $XeF_4$ square planar and lone pairs and above and below the plane. Therefore, no distortion.

d) $XeF_6$ has distorted octahedral shape.

**20. Correct Option: B) F-N-F angle is $NF_3$ is smaller than H-N-H angle in $NH_3$**

**Explanation**:

Both $NH_3$ and $NF_3$ have a pyramidal shape due to $sp^3$ hybridization of the central atom (N). However, fluorine (F) is more electronegative than hydrogen (H). This means F atoms pull electron density closer towards themselves, reducing the electron cloud size around the nitrogen atom. Smaller electron clouds experience less repulsion between each other, leading to a smaller bond angle in $NF_3$ compared to $NH_3$.

**INTEGER TYPE**

**21. Correct Option: 3**

**Explanation**:

In the $BCl_3$ molecule, the central boron atom (B) forms bonds with three chlorine atoms (Cl). Each bond involves a shared pair of electrons. Therefore, there are 3 bonding pairs in $BCl_3$.

**22. Correct Option: 3**

**Explanation**:

$BeCl_2$: Beryllium in $BeCl_2$ has sp hybridization, leading to a linear geometry with two Cl atoms bonded in a straight line.

$CO_2$: Carbon in $CO_2$ also has sp hybridization, resulting in a linear geometry with two oxygen atoms bonded to the carbon.

HCN: The carbon atom in HCN has sp hybridization, forming a linear molecule with one hydrogen and one nitrogen atom bonded linearly.

$SO_2$: Sulfur in $SO_2$ has $sp^2$ hybridization, forming a bent (V-shaped) geometry due to the presence of one lone pair.

$H_2O$: Oxygen in $H_2O$ has $sp^3$ hybridization, leading to a bent (V-shaped) geometry with two lone pairs causing repulsion.

# Previous year question paper

**1**. A) $CH_3$-$\underline{CH_2}$-$CH=CH_2$: The underlined carbon is bonded to two Carbons and two Hydrogens via single bonds. This is $sp^3$.
B) $CH_3$-$\underline{CH_2}$-$NH_2$: Similar to above, this carbon has 4 sigma bonds. This is also $sp^3$.
D) $CH_3$-$\underline{CH_2}$-CN: The underlined carbon has 4 sigma bonds. This is $sp^3$.

**2**. C**)** $O_2^-$ (1.5) + $N_2$ (3) + $O_2^+$ (2.5) = 7

**3**. A) $NH_3$ to $NH_4^+$: $NH_3$ is $sp^3$ (3 bond pairs + 1 lone pair). $NH_4^+$ is $sp^3$ (4 bond pairs). No change.

**4**. Total: 5 molecules. Correct Option: C
$ClF_3$: 10 electrons around Cl. (Yes)
$SF_6$: 12 electrons around S. (Yes)
$CH_4$: 8 electrons around C. (No)
$NH_3$: 8 electrons around N. (No)
$SF_4$: 10 electrons around S. (Yes)
$XeF_4$: 12 electrons around Xe. (Yes)
$PCl_5$: 10 electrons around P. (Yes)

**5. Correct Option: D (0 and -1)**
The structure of Ozone is a resonance hybrid. In a single Lewis structure:

- The central oxygen has a formal charge of +1.
- One terminal oxygen (double bonded) has a charge of 0.
- The other terminal oxygen (single bonded) has a charge of -1.

**6. Correct Option: A**
(ii) $SF_6$ is $sp^3d$ hybridised: Incorrect**.** SF6 is $sp^3d^2$ (octahedral).
(iii) $S_2O_3^{2-}$ is a linear ion: Incorrect**.** Thiosulfate is tetrahedral around the central sulfur.
(iv) There is no pi-bonding in $SO_4^{2-}$ ion: Incorrect. $SO_4^{2-}$ has $d\pi - p\pi$ bonding.

**7. Correct Answer: B**
sing Molecular Orbital Theory (Total electrons):

1. $NO^+$ (14 electrons): Same as $N_2$, Bond Order = 3.0
2. NO (15 electrons**):** Bond Order = 2.5
3. $NO^-$ (16 electrons): Bond Order = 2.0

**8. Correct Answer: B ($sp^2$, sp)**
Based on the structure provided in the prompt ($H_3C$-$CH=C=CH_2$):

- $C_2$**:** Involved in one double bond (CH=). Hybridization = $sp^2$
- $C_3$**:** Central carbon of the allene system bonded to two double bonds (=C=). Hybridization = sp

**9. Correct Answer: B) $H_2O$, $BeCl_2$, $BF_3$:**

- $H_2O$: Oxygen has **2 lone pairs** that repel the bond pairs strongly, pushing them closer together.( 104.5°)
- $BF_3$ : Boron has **0 lone pairs** and 3 bonding pairs that spread out equally in a flat plane.( $120^\circ$)
- $BeCl_2$: Beryllium has **0 lone pairs** and only 2 bonding pairs, which stay as far apart as possible ($180^\circ$)
- *Order:* 104.5° < $120^\circ$° < $180^\circ$ (This is increasing order).

**10. Correct Answer:C**
According to Valence Bond Theory (VBT) for coordination compounds:

1. $sp^3$: Involves one 4s and three 4p orbitals (e.g., in tetrahedral complexes like $[NiCl_4]^{2-}$.

2. $dsp^2$: Involves one inner 3d, one 4s, and two 4p orbitals (e.g., in square planar complexes like $[Ni(CN)_4]^{2-}$).

**3.** $d^2sp^3$: Involves two inner 3d, one 4s, and three 4p orbitals (e.g., in inner-orbital octahedral complexes like $[NH_3)_6]^{3+}$

While $sp^3d^2$ (outer-orbital octahedral) is also possible using 4d orbitals, the options provided specifically highlight the use of the 3d (inner) and 4s (outer) orbitals.

# 5. Metals and Non-Metals

## Solutions

### LEVEL – I

**1. Correct option: (B) Cations**

Metals are highly electropositive. Thus they tend to lose one electron from their valence shell to gain electronic configuration of Noble gas. Hence they form cations.

**2. Correct option: (C) Malleability**

Malleability is a physical property of metals which defines the ability of metals to be beaten into sheets without breaking.

Tenacity is the ability of metal to resist the effect of tensile forces.

Ductility is a physical property for which metal can be drawn into thin wire.

Porosing is a defect that can leave machinery to a wide range of vulnerable problems.

**3. Correct option: (C) Graphite**

Graphite is a crystalline allotrope of carbon.

It has layered structure which are held by weak Van der Waals force of attraction. The layers are composed of planar hexagonal rings of carbon. It has free electrons in its structure as the fourth valence electron of carbon atom remains unpaired.

Hence Graphite is a good conductor of heat & electricity due to presence of free electrons

**4. Correct option: (B) Positive**

Metals can donate electrons and form positive ions. Thus metals have electropositive valency

**5. Correct option: (B) Negative**

Non-metals can gain electrons and form negative ions. Thus non-metals have electronegative valency.

**6. Correct option: (C) Al**

Ductility is a physical property for which metal can be drawn into thin wire.

The metal which can be drawn into thin wire more easily is more ductile.

Metals are commonly more ductile.

Aluminium is a ductile metal due to it's low metal density.

**7. Correct option: (C) Hydrogen**

Metals are placed above hydrogen in reactivity series and displace it during reaction with acids.

**8. Correct option: (B) Acidic**

Non-metals oxides ($SO_2$ , $CO_2$ etc.) reacts with water to produce acidic compounds.

Example – $SO_2 + H_2O \rightarrow H_2SO_3$

Hence non-metals oxides are acidic in nature

**9. Correct option: (C) Basic**

Metal oxides ( BaO, MgO etc.) reacts with water to produce basic compounds.

Example : $MgO + H_2O \rightarrow Mg(OH)_2$ . So metals oxides are basic in nature.

**10. Correct option: (C) Hydrogen**

Hydrogen gas is liberated when metals reacts with acid as metals are placed above hydrogen in reactivity series.

Example : $Zn(s) + HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$

**11. Correct option: (C) Both 'A' and 'B'**

Metals are good conductor of heat because their particles are closely packed. Hence the vibrations of heat wave passed so quickly.

Metals also contain a large number of free electrons. Thus they are also good conductor of electricity.

**12. Correct option: (A) Non-metal**

Graphite is a allotrope of carbon with multiple hexagonal layers.

Hence it is a non-metal

It has free electrons in its structure as the fourth valence electron of carbon atom remains unpaired.

Hence Graphite is a good conductor of electricity due to presence of free electrons

**13. Correct option: (C) Form acidic oxides**

QMalleability is a physical property of metals which defines the ability of metals to be beaten into sheets without breaking.

Ductility is a physical property of metal for which metal can be drawn into thin wire.

Metals contain a large number of free electrons. Thus they are also good conductor of electricity.

When metals oxides (CaO, BaO etc.) react with dilute acid, they form salt and water.

Hence they are basic oxides.

**14. Correct option: (B) Homogeneous mixture**

Homogeneous mixture is a mixture in which composition is uniform throughout the mixture.

Heterogeneous mixture is a mixture in which composition is not uniform and contains smaller parts.

Brass is a mixture of copper and zinc. As the composition remains same throughout the mixture, it is a homogeneous mixture.

**15. Correct option: (C) Conc.HCl : Conc.$HNO_3$ = 3 : 1**

Aqua regia is mixture of Conc. HCl and Conc. $HNO_3$ at the ratio of 3:1.

It is a red-orange colour fuming liquid which is used to dissolve noble metals like gold and platinum.

**16. Correct option: (B) (ii) and (iii)**

In HCl and $CCl_4$ covalent bond is formed by mutual sharing of electron pairs.

Hence they are covalent compounds.

But in KCl and NaCl ionic bond is formed by donating and accepting of electrons.

Hence they are ionic compounds.

**17. Correct option: (B) Electrical conductivity in solid state**

Ionic compounds are highly soluble in water because their hydration energy is greater than lattice energy.

Ionic compounds break into cation and anion in molten or solution state.

Thus they conduct electricity in molten or solution state due to availability of free ions.

But in solid state they can't conduct electricity due to absence of free ions. Ionic bonds have high lattice energy. Hence they have high melting and boiling point.
So electrical conductivity in solid state is not exhibited by ionic compounds.

## LEVEL - II

**1. Correct option: (D) Carbon (Diamond)**
Diamond is an allotrope of carbon.
In diamond each carbon is covalently bonded with other four carbons. This arrangement makes a tetrahedral crystal in which the atoms are closely packed and make a rigid structure. Thus diamond is the hardest substance.

**2. Correct option: (D) Phosphorus**
In metals, there are free held electrons which creates a sea of electrons. Because of these free electrons metals are good conductors of electricity. But non-metal crystal doesn't contain any free electrons, hence they are not good conductors of electricity.

**3. Correct option: (D) Magnesium**
Substances that have one type of atoms is called pure substance.
Duralumin → Al + Cu + Mg
Magnalium → Mg + Al
Bell metal → Cu + Sn
Magnesium → element → pure substance

**4. Correct option: (D) Aluminium**
In metals, there are free held electrons which creates a sea of electrons. Because of these free electrons metals are good conductors of electricity. But non-metal crystal doesn't contain any free electrons, hence they are not good conductors of electricity.
As sulphur, diamond (carbon) , Phosphorus are non-metals and are not good conductors of electricity.
But Aluminium is metal and one of the best conductor of electricity.

**5. Correct option: (A) Diamond**
Diamond is an allotrope of carbon.
In diamond each carbon is covalently bonded with other four carbons. This arrangement makes a tetrahedral crystal in which the atoms are closely packed and make a rigid structure. Thus diamond is the hardest substance.

**6. Correct option: (D) Magnesium**
Magnesium is present in chlorophyll.
It is located at the centre of chlorophyll.
Magnesium usually helps for the proper functioning of chlorophyll and also stimulates some enzymes to carry photosynthesis.

**7. Correct option: (B) Iron**
Haemoglobin is a metal complex.
The heme group of haemoglobin contains iron as the central metal atom. This heme group is responsible for the binding and release of the oxygen atoms

**8. Correct option: (C) Graphite**

Graphite has layered structure which are held by weak Van der Waals force of attraction. The layers are composed of planar hexagonal rings of carbon. For these lars graphite is a slippery substance and used in pencil.

**9. Correct option: (A) Free electron**

Graphite is an allotrope of carbon. It has layered structure which are held by weak Van der Waals force of attraction. The layers are composed of planar hexagonal rings of carbon. It has free electrons in its structure as the fourth valence electron of carbon atom remains unpaired.
Hence Graphite is a good conductor of heat & electricity due to presence of free electrons.

**10. Correct option: (A) Phosphorous**

Red phosphorus is an allotrope of phosphorus which is actually used in match sticks.
It has low ignition temperature. Hence it can be burnt easily. Thus it is used in match sticks.
Ignition temperature of a substance is the minimum temperature that is required to start or cause combustion.

**One or more than one option correct type**

**11. Correct Options: A, B, D**

- **Explanation:** Potassium (K), Sodium (Na), and Calcium (Ca) are highly reactive alkali and alkaline earth metals that react vigorously with cold water. Magnesium (Mg) is less reactive and reacts only with hot water or steam.

**12. Correct Options: A, B, C**

- **Explanation:** Amphoteric oxides can react with both acids and bases to form salt and water. Zinc oxide (ZnO), Aluminum oxide ($Al_2O_3$), and Lead oxide (PbO) are all amphoteric. Sodium oxide ($Na_2O$) is purely basic.

**13. Correct Options: A, B, C**

- **Explanation:** Metals are ductile (can be drawn into wires), malleable (can be beaten into sheets), and sonorous (produce a ringing sound). They are excellent conductors, so "Non-conductivity" is incorrect.

**14. Correct Options: A, C, D**

- **Explanation:** Ionic compounds have high melting points due to strong electrostatic attractions. They are brittle because shifting the crystal layers brings like-charges together, causing repulsion. They conduct electricity in the molten (or aqueous) state because ions are free to move, but they are insulators in the solid state because the ions are locked in a fixed lattice.

**15. Correct Options: A, B, C**

- **Explanation:** Oxygen ($O_2$), Nitrogen ($N_2$), and Hydrogen ($H_2$) are gases at room temperature. Bromine ($Br_2$) is unique because it is a liquid at room temperature.

**16. Correct Options: A, C**

- **Explanation:** According to Coulomb's Law, lattice energy is directly proportional to the product of ionic charges and inversely proportional to the distance between them (ionic size). Therefore, smaller ionic sizes (bringing charges closer) and higher charges on the ions will increase lattice energy:

Lattice Energy $\propto \frac{Q_1 \cdot Q_2}{r_0}$

**Assertion and Reason Type**

**17. Correct Option: C (A is true, but R is false)**

- **Explanation:** The assertion is true; bromine is indeed the only liquid non-metal at room temperature. The reason is false because non-metals exist in all three states: gases (like $O_2$, $N_2$), liquids ($Br_2$), and solids (like Carbon, Sulfur).

**18. Correct Option: A (Both A and R are true, and R is the correct explanation of A)**

- **Explanation:** Metallic bonding consists of positive metal ions immersed in a delocalized "sea of electrons." Because these bonds are non-directional, atoms or layers can slide past one another under mechanical stress without shattering the crystal structure, allowing metals to be hammered into sheets (malleability).

**Match the Following**

**19. Matching Results:**

- **P** → **c** (Liquid metal is Mercury, Hg)
- **Q** → **a** (Hardest non-metal is Diamond, an allotrope of carbon)
- **R** → **b** (Conducting non-metal is Graphite, which has free electrons)
- **S** → **d** (Reddish metal is Copper, Cu)

**Comprehension Type**

**20. Correct Option: B (Transfer of electrons)**

- **Explanation:** As stated in the passage, ionic bonds are formed when electrons are completely transferred from a metal to a non-metal.

**21. Correct Option: C (Electrostatic forces)**

- **Explanation:** The oppositely charged ions (cation$^+$ and anion$^-$) attract each other via strong electrostatic (coulombic) forces.

**22. Correct Option: C (High melting point)**

- **Explanation:** A large amount of thermal energy is required to break the strong electrostatic forces holding the ionic crystal lattice together.

**Integer Type**

**23. Total number of elements in periodic table = 118**

- **Explanation:** There are currently 118 officially recognized and confirmed elements in the modern periodic table (up to Oganesson).

**24. Number of electrons lost by Na to form $Na^+$ = 1**

- **Explanation:** Sodium (Na, atomic number 11) has an electronic configuration of 2, 8, 1. It loses its single valence electron to achieve a stable octet configuration ($Na^+$: 2, 8).

**25. Charge on Ca ion in $CaCl_2$ = 2 (or +2)**

- **Explanation:** Calcium belongs to Group 2 and loses 2 valence electrons to form a $Ca^{2+}$ cation, balancing the two $Cl^-$ anions.

**26. Number of $H_2$ molecules formed in:** $2Na + 2H_2O \rightarrow 2NaOH + ___ H_2$ **= 1**

- **Explanation:** Balancing the hydrogen atoms on both sides: the reactant side has $2 \times 2 = 4$ hydrogens. The product side has 2 hydrogens in 2NaOH, leaving 2 hydrogens to form exactly 1 molecule of $H_2$.

**27. Maximum electrovalency in ionic compounds = 4**

- **Explanation:** Electrovalency represents the number of electrons lost or gained by an atom to form an ion. The maximum typical electrovalency observed is 4 (e.g., in certain transition metal ions like $Sn^{4+}$ or $Pb^{4+}$), as losing or gaining more than 4 electrons requires an unsustainably high amount of ionization energy.

**28. Total number of non-metals = 20**

- **Explanation:** In the standard classification of the periodic table, there are 20 non-metal elements: 11 gases (H, He, N, O, F, Ne, Cl, Ar, Kr, Xe, Rn), 1 liquid (Br), and 8 solids (C, P, S, Se, I, At, Ts, Og). *(Note: Depending on the curriculum syllabus, some textbooks omit the heavy synthetic elements At/Ts/Og and list it as 17 or 18, but 20 is the modern chemistry convention).*

**29. Number of electrons gained by O to form $O^{2-}$ = 2**

- **Explanation:** Oxygen (atomic number 8) has a configuration of 2, 6. It needs to gain 2 electrons to fill its valence shell and attain a stable octet configuration ($O^{2-}$: 2, 8).

**30. Number of nearest neighbours of $Na^+$ in NaCl crystal = 6**

- **Explanation:** NaCl possesses a face-centered cubic (fcc) crystal structure where the coordination number of both $Na^+$ and $Cl^-$ is 6. Each sodium ion is directly surrounded by 6 chloride ions in an octahedral geometry.

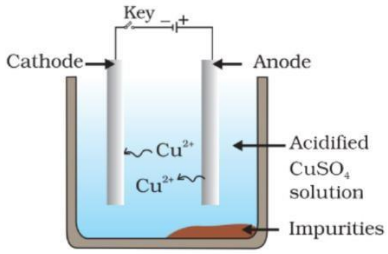
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# Metallurgy

## Level-1

**1. Correct option: (C)**

Electrolytic refining is used for extraction and purification of metals which are obtained by refining methods.
Cathode is negative where positive ions are attached.
Anode is positive which deposit positive ions.




**2. Correct option: (D) $CuSO_4$+Fe**

In reactivity series, the metal placed above is regarded as more reactive.
And a more reactive metal can displace less reactive metal.
Fe placed above Cu in reactivity series. So it can replace Cu and displacement reaction will occur.
But Mg and Zn placed above Fe in reactivity series. Hence reaction isn't possible.
Similarly Mg placed above Pb and reaction doesn't occur.

**3. Correct option: (A) Deposited on cathode**

Electrolytic refining is used for extraction and purification of metals which are obtained by refining methods.
For Zinc
Anode: Impure zinc
Cathode: Pure zinc

Electrolyte: Zinc sulphate solution
On passing current, zinc from the electrolyte is deposited at the cathode.

**4. Correct option: (C) A homogenous mixture**

An alloy is a mixture of two or more metal or a metal and a nonmetal.
Example – brass ( Cu +Zn)
Alloy is a homogeneous mixture as it's composition is uniform throughout the whole mixture.

**5. Correct option: (D) (i) and (ii)**

Electrolysis is a process that leads to chemical change when current passes through an electrolyte.
Sodium and calcium are obtained by electrolysis of their chlorides in molten state as they are highly reactive.
But iron and copper are moderately reactive thus they can't be obtained by electrolysis of their chlorides.

**6. Correct option: (C) Zinc**

Iron in presence of air and moisture gets rusted. The rusting can be prevented by a coating with thin layer of zinc. This process of coating zinc on iron is called galvanisation. Hence zinc is the right answer.

**7. Correct option: (B) $CuCO_3$**

Copper is moderately reactive metal. When it is exposed to air it reacts with gases like carbon dioxide and vapour. This reaction leads to form copper carbonate.
It makes the shiny brown surface dull and green.
$2Cu(s) + CO_2(g) + H_2O(g) \rightarrow CuCO_3(s) + Cu(OH)_2(s)$
Since sulphur is not present in the air. It cannot react with sulphur to make $CuSO_4$.
Copper doesn't react with nitrogen. Thus $Cu(NO3)_2$ is not formed.
Copper first reacts with oxygen to form CuO. But further it reacts with carbon dioxide to form copper carbonate. So $CuCO_3$ is the correct answer.

**8. Correct option: (C) $Ag_2S$**

Silver articles become black on prolonged exposure to air due to thin coating of silver sulphide bring formed on the surface of silver.
Silver is not affected by dry or moist air.
But in the presence of hydrogen sulphide in the surroundings silver is affected . It reacts with the silver to form black coloured silver sulphide.
$2Ag(s) + H_2S(g) \rightarrow Ag_2S(s) + H_2(g)$

**9. Correct option: (A) (i) and (ii)**

Electrolytic refining is a process in which the extraction and purification of metals is done.
The impure metal is used as an anode and the pure metal is used as a cathode.
Copper and Gold is refined electrolytically as they are less reactive.
Sodium and potassium are highly reactive metals and are extracted by the electrolytic reduction of molten chloride salts.

**10. Correct option: (B) chemical; bauxite**

Bayer's method for the extraction of Alumina from bauxite and gangue.
The difference in chemical affinity between bauxite and gangue is utilized for concentrating the ore.

**11. Correct option: (A) iron forms ferrous sulphate solution**

Iron is more reactive than copper. When nail (iron) is dipped in copper sulphate solution, it displaces copper from copper sulphate and forms greenish blue ferrous sulphate. Thus the solution turns light green colour.

$CuSO_4(aq) + Fe(s) \rightarrow FeSO_4(aq) + Cu(s)$

**12. Correct option: (A) $CuCO_3.Cu(OH)_2$**

Copper reacts with oxygen and vapour in the air to form copper oxide.

Copper oxide then combines with carbon dioxide to make copper carbonate, which gives it a green colour.

The green material is a mixture of copper hydroxide and copper carbonate.

**13. Correct option: (D) Covering with glass**

Corrosion is a process in which refined metals are converted to their more stable oxides.

It can be prevented if a barrier between metal and oxygen can be created.

Anodizing is process used to increased the natural oxide.

Alloying is a process in which two or more metals are combined together.

Painting , Anodizing and alloying can prevent corrosion.

But covering with glass can't prevent metal to get touch from oxygen. So it will not be able to prevent corrosion.

**14. Correct option: (C) (ii) and (iv)**

Au and Ag are placed below in the reactivity series. They are highly non reactive as they do not lose electrons more easily.

Thus Au and Ag exists in their native state in Nature.

But Cu and Zn are highly reactive. And they react with other elements.

**15. Correct option: (A) oxygen**

Non-metals are the species which are electronegative and have the tendency to gain electron to form negative ions.

Oxygen is the most abundant non-metal in the earth's crust.

**16. Correct option: (A) decreasing order of reactivity**

Reactivity series is a list of metals arranged in decreasing order of their reactivity.

Most reactive metals are at the top while the least reactive metals at the bottom.

Hence option A is correct.

## Level-II

**1. Correct option: (C) Gold, Mercury**

An amalgam is a homogeneous mixture of metal and mercury.

If in alloy one of the components is mercury then it is called and amalgam.

Example *:* gold amalgam → Au + Hg → gold + mercury

**2. Correct option: (A) isolation**

An ore is full with many undesired earthly substances that are called gangue. The process of removing unwanted materials such as clay, sand etc. from the ore is termed as zone refining, benefaction or dressing.

Whereas isolation is used for conversion of oxide of the ore.

**3. Correct option: (C) zinc with iron**

Iron in presence of air and moisture gets rusted. The rusting can be prevented by a coating with thin layer of zinc. This process of coating zinc on iron is called galvanisation

**4. Correct option: (A) alloy of mercury**

An amalgam is a homogeneous mixture of metal and mercury.
If in alloy one of the components is mercury then it is called and amalgam.
Example – (Na + Hg) is called sodium amalgam.

**5. Correct option: (B) Cu**

Fe , Mg and Zn are placed above Cu in reactivity series. Thus they can react with dilute HCl and produce $H_2$ gas.

$Fe + 2HCl\ (dil.) \rightarrow FeCl_2 + H_2 \uparrow$

$Zn + 2HCl\ (dil.) \rightarrow ZnCl_2 + H_2\uparrow$

$Mg + 2HCl\ (dil.) \rightarrow MgCl_2 + H_2\uparrow$

But Cu is less reactive and cannot react with dilute HCl

$Cu + 2HCl\ (dil.) \rightarrow$ No reaction

**6. Correct option: (A) Gold , Platinum, B) Gold, Silver**

Silver, gold and platinum come downwards in the reactivity series. They are less reactive and called inert metal. Even if in high temperature these elements do not react with oxygen.
But Zinc , lead and copper come upwards in the reactivity series. Hence they are more reactive and can react with oxygen.

**7. Correct option: (D) electrolytic reduction**

The oxides of highly reactive metals like K, Na cannot be oxidised by carbon. Because they have high affinity for oxygen then carbon.
Thus carbon is unable to remove oxygen from metal oxides.
Thus oxides of highly reactive metals are reduced by a process called electrolytic reduction.
Electrolytic reduction is a process where the metal ions comes in the touch of cathode, accepts electrons from it and get reduced .

**8. Correct option: (B) ores**

A mineral is called an ore if a metal is profitably extracted from it.
Through metallurgy metals are extracted from ore.

**9. Correct option: (B) refining**

The process used in purifying of metals is called refining.
Removal of impurities from metal ores are called enrichment of ore.
Cracking is a process in which the minerals are dissolved in acid and gets simplified
Roasting is process in which ore is converted into its oxide by heating it below its melting point in presence of excess air.

**10. Correct option: (C) aluminothermic**

Thermite reaction is a highly exothermic reaction which involves reaction between iron oxide with aluminium to produce molten iron.
This is also known as aluminothermic process.
Hall's process is the process for smelting aluminium.
Bayer's process is used to produce alumina from refined bauxite.
Lead chamber is used to produce sulfuric acids industries

**11. Correct option: (C) calcinations**

The process in which carbonate ores are converted to oxides is known as calcination.
In this process the ore is heated to high temperature in presence of absence of air.

**12. Correct option: (A) carbon**

Alloy is a homogeneous mixture of two or more metals or a metal and a nonmetal.

Carbon is non-metal which forms an alloy.

Example of carbon alloy – Carbon steel

**13. Correct option: (C) zinc is more reactive than hydrogen**

Zinc is situated above hydrogen in the reactivity series. So zinc is more reactive than hydrogen.

Hence it can react with HCl and replace hydrogen to produce $H_2$ gas.

$Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$

**14. Correct option: (D) collectors; froth stabilizers**

Froth floatation is the method employed in sulphide ores to remove the gangue.

To enhance the mineral particle non-wettability, fatty acids, xanthates, pine oils, etc are used as collectors.

For the stabilization of froth, cresols, aniline, etc can be used as froth stabilizers.

**15. Correct option: (C) depressants**

Froth floatation is the method employed in sulphide ores to remove the gangue.

It Is possible to separate two sulphide ores by adjusting the proportion of oil to water or by using depressants.

**16. Correct option: (A) an alloy**

Alloy is a homogeneous mixture of two or more metals or a metal and a nonmetal.

Minerals is a substance that is formed in the earth.

A mineral is called an ore if a metal is profitably extracted from it.

Flux is a chemical that is added to molten metal to connect with impurities that may then be removed.

**17. Correct option: (B) hydrogen**

Hydrogen is placed in the reactivity series. Because it has one electron in its valence shell. It can gain one electron to become a negative ion ($H^-$) and acts like a nonmetal.

And also can donate one electron to become a positive ion ($H^+$) and acts like a metal.

**18. Correct option: (C) Ag**

Corrosion is process which converts a refined metal into a more chemically stable oxide.

The metals that are less reactive does not corrode when exposed to air.

As silver (Ag) comes downwards in the reactivity series, it is less reactive and does not corrode.

Whereas Cu, Al and Fe come upwards in the reactivity series and are highly reactive. Thus corrosion occurrs.

**19. Correct option: (B) reduction**

The process of removal of unwanted materials from the ore is known as ore concentration.

Ore concentration can be accomplished through a variety of methods, including hydraulic washing, magnetic separation, froth flotation.

Hydraulic washing is used when the impurities are lighter and ore particles are heavier.

Froth floatation is the method employed in sulphide ores to remove the gangue. Magnetic separation is a process in which solid ores are separated by a magnet.

**20. Correct option: (B) Magnesium**

Magnesium comes above zinc in reactivity series. Whereas Lead, copper and silver comes below zinc in reactivity series.

Thus Magnesium is more reactive than zinc and it can replace zinc in zinc sulphate solution.

$Mg(s) + ZnSO_4(aq) \rightarrow MgSO_4(aq) + Zn(s)$

**One or more than one answer type:**

**21. Correct Options: A, B, D**

- **Explanation:** Ores are minerals from which metals can be extracted profitably and conveniently. Bauxite (for Aluminum), Haematite (for Iron), and Zinc blende (for Zinc) are classic examples of industrial ores. Clay contains aluminum, but it is not considered an ore because extracting aluminum from it is commercially unprofitable.

**22. Correct Options: A, B, D**

- **Explanation:** In the reactivity series, Potassium (K), Sodium (Na), and Calcium (Ca) sit right at the top. They react vigorously even with cold water and air. Gold (Au) sits at the very bottom and is highly unreactive (noble).

**23. Correct Options: A, B, D**

- **Explanation:** Highly unreactive metals are found in their elemental (native or free) state because they do not easily combine with oxygen, moisture, or sulfur. Gold (Au) and Platinum are almost exclusively found in a free state. Silver (Ag) and Mercury (Hg) are also occasionally found in their native states alongside their sulfide compounds. Sodium is too reactive to ever exist in a free state.

**24. Correct Options: A, B, C**

- **Explanation:** Gangue refers to the unwanted, commercially worthless earthy, rocky, or sandy impurities associated with an ore in a deposit. This includes sand (silica), clay, and rocky limestone. It explicitly excludes the pure metal itself.

**25. Correct Options: A, C**

- **Explanation:** A flux is a chemical substance added to the furnace to remove infusible gangue impurities. The flux reacts chemically with the gangue to form a fusible, easily removable waste compound called slag (Gangue + Flux → Slag).

**26. Correct Options: A, B, C**

- **Explanation:** "Pyro" refers to high heat. Pyrochemical operations are metallurgical processes that use high temperatures to bring about chemical alterations in the ore. Smelting, Roasting, and Calcination are all performed at elevated temperatures in furnaces. Electrolysis relies on electrical energy rather than high thermal chemical reactions.

**27. Correct Options: A, B, C**

- **Explanation:** Smelting is the process of heating a roasted or calcined ore with a reducing agent (like coke/carbon) and a flux past its melting point. During this, the ore is reduced to its free element, the metal melts down to the bottom in a molten state, and the flux binds with impurities to form a floating layer of slag. High heat is mandatory.

**28. Correct Options: A, B, C**

- **Explanation:** Metals of moderate reactivity populate the middle of the activity series. These include Zinc (Zn), Iron (Fe), Lead (Pb), and Copper (Cu). Potassium sits right at the top of the series.

**Assertion and Reason Type**

**29. Correct Option: C (A is true, R is false)**

- **Explanation:** Every ore is a mineral because it occurs naturally in the earth's crust. However, not every mineral is suitable for metal extraction; many minerals contain too few metal percentages or too many stubborn impurities to make extraction economically viable.

**30. Correct Option: A (Both A and R are true, and R is the correct explanation)**

- **Explanation:** Roasting is defined as heating concentrated sulfide ores strongly below their melting point in a controlled excess of air (oxygen). The primary objective of this process is to chemically transform sulfide minerals into easily reducible metal oxides while burning away volatile impurities like sulfur dioxide.

**Match the Following**

**31. Matching Results:**

- P → b (Froth flotation is specifically used to concentrate hydrophobic Sulphide ores)
- Q → a (Magnetic separation uses rollers to separate Magnetic ores like magnetite from non-magnetic gangue)
- R → d (Hand picking is used to discard easily Visible impurities based on color or shape differences)
- S → c (Washing/Gravity separation relies heavily on Density differences to wash away lighter gangue from heavy ore particles)

**Comprehension Type**

**32. Correct Option: C (Air and water)**

- **Explanation:** For iron to rust, both oxygen (from the air) and moisture/water are essential components of the electrochemical corrosion cell. If either is missing, rusting stops.

**33. Correct Option: B (Chemical change)**

- **Explanation:** Corrosion involves a true chemical reaction where elemental metal reacts with atmospheric components to produce entirely new compound substances like oxides, hydroxides, or sulfides.

**34. Correct Option: B (Iron oxide)**

- **Explanation:** Chemically, rust is composed of hydrated ferric oxide, represented by the formula $Fe_2O_3 \cdot xH_2O$.

**Integer Type**

**35.** Number of oxygen atoms in $Fe_2O_3$ = 3

- Explanation: Looking directly at the chemical formula for iron(III) oxide ($Fe_2O_3$), there are 2 Iron atoms bound to 3 Oxygen atoms.

**36.** Number of electrons gained by $Na^+$ in electrolysis = 1

- Explanation: During the electrolytic reduction of a molten sodium salt (like NaCl), the positive sodium cation ($Na^+$) travels to the negative cathode where it gains exactly one electron ($e^-$) to reduce into neutral metallic sodium: $Na^+ + e^- \rightarrow Na$

# 6. General Organic Chemistry

## Solutions

### LEVEL – I

**1. Correct option: (B)**

$H_2N-CO-NH_2$(urea/carbamide) is considered an organic compound. Organic compounds are primarily composed of carbon and hydrogen atoms, often with other elements like oxygen, nitrogen, sulfur, and more. In this case, $H_2N-CO-NH_2$ is a compound that contains carbon and hydrogen atoms.

**2. Correct option: (D)**

Friedrich Wohler is famous for synthesizing urea from ammonium cyanate in 1828, marking a pivotal moment in the history of chemistry and demonstrating that organic compounds could be synthesized from inorganic substances.

**3. Correct option: (D)**

Friedrich Wohler synthesized urea from ammonium cyanate in 1828, and this is considered a landmark achievement in the history of chemistry, as it demonstrated that organic compounds could be artificially produced from inorganic materials.

**4. Correct option: (A)**

In the early days of chemistry, organic compounds were defined as those compounds that contained carbon, and urea contains carbon atoms in its chemical structure. So, it is an organic compound.

**5. Correct option: (C)**

For the first time in 1845, German chemist Hermann Kolbe synthesized acetic acid from inorganic compounds.

**6. Correct option: (D)**

Organic molecules are composed of carbon and hydrogen atoms, often with other elements like oxygen, nitrogen, sulfur, and more. Based on this definition, the following options are organic molecules:

**Proteins** (composed of carbon, hydrogen, oxygen, nitrogen, and sometimes sulfur)
**Rubber** (composed of carbon and hydrogen atoms, typically in the form of long hydrocarbon chains)

**7. Correct option: (A)**

Organic compounds are defined by the presence of carbon atoms in their chemical structure. While organic compounds can also contain other elements like hydrogen, oxygen, nitrogen, and others, carbon is the key element that distinguishes them from inorganic compounds.

**8. Correct option: (B)**

Hydrogen is the second most abundant element in organic compounds, as it frequently forms covalent bonds with carbon in organic molecules.

**9. Correct option: (D)**

Compounds that contain carbon and hydrogen only are called Hydrocarbons.

**10. Correct option: (D)**

The radioactive isotope of carbon widely used in carbon dating to determine the age of fossils is C-14 (Carbon-14).

**11. Correct option: (C)**

Carbon has the unique ability to form long chains and rings with other carbon atoms, creating a vast variety of organic compounds. This property is known as catenation,

and it allows for the formation of a wide range of organic molecules, contributing to the diversity of carbon compounds.

**12. Correct option: (C)**

Carbon has the strongest catenation ability among the listed elements. Carbon is well-known for its ability to form long chains and rings with other carbon atoms, resulting in the vast diversity of organic compounds. While other elements like silicon, sulfur, and phosphorus can also form chains, they are not as versatile or extensive in their catenation as carbon.

**13. Correct option: (C)**

Carbon-nitrogen triple bonds are found in compounds like cyanides, where carbon is bonded to a nitrogen atom with a triple covalent bond. Each carbon and nitrogen atom will share three electrons. As a result, six electrons are shared, and a triple covalent bond is formed.

**14. Correct option: (B)**

Different compounds having the same molecular formula are called **Isomers**.

**15. Correct option: (C).**

$C_2H_6$ is a saturated compound. Saturated hydrocarbons have only single covalent bonds between carbon atoms, and ethane ($C_2H_6$) is a saturated hydrocarbon consisting of single carbon-carbon and carbon-hydrogen bonds.

**16. Correct option: (B)**

$C_3H_6$ is an unsaturated compound. Unsaturated hydrocarbons have at least one or more carbon-carbon double or triple bonds. In this case, $C_3H_6$ (propene) has a carbon-carbon double bond, making it an unsaturated compound.

**17. Correct option: (A)**

Cyclohexane is an alicyclic compound. Alicyclic compounds are organic compounds that have carbon atoms arranged in a closed ring structure, and cyclohexane consists of a ring of six carbon atoms. It is a saturated alicyclic hydrocarbon with single bonds between all carbon atoms in the ring.

**18. Correct options: (A, B, C.)**

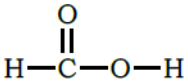
The non-benzenoid aromatic compounds are those which do not contain any benzene ring in their structure. So, options A, B, C are non-Benzenoid compounds.

**19. Correct option: (B)**

$H_3\underset{1}{C}-\underset{2}{C}\equiv\underset{3}{C}-\underset{4}{C}H_3$ in this compound 1st,4th carbon atom produce 4 $sp^3$ hybrid orbitals.

**20. Correct option: (B)**

Kekule structures are similar to Lewis Structures, but instead of covalent bonds being represented by electron dots, the two shared electrons are shown by a line.



## LEVEL – II

**1. Correct option: (C)**

$\underset{1}{C}H_3\text{-}\underset{2}{C}H=\underset{3}{C}H\text{-}\underset{4}{C}H_2\text{-}\underset{5}{C}H_3$

A carbon atom attached to a double bond is $sp^2$ hybridized. Here $C_2$ is attached to double bond it is $sp^2$ hybridized and $C_4$ is $sp^3$ hybridized.

**2. Correct option: (A)**

A carbon atom attached to a double bond is $sp^2$ hybridized. A carbon atom attached to two double bonds or one triple bond is $sp$ hybridized.

So here $C_2$ carbon atom is $sp^2$ hybridized. $C_4$ Carbon atom is $sp^3$ hybridized.

$C_5$ Carbon atom is $sp$ hybridized.

**3. Correct option: (B)**

$(CH_3-CH_2-CH_2-CH_3)$ It can be written in form of $C_4H_{10}$. Here all C atoms are $sp^3$ hybridized so each gives 4 hybrid orbitals i.e. $4\times4=16$ in total. All hydrogen molecules can be calculated as $10\times1=10$.

Ratio can be written as $\begin{matrix}=10:16\\=5:8\end{matrix}$

**4. Correct option: (C)**

Neopentane (also known as 2, 2-dimethylpropane) has the molecular formula $C_4H_{12}$.

All bonds in neopentane are sigma bonds. Bonds between C-C are4, bonds between C-H are 12 in total 4+12=16.

**5. Correct option: (C)**

The first homologue of an alkyne is $C_2H_2$, which is ethyne.

**6. Correct option: (D)**

The compounds in a particular homologous series differ by a $-CH_2$ unit.

**7. Correct option: (B)**

Given formula is $C_nH_{2n+1}OH$ place n=1 then we get $C_2H_5OH$ (ethyl alcohol).

**ONE OR MORE THAN ONE OPTION CORRECT TYPE**

**8. Correct options: (A, B, D)**

$C_1-C_2\rightarrow2\pi,1\sigma$, $C_3-C_2\rightarrow0\pi,1\sigma$ , $C_3-C_4\rightarrow1\sigma,1\pi$ bonds are present.

**9. Correct options:B,C**

B) Cyclopropenyl cation ($C_3H_3^+$)

- This is the cyclopropenyl cation.
- It is a non-benzenoid aromatic compound because:
  - It is cyclic.
  - It is planar.
  - It has 2 π electrons, satisfying Hückel's rule ($4n + 2$, $n = 0$).
  - The positive charge leaves one empty p-orbital, allowing continuous conjugation.
- Aromatic and non-benzenoid.

C) Tropylium cation ($C_7H_7^+$)

- This is the tropylium ion.
- It is also non-benzenoid aromatic because:
  - It is cyclic and planar.
  - It has 6 π electrons, satisfying Hückel's rule.
- Aromatic and non-benzenoid.

**10. Correct options: (A, B, D)**

Compounds which obey huckels rule $(4n+2)\pi$ electrons and have a conjugated electron system are called as aromatic compounds.

-6$\pi$ electrons.

-2$\pi$ electrons

-2$\pi$ electrons.

**11. Correct options: (A, C)**

**Assertion and reason type:**

**12. Correct Option :D**

- Assertion (A): Carbon compounds are mostly ionic in nature.
  - *Evaluation:* False. Carbon has 4 valence electrons and a medium electronegativity, meaning it prefers to share electrons rather than transfer them. Therefore, carbon compounds are predominantly covalent in nature.
- Reason (R): Carbon readily loses or gains electrons.
  - *Evaluation:* False. Carbon cannot easily lose 4 electrons (requiring an enormous amount of ionization energy to form $C^{4+}$) nor can it easily gain 4 electrons (as it is energetically unstable for a nucleus with 6 protons to hold 10 electrons to form $C^{4-}$). Instead, it shares electrons.

  Since both statements are incorrect

**13. Correct Option :A) Both A and R are true and R is the correct explanation of A**

Assertion (A): Bond length decreases as bond order increases.

- *Evaluation:* True. A higher bond order means more shared electron pairs between the two nuclei (e.g., a triple bond has a bond order of 3, while a single bond has a bond order of 1). This pulls the atoms closer together, shortening the bond length.

  Reason (R): Higher bond order means greater attraction between bonded atoms.
- *Evaluation:* True. More shared electrons result in a stronger electrostatic pull between the positively charged nuclei and the shared negative electrons, which is exactly why the atoms are drawn closer together.

**MATRIX MATCHING TYPE**

**14. Correct Matching is: a-P; b-R; c-S; d-Q**

a) $C_2H_6$ - In ethane, each carbon atom forms sigma bonds with three hydrogen atoms and a sigma bond with the other carbon atom. These sigma bonds are formed by the overlap of atomic orbitals. To accommodate this bonding, carbon atoms in ethane undergo $sp^3$ hybridization, which means that one s orbital and three p orbitals on each carbon atom combine to form four $sp^3$ hybrid orbitals. These four $sp^3$ hybrid orbitals then overlap with the hydrogen 1s orbitals to form four sigma bonds.

**Correct option is P.**

b) $C_2H_2$ - Acetylene has a linear molecular structure, and the carbon atoms are bonded together by a triple bond. In acetylene, each carbon atom forms two sigma (σ) bonds and one pi (π) bond. To accommodate this bonding arrangement, carbon atoms in acetylene use sp hybridization.

**Correct option is R.**

c) $C_3H_6$ - C3H6 can represent two different isomers Propene (or Propylene). Molecular Structure: $CH_2-CH_2-CH_3$

In propene, the carbon-carbon double bond (C=C) is present. The carbon atoms participating in the double bond are sp2 hybridized. The carbon atom bonded to only one other carbon and one hydrogen is sp3 hybridized.

Cyclopropane- In Cyclopropane, all three carbon atoms are bonded to each other in a ring. Each carbon atom in Cyclopropane is $sp^3$ hybridized. So, the hybridization of carbon atoms in propene is $sp^2$, while in Cyclopropane, it is $sp^3$.

**Correct option is S.**

d) The molecular formula $C_7H_8$ suggests that the compound likely contains a benzene ring, which is a six-membered ring of carbon atoms with alternating single and double bonds. The carbon atoms in a benzene ring are $sp^2$ hybridized. So, in $C_7H_8$, the carbon atoms are $sp^2$ hybridized due to the presence of the benzene ring.

**Correct option is Q.**

**15. a) Benzene ($C_6H_6$)**

- **Hybrid Orbitals:** All 6 carbons are $sp^2$ hybridized.
  Total Hybrid Orbitals = $6 \times 3 = 18$
- **Pure Orbitals:** 6 hydrogen atoms (1$s$ orbitals) + 6 unhybridized $p$-orbitals from the 6 carbons (forming the $\pi$-system).
  Total Pure Orbitals = $6 + 6 = 12$
- **Ratio:** $18:12 = 3:2$
- **Match: a → R**

**b) Ethane ($C_2H_6$)**

- **Hybrid Orbitals:** Both carbons are $sp^3$ hybridized.
  Total Hybrid Orbitals = $2 \times 4 = 8$
- **Pure Orbitals:** 6 hydrogen atoms (1$s$ orbitals) and no $\pi$-bonds.
  Total Pure Orbitals = 6
- **Ratio:** $8:6 = 4:3$
- **Match: b → S**

**c) Ethyne ($C_2H_2$)**

- **Hybrid Orbitals:** Both carbons are $sp$ hybridized.
  Total Hybrid Orbitals = $2 \times 2 = 4$
- **Pure Orbitals:** 2 hydrogen atoms (1$s$ orbitals) + 4 unhybridized $p$-orbitals (2 from each carbon forming two $\pi$-bonds).
  Total Pure Orbitals = $2 + 4 = 6$
- **Ratio:** $4:6 = 2:3$
- **Match: c → P**

**d) Ethene ($C_2H_4$)**

- **Hybrid Orbitals:** Both carbons are $sp^2$ hybridized.
  Total Hybrid Orbitals = $2 \times 3 = 6$
- **Pure Orbitals:** 4 hydrogen atoms (1$s$ orbitals) + 2 unhybridized $p$-orbitals (1 from each carbon forming one $\pi$-bond).
  Total Pure Orbitals = $4 + 2 = 6$
- **Ratio:** $6:6 = 1:1$
- **Match: d → Q**

## COMPREHENSION TYPE

### Passage – I:

**16. Correct option: (A)**

Primary carbon is those which are attached to one carbon atom only. So $C_1$ carbon $1^0$ atom because it is attached to one carbon atom only. $C_2$ Carbon $1^0$ atom because it is attached to one carbon atom only.

**17. Correct option: (C)**

Primary carbon are which is attached to one carbon atom only, secondary carbon are those which are attached to two carbon atoms, tertiary carbon are those which

is attached to three carbon atoms, and quaternary carbon are those which is attached to three carbon atoms.

In the given compound carbons 1 & 2 are $1^0$ and $4^0$.

**18. Correct option: (C)**

Tertiary carbon is those which is attached to three carbon atoms. In the given question only $C_3$ fulfill the above statement.

**19. Correct option: (B)**

Tertiary carbon is those which are attached to three carbon atoms. Carbons 2 and 4 are tertiary carbon atoms.

**Integer**

**20. Answer: 2**

The two methyl carbon atoms ($\underline{C}H_3$) are attached to 4 single bonds (3 C-H and 1 C-C). They are $sp^3$ hybridized.

The two central carbon atoms ($\underline{C} \equiv \underline{C}$) are involved in a triple bond. They form 2 $\sigma$-bonds each. They are $sp$ hybridized.

- Types of hybridization present = $sp^3$ and $sp$

**21. Answer: 9**

**Central double bond:** C = C consists of 1 $\sigma$-bond and 1 $\pi$**-bond**.

**Cyano groups (**-C ≡ N**):** There are 4 cyano groups. Each triple bond (C ≡ N) contains 1 $\sigma$-bond and 2 $\pi$**-bonds**.

Contribution from cyano groups = $4 \times 2 = 8$ $\pi$-bonds

- Total $\pi$-bonds = 1 (from C=C) + 8 (from four $-C \equiv N$ groups) = 9

**22. Answer: 4**

**Naphthalene:** Cyclic, planar, fully conjugated, 10 $\pi$-electrons ($4n + 2$, where $n = 2$). → **Aromatic**

**Furan:** Cyclic, planar, fully conjugated heterocyclic ring, 6 $\pi$-electrons (includes one lone pair from oxygen; $4n + 2$, where $n = 1$). → **Aromatic**

**Pyridine:** Cyclic, planar, fully conjugated heterocyclic ring, 6 $\pi$-electrons ($4n + 2$, where $n = 1$). → **Aromatic**

**Cyclohexane:** Saturated ring, non-planar ($sp^3$ carbons). → **Non-aromatic**

**Benzene:** Cyclic, planar, fully conjugated, 6 $\pi$-electrons ($4n + 2$, where $n = 1$). → **Aromatic**

**Cyclopropane:** Saturated ring ($sp^3$ carbons). → **Non-aromatic**

**Cyclobutadiene:** Cyclic, planar, fully conjugated, but has 4 $\pi$-electrons ($4n$). → **Anti-aromatic**

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# Nomenclature

## LEVEL - I

**1. Answer: B) Grain alcohol**

- **Explanation:** Ethanol ($C_2H_5OH$) is traditionally produced by fermenting sugars from grains like corn, wheat, or barley, which is why its common/trivial name is grain alcohol. (Wood spirit is methanol, $CH_3OH$).

**2. Answer: D) Formic acid**

- **Explanation:** The name "formic acid" comes from the Latin word for ant, *formica*. Red ants inject this acid when they bite, causing a stinging sensation.

**3. Answer: C) Apple**

- **Explanation:** Malic acid gets its name from the Latin word *malum*, meaning apple. It is the primary organic acid responsible for the sour taste in apples and many other fruits.

**4. Answer: D) -yne**

- **Explanation:** The primary suffix indicates the type of carbon-carbon bond in the principal chain. Because this molecule contains a carbon-carbon triple bond (C ≡ C), the suffix used is **-yne**.

**5. Answer: B) Pent**

- **Explanation:** Let's count the continuous chain of carbon atoms from one end to the other along the zig-zag line. The structure is an alkene chain with a total of **5 carbon atoms** ($CH_3 - CH = CH - CH_2 - CH_3$). A 5-carbon chain corresponds to the root word **Pent**.

**6. Answer: A) Methyl, hex, alcohol**

- **Explanation:**
  - **Longest Carbon Chain containing the principal functional group (**-OH**):** Starting from the -$CH_2$OH carbon and moving through the longest horizontal path gives a total of **6 carbons**. Thus, the root word is **hex**.
  - **Prefix (Substituent):** There is a methyl group (-$CH_3$) attached to carbon-3 of the principal chain. Thus, the prefix is **methyl**.
  - **Suffix (Functional Group):** The principal functional group is an alcohol (-OH), so the suffix nature is **alcohol** (or -ol).

**7. Answer: B)** $C_nH_{2n+1}$

- **Explanation:** An alkyl group is formed by removing one hydrogen atom from an alkane. Since alkanes have the general formula $C_nH_{2n+2}$, removing one hydrogen gives $C_nH_{2n+1}$.

**8. Answer: A)** $C_5H_{12}$

- **Explanation:** Pentyl is an alkyl group containing 5 carbon atoms. As established above, an alkyl group is obtained by removing a hydrogen atom from its corresponding alkane. Therefore, pentyl (-$C_5H_{11}$) is derived directly from pentane ($C_5H_{12}$).

**9. Answer: C) 2,3-Dimethylbutane**

- **Explanation:**

1. The longest continuous carbon chain has **4 carbon atoms**, so the root word is **butane**.
2. Numbering from either side gives substituents at positions **2** and **3**.
3. The two methyl groups at these positions make it **2,3-dimethylbutane**.

**10. Answer: B) 2,2-Dimethyl propane**

- **Explanation:**

1. The longest continuous carbon chain consists of **3 carbon atoms**, so the root word is **propane**.
2. The central carbon (carbon-2) holds two methyl groups.
3. Therefore, the IUPAC name is **2,2-dimethylpropane**. (Note: *Neopentane* is its common/trivial name).

**11. Answer: D) 2,3,4-trimethyl hexane**

- **Explanation:**

1. Let's find the longest continuous carbon chain. Starting from the left end and tracing to the rightmost end gives a chain of **6 carbon atoms** (**hexane**).

2. Numbering the chain from left to right puts the methyl substituents at carbons **2, 3, and 4**. (Numbering from right to left would give positions 3, 4, 5, which is higher).
3. Combining the positions and names gives **2,3,4-trimethyl hexane**.

**12. Answer: A)**

- **Explanation:**

1. The root word **pentane** means the main horizontal chain must have 5 carbons: $C_1 – C_2 – C_3 – C_4 – C_5$.
2. At carbon-2, there must be two methyl groups ($-CH_3$).
3. At carbon-4, there must be one methyl group ($-CH_3$).
4. Looking at option **A**: $H_3C – C(CH_3)_2 – CH_2 – CH(CH_3) – CH_3$, numbering from the left side gives two methyls at position 2 and one at position 4.

**13**. **Correct Option: D) 4-Ethyl-3-methyloctane**

**Explanation:**

1. **Find the longest carbon chain:** Trace a continuous path starting from the ethyl group on the left, going through the backbone, and extending down the long branch on the right. This paths gives the maximum length of **8 carbon atoms**, making the root word **octane**.
2. **Numbering the chain:** Number from left to right to give the substituents the lowest possible positions:
   - Carbon-3 has a **methyl** group.
   - Carbon-4 has an **ethyl** group.
3. **Alphabetical ordering:** When writing the final name, "ethyl" comes before "methyl" alphabetically, which yields **4-Ethyl-3-methyloctane**.

**14**. **Correct Option: B) Butanedial**

**Explanation:**

Let's check the molecular formulas of all the given functional isomers:

- **Butanoic acid:** $CH_3CH_2CH_2COOH \rightarrow C_4H_8O_2$
- **Ethylethanoate:** $CH_3COOCH_2CH_3 \rightarrow C_4H_8O_2$
- **Methyl propanoate:** $CH_3CH_2COOCH_3 \rightarrow C_4H_8O_2$
- **Butanedial:** A 4-carbon chain containing two aldehyde groups ($CHO-CH_2-CH_2-CHO$). Its molecular formula is $C_4H_6O_2$. Because it has two double bond equivalents (two carbonyl groups), it lacks two hydrogens compared to the target formula.

**15**. **Correct Option: B) Pentanone *(Pentan-3-one)***

**Explanation:**

1. The longest continuous carbon chain containing the functional group contains **5 carbon atoms**, so the root name is **pentane**.
2. The principal functional group is a ketone (-CO-), which takes the suffix **-one**.
3. Numbering from either side places the carbonyl carbon at position **3**, giving **pentan-3-one** (or 3-pentanone), which simplifies to the choice **Pentanone**.

**16**. **Correct Option: D) 2-chloropropenoyl chloride**

**Explanation:**

1. **Identify the principal functional group:** The highest priority functional group here is the acyl chloride (-COCl), which marks carbon-1.
2. **Find the parent chain:** The chain contains 3 carbons ($C_1(=O)Cl – C_2H(Cl) – C_3H_3$), so the parent name is **propanoyl chloride**.
3. **Add the substituent:** A chlorine atom is located at carbon-2, making the exact IUPAC name **2-chloropropanoyl chloride**.

**17**. **Correct Answer:C) But-2-enal**
**Explanation:**

1. **Common structure:** Crotonaldehyde is the common name for the unsaturated aldehyde:
   $CH_3 - CH = CH - CHO$
2. **Parent chain & numbering:** The chain consists of **4 carbon atoms** (**but-**), numbered starting from the aldehyde group (-CHO) at position 1.
3. **Combining elements:** The carbon-carbon double bond begins at carbon-2 (**-2-en-**), and the primary suffix for the aldehyde is **-al**. Combining these gives **but-2-enal**.

**18**. **Correct Option: D) 2-hexyne**
**Explanation:** A carbon involved in a triple bond (-C ≡ C-) is *sp* hybridized and has a **linear geometry** (180° bond angle). This linear arrangement extends to the adjacent single bonds directly attached to the triple-bonded carbons (C-C ≡ C-C).

- In **2-hexyne** ($CH_3$-**C ≡ C**-**$CH_2$**-$CH_2$-$CH_3$), carbons 1, 2, 3, and 4 form a continuous, straight linear geometry ($C_1$-$C_2$-$C_3$-$C_4$).

**19**. **Correct Option: C) 3, 3-Dimethyl-1-Butene**
**Explanation:**

1. The longest carbon chain containing the double bond has **4 carbon atoms**, so the root name is **butene**.
2. Number the chain from right to left to give the double bond the lowest locant: $C_1$=$C_2$-$C_3$-$C_4$. Thus, it is a **1-butene**.
3. Carbon-3 has two methyl substituents, yielding **3,3-dimethyl-1-butene**.

**20**. **Correct Option: A) 3-Penten-1-yne**
**Explanation:**

1. The continuous chain has **5 carbon atoms**, so the root word is **pent**.
2. When a double bond and a triple bond are at equidistant positions from the two ends, the **double bond gets lower numbering priority** by rule. However, here, numbering from the right end gives the triple bond position **1** ($C_1 \equiv C_2$-$C_3$=$C_4$-$C_5$), whereas numbering from the left gives the double bond position **2**. We follow the **lowest locant set rule**, which favors numbering from the right end (1,3) over the left end (2,4).
3. The double bond starts at position 3 (**3-en**) and the triple bond at position 1 (**1-yne**). Combining them alphabetically gives **pent-3-en-1-yne** (or 3-penten-1-yne).

**21**. **Correct Option: A)** HO - $CH_2$ - CH = $CH_2$
**Explanation:** In carbinol nomenclature, the methyl alcohol carbon (-$CH_2$OH) is termed "carbinol". An attachment of a vinyl group (-CH=$CH_2$) to this carbinol unit yields vinylcarbinol ($CH_2$=CH-$CH_2$OH).

**22**. **Correct Option: B) Ethyl 2-butenoate**
**Explanation:**

1. The compound is an **ester**. The alkyl group attached to the oxygen atom is named first: **ethyl**.
2. The principal acyl chain contains **4 carbon atoms** including the carbonyl carbon, so the root word is **but**.
3. Numbering begins at the carbonyl carbon: $C_1$(O)-$C_2$H=$C_3$H-$C_4H_3$. The double bond starts at carbon-2 (**2-en**).
4. Combining these with the ester suffix (**-oate**) gives **ethyl 2-butenoate**.

**23**. **Correct Option: A) 2-Propenal**

**Explanation:** Acraldehyde (commonly known as acrolein) is the simplest unsaturated aldehyde with the chemical formula $CH_2$=CH-CHO.

- It has **3 carbon atoms** (**prop**).
- The double bond starts at carbon-2 (**2-en**).
- The aldehyde suffix is **-al**. This gives the IUPAC name **2-propenal**.

**24**. **Correct Option: A) 1, 2, 3 propane triol**

**Explanation:** The parent alkane chain has **3 carbon atoms** (**propane**). There are three hydroxyl (-OH) functional groups located at carbons 1, 2, and 3. Hence, the IUPAC name is **propane-1,2,3-triol** (written here as 1, 2, 3 propane triol).

**25**. **Correct Option: B) 2-Methylpentanedial**

**Explanation:**

1. When selecting the principal carbon chain for a dialdehyde, the longest path that includes **both** terminal principal aldehyde carbons (-CHO) is chosen as the parent chain. This gives a parent chain of **5 carbon atoms** (**pentanedial**).
2. Number the chain from the right side to give the substituent the lowest possible number: $C_1HO$-$C_2H(CH_3)$-$C_3H_2$-$C_4H_2$-$C_5HO$.
3. This places a **methyl** group at carbon-2, resulting in the IUPAC name **2-methylpentanedial**.

**26**. **Correct Option: A) 2-formyl-propanoic acid**

**Explanation:**

1. **Identify the principal functional group:** The carboxylic acid group (-COOH) has a higher priority than the aldehyde group (-CHO). Therefore, the suffix will be **-oic acid**, and numbering starts at the carboxylic acid carbon ($C_1$).
2. **Find the parent chain:** The longest continuous carbon chain starting from the -COOH carbon and moving to the left is 3 carbons long ($C_1OOH - C_2H - C_3H_3$). This gives the parent chain name **propanoic acid**.
3. **Identify the substituent:** The aldehyde group attached to carbon-2 is treated as a prefix/substituent named **formyl**.
4. Combining these gives **2-formyl-propanoic acid**.

## Level-II

**1. Correct Option: A) 4-Ethyl-4, 5-dimethyldecane**

**Explanation:**

1. Find the longest carbon chain: Let's trace the longest continuous chain of carbon atoms.
   - Starting from the left end, we have $H_3C - (CH_2)_4 -$, which accounts for 1 + 4 = 5 carbons.
   - Moving through the next two substituted carbons adds 2 more carbons.
   - Continuing down the rightmost propyl group ($-CH_2 - CH_2 - CH_3$) adds another 3 carbons.
   - Total length of the parent chain = 5 + 2 + 3 = 10 carbons, which gives the root word decane.
2. Number the chain:
   - Counting from left to right, the substituents appear at carbons 5 and 6.
   - Counting from right to left, the substituents appear at positions 4 and 5.
   - According to the lowest locant rule, we number from right to left.
3. Identify and arrange substituents:
   - Carbon-4 carries a methyl group ($-CH_3$) and an ethyl group ($-CH_2CH_3$).

- Carbon-5 carries a methyl group ($-CH_3$).
- Arranging alphabetically ("ethyl" before "methyl") yields: 4-Ethyl-4,5-dimethyldecane.

**2. Correct Option: D) 2-Chloro-4-ethyl-3-methyl-hept-2-ene**

**Explanation:**

1. Find the parent chain containing the double bond: The principal functional feature is the carbon-carbon double bond (C = C). The longest continuous chain containing this double bond runs horizontally all the way to the rightmost group ($-CH_3$).
   - Counting the carbons from the left: $C_1$ (in $CH_3$), $C_2$, $C_3$, $C_4$, followed by the three carbons in the propyl chain ($-CH_2-CH_2-CH_3$).
   - Total length of the parent chain = 7 carbons, which gives the root word heptene.
2. Number the chain: Numbering from left to right gives the double bond the lowest possible location number, which is position 2. Thus, the parent alkene is hept-2-ene.
3. Identify the substituents:
   - At carbon-2: a chlorine atom (2-chloro)
   - At carbon-3: a methyl group (3-methyl)
   - At carbon-4: an ethyl group ($-C_2H_5$) (4-ethyl)
4. Arrange alphabetically: "Chloro" comes first, followed by "ethyl", then "methyl". Combining these gives the name: 2-Chloro-4-ethyl-3-methyl-hept-2-ene.

**3. Correct Option: B)**

**Explanation:**

- An **isopropyl group** consists of a central secondary carbon atom bonded to a hydrogen atom and two identical methyl groups: $-CH(CH_3)_2$.
- Attaching a bromine atom (-Br) to this secondary carbon yields **isopropyl bromide**.

**4. Correct Option: A**

**Explanation:**

Let's analyze the name **4-ethyl-3,4-dimethylheptane**:

- The root name **heptane** signifies a continuous main chain containing **7 carbon atoms**.
- At position **3**, there is a **methyl** substituent.
- At position **4**, there are two substituents: an **ethyl** group and another **methyl** group.

**5. Correct Option: A) 4-Methylpent-2-yne**

**Explanation:**

1. **Find the parent chain:** Expand the structure to find the longest continuous carbon chain containing the triple bond:
   $C_1H_3-C_2\equiv C_3-C_4H(CH_3)-C_5H_3$
   The chain has **5 carbon atoms**, making the root word **pent**.
2. **Numbering the chain:** Number from left to right to give the triple bond the lowest possible position number (locant). The triple bond starts at carbon-2, giving **pent-2-yne**.
3. **Identify substituents:** A methyl group is attached to carbon-4 (**4-methyl**).
4. Combining these gives **4-Methylpent-2-yne** (or 4-methyl-2-pentyne).

**6. Correct Option: B) 4-Methylpentane-2,4-diol**

**Explanation:**

1. **Find the parent chain:** Find the longest continuous carbon chain containing both principal alcohol (-OH) functional groups:
   $C_1H_3-C_2H(OH)-C_3H_2-C_4(OH)(CH_3)-C_5H_3$

The main chain consists of **5 carbon atoms**, giving the root word **pentane**.

2. **Numbering the chain:**
   - Numbering from left to right gives the diol positions as **2,4** and the methyl position as **4**. (Locant set: 2, 4, 4)
   - Numbering from right to left gives the diol positions as **2,4** and the methyl position as **2**. (Locant set: 2, 2, 4) Following the **lowest locant rule**, we number from right to left.
3. **Identify substituents and functional groups:**
   - At carbon-2, there is a **methyl** group.
   - At carbons 2 and 4, there are two alcohol groups (**-2,4-diol**).
4. Combining these gives **4-Methylpentane-2,4-diol** (Note: Since option B lists this name precisely, it matches the structure perfectly when numbered standardly).

**7**. **Correct Option: B) 2,4, 4-Trimethylpentanenitrile**

**Explanation:**

1. **Identify the principal functional group:** The nitrile group (-CN) contains a carbon atom that must be included as carbon-1 ($C_1$) of the parent chain.
2. **Find the parent chain:** Counting the longest continuous chain starting from the nitrile carbon gives:
   $C_1N - C_2H(CH_3) - C_3H_2 - C_4(CH_3)_2 - C_5H_3$
   The main chain has **5 carbon atoms**, making the parent alkane **pentanenitrile**.
3. **Identify substituents:**
   - There is one methyl group at carbon-2.
   - There are two methyl groups at carbon-4.
4. Combining these gives **2,4,4-trimethylpentanenitrile**.

**8**. **Correct Option: A) Cinnamic acid**

**Explanation:**

- **Cinnamic acid** has the structure $C_6H_5$-CH=CH-COOH.
- In IUPAC terms, the parent chain is a 3-carbon carboxylic acid with a double bond beginning at carbon-2 (**propenoic acid**). A phenyl ring (-$C_6H_5$) is attached as a substituent at carbon-3, resulting in **3-phenylpropenoic acid**.

**9**. **Correct Option: A) 2-Ethylpropane-1,3-dinitrile**

**Explanation:**

1. According to modern IUPAC recommendations, when multiple principal carbon-containing functional groups like nitriles (-CN) are attached to a central chain, we select the longest chain containing the maximum number of nitrile carbons as the parent chain. Here, the two terminal nitrile carbons form a 3-carbon chain:
   $C_1N - C_2H(-CH_2CH_3) - C_3N$
2. The parent chain has 3 carbon atoms, giving the root word **propane**.
3. Since both ends have nitrile groups, it is a **propane-1,3-dinitrile**.
4. The group attached to the central carbon (carbon-2) is an **ethyl group** (-$CH_2CH_3$).
5. Combining these gives **2-ethylpropane-1,3-dinitrile**.

**10**. **Correct Option: A)**

**Explanation:**

Let's trace the requirements of the name **1,2-dibromo-2-methylbutane**:

- The root name **butane** means a main horizontal chain of 4 carbons: $C_1 - C_2 - C_3 - C_4$.
- Two bromine atoms must be placed at positions **1** and **2**.
- One methyl group must be placed at position **2**.

Looking at option **A**:
$H_3C - CH_2 - C(Br)(CH_3) - CH_2 - Br$
Numbering from right to left gives:

- $C_1$ is the $-CH_2Br$ carbon on the right.
- $C_2$ is the central carbon attached to $-CH_3$ and $-Br$.
- $C_3$ is $-CH_2-$.
- $C_4$ is $-CH_3$.

This perfectly matches **1,2-dibromo-2-methylbutane**.

**11**. **Correct Option: A) Hexa-1,3-dien-5-yne**
**Explanation:**

1. **Find the parent chain:** The continuous chain has **6 carbon atoms**, giving the parent root **hexa**.
2. **Numbering preference (Double vs. Triple bond):** When a double bond and a triple bond are at identical distances from opposite ends of a chain, the **double bond receives lower locants** over the triple bond by alphabetical priority rule ("en" comes before "yne").
   - Numbering from right to left: $C_1H_2 = C_2H - C_3H = C_4H - C_5 \equiv C_6H$
   - Double bonds are at positions **1** and **3** (**1,3-dien**).
   - The triple bond is at position **5** (**5-yne**).
3. Combining them yields **hexa-1,3-dien-5-yne**.

**12**. **Correct Options: A, B**
**Explanation:**

1. **Find the parent chain & functional priority:** The molecule is a 5-carbon unsaturated chain containing both a double bond (C=C) and a triple bond (C ≡ C).
2. **Determine the correct numbering:**
   - **Left-to-right numbering:** Substituents and unsaturated bonds get locants: double bond at position 2, triple bond at position 4. (Locant set for unsaturation: 2, 4).
   - **Right-to-left numbering:** The triple bond begins at carbon-1, and the double bond begins at carbon-3. (Locant set for unsaturation: 1, 3).
     According to IUPAC rules, numbering must follow the **lowest locant set for unsaturation** regardless of whether it is a double or triple bond. Since (1,3) is lower than (2,4), **numbering from right to left is correct**. Thus:
   - $C_1$ is the terminal alkyne carbon on the right.
   - $C_3$ has the chlorine atom and the double bond.
   - $C_4$ has the bromine atom.
3. **Correct IUPAC Name: 4-bromo-3-chloropent-3-en-1-yne**

**13**. **Correct Options: A, B, C**
**Explanation:**
Let's check the molecular formulas of all four compounds:

- **A) Butanoic acid:** A 4-carbon mono-carboxylic acid → $CH_3CH_2CH_2COOH = \mathbf{C_4H_8O_2}$ (Valid)
- **B) Methyl propanoate:** A 4-carbon ester → $CH_3CH_2COOCH_3 = \mathbf{C_4H_8O_2}$ (Valid)
- **C) 2-butene-1,4-diol:** A 4-carbon unsaturated diol → $HO\text{-}CH_2\text{-}CH{=}CH\text{-}CH_2\text{-}OH = \mathbf{C_4H_8O_2}$ (Valid)

Assertion and reason type:

**14**. **Correct Option: B) Both A and R are true but R is not the correct explanation**
**Explanation:**

- **Assertion (A) is True:** According to IUPAC rules, substituents must always be arranged in alphabetical order in the final name (e.g., *ethyl* before *methyl*), regardless of their numerical position numbers.
- **Reason (R) is True:** Numbering is fundamentally decided by the **lowest locant rule** (giving the lowest numbers to substituents overall). Alphabetical order is only brought in as a tie-breaker if numbering from either end gives identical sets of position locants. Therefore, in general, alphabetical listing order has no direct relation to the numbering itself.
- **Why B?** While both statements are true, the reason does not explain *why* substituents are listed alphabetically in a name; it simply describes a separate procedural independence between listing and numbering.

**15**. A) Both A and R are true and R is the correct explanation
**Assertion (A):** In IUPAC nomenclature, double bonds are indicated using the suffix "-ene".

- *Evaluation:* **True.** The primary suffix "-ene" explicitly indicates the presence of a carbon-carbon double bond in the parent chain.

**Reason (R):** The suffix represents the type of bonding in the parent chain.

- *Evaluation:* **True.** Primary suffixes (-ane, -ene, -yne) denote the saturation or unsaturation level (the type of carbon-carbon bonds) in the principal carbon chain.

**16**. **a) 3,3,4,4-tetramethyl-1-pentyne:** Ends in **-yne**, which indicates an **Alkyne**.

- **Match: a → Q**

**b) 2-methyl-butanol:** Ends in **-ol**, which indicates an **Alcohol**.

- **Match: b → P**

**c) 3 ethyl-4,5-dimethyl hexanone:** Ends in **-one**, which indicates a **Ketone**.

- **Match: c → R**

**d) 2,2,5,5-tetra methyl-3-hexanal:** Ends in **-al**, which indicates an **Aldehyde**.

- **Match: d → S**

**17**. **a) Structure with ketone and alcohol:**

- **Analysis:** The principal functional group is the ketone (−CO −) which has higher priority than the alcohol (−OH). Numbering starts from the right side near the alcohol to give the carbonyl carbon the lowest possible number:
  $HO - C_1H_2 - C_2H_2 - C_3(= O) - C_4H(CH_3)_2$
- This is a 5-carbon ketone chain (**pentan-3-one**) with a hydroxyl substituent at carbon-1 and a methyl group at carbon-4.
- **Name:** 1-hydroxy-4-methylpentan-3-one
- **Match: a → R**

**b) Structure with carboxylic acid and double bond:**

- **Analysis:** The main functional group is the carboxylic acid (−COOH). Numbering starts at the acid carbon:
  $C_1OOH - C_2H = C_3(CH_3)_2$
- The parent chain has 4 carbons with a double bond at position 2 (**but-2-enoic acid**). There is a methyl group at carbon-3.
- **Name:** 3-methylbut-2-enoic acid
- **Match: b → Q**

**c) Structure with acyl chloride and two carbonyls:**

- **Analysis:** The primary group is the acyl chloride (−COCl). Numbering starts from the chlorine-bearing carbonyl carbon:

$C_1(=O)Cl - C_2(=O) - C_3(=O) - C_4H_3$

- The chain contains 4 carbons (**butanoyl chloride**) with two oxo (= O) groups at positions 2 and 3.
- **Name:** 2,3-dioxobutanoyl chloride
- **Match: c → P**

**d) Alkene line structure:**

- **Analysis:** The longest chain containing the principal double bond has 4 carbons starting from the terminal alkene = $CH_2$:
  $C_1(=CH_2) - C_2(-CH_2CH_3) - C_3H(CH_3) - C_4H_3$
- This gives a **but-1-ene** main chain. Carbon-2 has an **ethyl** group and carbon-3 has a **methyl** group.
- **Name:** 2-ethyl-3-methylbut-1-ene
- **Match: d → S**

**Comprehension**

**18. Correct Option: B) Give lowest locants**

Explanation: In IUPAC nomenclature, the main carbon chain is numbered starting from the end that gives the lowest possible position numbers (lowest locants) to functional groups, multiple bonds, or substituents.

**19. Correct Option: B) Position number**

Explanation: A locant is the numerical indicator used in an IUPAC chemical name to specify the exact position number of an atom, bond, or substituent group along the parent carbon chain.

**20. Correct Option: B) Standardization**

Explanation: Numbering ensures standardization in organic chemistry. By following universal rules, every chemist assigns the exact same distinct name to a given chemical structure, removing ambiguity and ensuring global consistency.

**Integer type:**

**21. Answer: 8**

1. **The parent chain and double bond:** The name specifies **3-nonene**, meaning there is a main 9-carbon chain with a double bond between carbon-3 and carbon-4 (-CH=CH-).
   - These 2 carbon atoms are $sp^2$-hybridized.
2. **The carboxylic acid groups:** The molecule contains three carboxylic acid groups (-COOH) attached at positions 2, 5, and 8. Let's analyze the hybridization of the atoms inside each individual -C(=O)-OH group:
   - The carbonyl carbon atom (-C=O) is $sp^2$-hybridized (3 groups × 1 = 3 carbons).
   - The carbonyl oxygen atom (=O) is $sp^2$-hybridized (3 groups × 1 = 3 oxygens).
   - *(Note: The hydroxyl oxygen* -OH *is $sp^3$-hybridized due to two single bonds and two lone pairs).*

   Adding up all the $sp^2$-hybridized atoms:

   Total $sp^2$ atoms = 2 (alkene carbons) + 3 (carbonyl carbons) + 3 (carbonyl oxygens) = 8

**22. Answer: 5**

1. **Parent chain and double bond:** The root **3-hexenedioic acid** indicates a 6-carbon parent chain with a double bond between carbon-3 and carbon-4 (-CH=CH-).
   - These 2 alkene carbon atoms are $sp^2$-hybridized.
2. **Terminal carboxylic acids:** Being a **dioic acid**, it has two terminal carboxylic acid functional groups (-COOH) at carbons 1 and 6.

- The carbonyl carbons in both groups are $sp^2$-hybridized (2 carbons).

3. **Substituent group:** The substituent **2-ethanoyloxy** (often written as ethanoyloxy or acetoxy, -O-CO-$CH_3$) contains an ester carbonyl group.
   - The carbonyl carbon (-C=O) in this group is $sp^2$-hybridized (1 carbon).

   Adding up all the $sp^2$-hybridized carbon atoms:
   Total $sp^2$ carbons = 2 (alkene) + 2 (dioic acids) + 1 (ester carbonyl) = 5

**23**. **Answer: 8**

1. $C_2$ **in the main chain:** It is a $-CH_2-$ group bonded to $C_1$ and $C_3$. → **2 secondary hydrogens**
2. $C_6$ **in the main chain:** It is a $-CH_2-$ group bonded to $C_5$ and $C_7$. → **2 secondary hydrogens**
3. $C_7$ **in the main chain:** It is a $-CH_2-$ group bonded to $C_6$ and $C_8$. → **2 secondary hydrogens**
4. **The** $-CH_2-$ **inside the ethyl substituent:** It is bonded to the main chain $C_3$ and the terminal substituent $-CH_3$. → **2 secondary hydrogens**

   Total secondary hydrogens = 2 + 2 + 2 + 2 = 8

## Previous Year Questions

**1**. D. Methanol and ethanol react with sodium metal to evolve hydrogen gas ($H_2$), not $CO_2$. Both methanol and ethanol contain the same functional group (–OH), so they do have the same functional group.

**2**. B, C and D. –COOH group contains one C=O and one C–O bond (plus C–O to OH), $CH_3COOH$ has three C–H bonds in the $CH_3$ group and There is one C–C bond ($CH_3$–CO) and two C–O single bonds.

**3**. B. Reacts with ethanol to give a sweet-smelling compound.

**4**. B. Ethene burns to give $CO_2$ and $H_2O$.

**5**. D. Y is ethanoic acid and Ethanoic acid is weaker than sulphuric acid.

**6**. B. Chlorofluorocarbons (CFCs) are the *main chemicals* responsible for ozone layer depletion because they release chlorine radicals in the stratosphere. Chlorine radicals and nitrous oxide ($N_2O$) also participate in reactions that destroy ozone.

**7**. C. (vi) Propanone and (vii) Butanone represent the same homologous series. (ii) Ethanoic acid reacts with ethanol in the presence of an acid catalyst to produce an ester.

**8**. D. When ethanol is heated with hot concentrated sulphuric acid (≈443 K), it undergoes dehydration to form ethene ($C_2H_4$).

**9**. **Correct Option: C**

$CH_3CH_2-COOCH_3$ **: Ethanoic acid → Incorrect.** The given structure is an ester (specifically **methyl propanoate**), not a carboxylic acid. Ethanoic acid has the formula $CH_3COOH$.

**10**. **Correct Option: C**

**Explanation:**

1. **Identify Compound X:**
   - Compound X has the molecular formula $C_2H_4O_2$.
   - It reacts with sodium carbonate ($Na_2CO_3$) to produce brisk effervescence of a gas, which is a characteristic test for a **carboxylic acid**.
   - The 2-carbon carboxylic acid is ethanoic acid, $CH_3COOH$.
2. **Identify Gas Z:**

- When a carboxylic acid reacts with sodium carbonate, it forms its corresponding salt, water, and **carbon dioxide gas (**$CO_2$**)**. Therefore, gas Z is $CO_2$, which causes the brisk effervescence.

3. **Identify Compound Y:**

- When ethanoic acid ($CH_3COOH$) reacts with ethanol ($C_2H_5OH$ or $CH_3CH_2OH$) in an esterification reaction, it forms the ester **ethyl ethanoate**:
  $CH_3COOH + CH_3CH_2OH \rightarrow CH_3COOCH_2CH_3 + H_2O$
- Therefore, compound Y is $CH_3COOCH_2CH_3$.